

Hybrid Modelling through Simplified Physics and Data-Driven Correction

A Comparative Modelling Study of Lithium Diffusion in a Spherical Particle

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Executive Summary

Purpose and Decision Context

This report documents the development and evaluation of a family of numerical, physics-informed, hybrid, and data-driven models for lithium diffusion and temperature evolution in a spherical battery particle. Its purpose is to support an engineering decision that arises when a simulation tool must be both reliable and sufficiently fast for its intended use:

When is a reduced representation preferable to a higher-fidelity physical simulator, which reduced-modelling strategy is appropriate, and when is a separately trained data-driven correction technically justified?

The report treats accuracy, physical consistency, offline cost, online latency, robustness, stochastic variability, model complexity, and engineering effort as separate quantities. It does not assume that the model with the lowest error, the shortest inference time, or the smallest training cost is automatically suitable for every industrial application.

The selected case study is solid-state lithium diffusion in a spherical electrode particle. A conservative finite-volume formulation is first derived, implemented, and numerically validated. The model is then extended by adding a lumped thermal state and a diffusion coefficient that depends on concentration and temperature. This thermo-diffusive finite-volume model generates the higher-fidelity numerical reference used in the learning experiments. The reference is an internal synthetic benchmark and is not presented as experimental ground truth or as a complete electrochemical cell model.

Implemented Modelling Strategies

Two reduced-modelling streams are investigated.

The first stream retains an explicit physical solver. A constant-diffusivity finite-volume model is used as the simplified baseline. It preserves the conservative discretization but omits the state dependence of the diffusion coefficient and the thermal coupling. A feedforward neural network is then trained to predict the radial discrepancy between this simplified solution and the thermo-diffusive reference. The resulting FVM-FNN model applies an affine projection to the correction so that the prescribed lithium-inventory relationship is retained.

The second stream uses a Physics-Informed Neural Network constrained by a deliberately simplified diffusion equation and surface-flux representation. The PINN combines reference observations with reduced physical constraints. A post-training Hybrid-PINN is subsequently constructed by freezing one trained PINN checkpoint and training a separate residual network. The residual architecture retains the PINN initial condition and centre symmetry by construction, while mass balance, surface-flux consistency, and PDE consistency are assessed a posteriori.

The omitted thermal dynamics are also studied independently. SINDyC identifies a sparse controlled temperature equation from multiple thermal trajectories, whereas a thermal feedforward network learns the corresponding temperature derivative without a predefined candidate library. Both approaches are evaluated through trajectory rollout under interpolation and extrapolation conditions.

Publication Benchmark at a Glance

The final publication preset executed seven models using five seeds and two complete repeats per seed, producing 70 successful runs with no failures and no skipped models. Every run was performed in an isolated Python process. Five warm-up evaluations and repeated synchronized timing measurements were used for the reported online latencies. The complete sequential benchmark required 4.63 h on an Apple Silicon workstation.

Table 1 reports the principal concentration-model outputs. The FVM and FVM-FNN times are end-to-end values. The PINN and Hybrid-PINN values correspond to complete neural evaluation of the 20,301-point concentration field. The models were executed on their configured backends: CPU for the finite-volume and FVM-FNN pipelines and Apple MPS for the PINN-based pipelines.

Table 1: Principal publication-benchmark results for the concentration models. Values are run-level means across ten seed-repeat executions.

Model	Field RMSE	Latency [ms]	Speedup	Training [s]	Maximum mass residual
Reference FVM	≈ 0	108.679	1.000	–	4.467×10^{-5}
Simplified FVM	8.208×10^{-3}	80.767	1.346	–	4.552×10^{-5}
FVM-FNN	2.964×10^{-4}	83.793	1.297	11.797	4.500×10^{-5}
Standalone PINN	3.870×10^{-2}	4.648	23.383	590.203	1.326×10^{-1}
Hybrid-PINN	1.241×10^{-3}	7.031	15.457	121.328 [†]	5.201×10^{-3}

[†]Residual-network training only. The value excludes the previous training of the frozen PINN; the measured residual-dataset generation and residual-training pipeline required 121.551 s.

The FVM-FNN model reduced the complete-field RMSE of its common simplified baseline by 96.39%. Its mean relative L_2 error was 3.978×10^{-4} . All reference-FVM, simplified-FVM, and FVM-FNN runs remained within the normalized concentration bounds and passed the 10^{-4} mass-balance criterion.

The standalone PINN had a mean relative L_2 error of 5.194×10^{-2} . Its predictions contained an average of 123.8 out-of-bounds coordinates, equal to 0.610% of the evaluated field, and none of the ten runs passed the mass-balance threshold. The Hybrid-PINN branch used the same frozen PINN checkpoint in every run and trained ten residual models. It produced no concentration-bound violation and had a mean relative L_2 error of 1.666×10^{-3} . Its maximum mass residual remained above the adopted tolerance in all ten runs. Because the standalone benchmark PINNs were trained separately for each seed, the standalone-PINN and Hybrid-PINN rows are not a paired before-and-after comparison of identical PINN realizations.

The compute-only break-even values reported by the benchmark were 474 calls for the FVM-FNN model, 5,673 calls for the standalone PINN, and 1,194 calls for the incremental Hybrid-PINN correction. These values compare machine training time with online time saved relative to the reference FVM. They do not include engineering effort, maintenance, all reference-data generation, or, for the Hybrid-PINN, the original frozen-PINN training.

The frozen nominal FVM-FNN network was also evaluated on five stored simplified-model variants without retraining. The mean RMSE changes relative to the corresponding uncorrected variants were +9.40%, +11.60%, −220.12%, −58.26%, and −34.00%, where a positive value denotes

a reduction in error. No concentration-bound violation was recorded for these corrected variants.

For the independent thermal comparison, SINDyC produced interpolation and extrapolation temperature RMSE values of 3.40×10^{-4} K and 4.94×10^{-4} K, respectively. The thermal FFN produced corresponding mean RMSE values of 3.808×10^{-3} K and 6.549×10^{-2} K. SINDyC identified two active terms and required 5.454 s for model identification. The thermal FFN contained 1,185 trainable parameters and required 34.437 s for training. Their mean interpolation rollout times were 1.017 s and 2.722 s, respectively. These thermal results refer to a separate scalar-output task and are not direct latency comparisons with the concentration models.

The recorded project effort was 152 person-hours, equivalent to 19 person-days at eight hours per day. The largest recorded components were documentation (37 h), problem definition (36 h), validation (28 h), surrogate development (26 h), and reference-model development (16 h). Dataset engineering accounted for 6 h and the activity recorded specifically as training accounted for 3 h.

How to Use This Report

The document is organized so that an internal reader can enter at the section most relevant to the decision or task at hand.

For the governing equations and physical assumptions: consult *Physical Model of Lithium Diffusion in a Spherical Particle* and *Non-Dimensional Formulation*. These sections define the state variable, conservation law, boundary conditions, characteristic scales, and quantities of interest.

For numerical implementation and solver verification: consult *Spatial Discretization*, *Conservative Finite-Volume Discretization*, *Treatment of the Particle Centre*, *Treatment of the Particle Surface*, *Numerical Time Integration*, *Python Implementation*, and *Baseline Solver Validation Results*. These sections document the conservative numerical scheme, analytical comparison, mesh convergence, and numerical checks.

For reference-model and dataset generation: consult *Model Extension* and *Results of the Extended Reference Model*. They describe state-dependent diffusivity, thermal coupling, reference trajectories, conservation checks, and the synthetic datasets used by the learned models.

For the standalone physics-informed surrogate: consult *Physics-Informed Neural Network with Simplified Physics* and *PINN Results and Comparison with the Full-Physics Model*. These sections specify the reduced physics, exact architectural constraints, training objective, held-out tests, full-grid accuracy, and physical diagnostics.

For correcting an explicit simplified physical solver: consult *Hybrid Physics-ML Modelling through Residual Learning*, *Hybrid Model Results*, and *Comparison of the Simplified, PINN and Hybrid Models*. They document residual-target construction, the radial FNN, the inventory projection, nominal accuracy, and the frozen-network variant tests.

For identifying or learning the omitted thermal dynamics: consult *Data-Driven Reconstruction of the Thermal Dynamics: SINDyC and Feedforward Neural Networks*. It contains trajectory generation, sparse-library selection, FFN training, interpolation, extrapolation, rollout timing, and model-complexity measurements.

For correcting an already trained PINN: consult *Post-Training Residual Correction of the Physics-Informed Neural Network*. This section describes checkpoint freezing, residual-data generation, correction training, inherited hard constraints, and the physical properties evaluated only through diagnostics.

For a direct operational comparison: consult *Operational Benchmark and Model-Selection Guidance*. This section defines the common metrics, reports aggregated publication results, separates model-specific and end-to-end timing, and provides the evidence used by the final scenario-based discussion.

For implementation-oriented summaries: consult the appendices containing the numerical algorithms and project flowcharts. These provide compact views of dataset generation, baseline construction, and residual-learning connections.

Scope of the Reported Evidence

The benchmark compares the implemented software pipelines rather than idealized model classes. Hardware assignments, data availability, training protocols, and output dimensionality are part of the reported configurations. The concentration models and thermal models solve different prediction tasks and are presented in separate comparisons.

The FVM-FNN robustness experiment changes stored simplified-model variants while keeping the correction network frozen. The thermal study uses multiple current trajectories and includes separate interpolation and extrapolation tests. The standalone PINN and Hybrid-PINN use point-wise splits drawn from one operating trajectory; those results do not constitute a trajectory-wise transfer test to a new current history.

Finally, predictive errors are measured relative to the internal thermo-diffusive numerical reference. The report establishes numerical and methodological evidence within this controlled setting; experimental validation, deployment monitoring, energy consumption, memory footprint, and long-term software-maintenance costs remain outside the measured scope.

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1 Introduction

1.1 Industrial Motivation

High-fidelity physical models provide detailed and physically interpretable descriptions of complex engineering systems. Their numerical solution, however, may require computational resources and execution times that are not compatible with applications involving repeated simulations, optimization, control, uncertainty quantification, or rapid design exploration.

Reduced descriptions offer an alternative. A physical model may be simplified by neglecting secondary mechanisms, reducing the constitutive complexity, or adopting less demanding numerical assumptions. Alternatively, a data-driven surrogate may learn the input–output behaviour of the system without solving the complete governing equations during online operation. Both strategies can substantially reduce computational cost, but neither is universally preferable.

A simplified physical model may preserve interpretability, conservation properties, and a direct connection with engineering parameters, while introducing a systematic modelling error. A learned surrogate may provide a larger reduction in online latency, but its reliability depends on the representativeness of the training data, on the imposed constraints, and on its behaviour outside the training domain. Consequently, model selection cannot be based on inference speed or predictive accuracy alone.

From an industrial perspective, the relevant question is not simply whether a reduced model can approximate a more detailed simulator. It is necessary to determine:

- whether the computational saving justifies introducing a reduced representation;
- which type of reduced approach is appropriate for the intended application;
- whether the resulting predictions are sufficiently accurate and physically admissible;
- whether the model remains reliable when operating conditions change;
- whether the remaining discrepancy can be corrected without replacing or retraining the complete baseline model;
- how training, validation, deployment, monitoring, and possible retraining affect the total lifecycle cost.

The present work addresses these questions through a controlled modelling study. The objective is not to declare a universally superior model, but to construct a reproducible framework through which different simulation strategies can be evaluated and assigned an appropriate operational role.

1.2 Scientific Question

The central scientific question of the study is formulated as follows:

When should a reduced model be preferred to a higher-fidelity physical simulator, which modelling approach provides the most appropriate balance between computational efficiency and reliability, and under which conditions should the reduced prediction be corrected through a separately trained data-driven model?

This question combines model reduction, physics-guided learning, discrepancy modelling, and operational model selection. It is addressed by comparing two principal reduced-modelling streams and by applying the same additive correction principle to both of them.

The first stream is based on an independently executable simplified physical model. The second is based on a Physics-Informed Neural Network (PINN) constrained by deliberately simplified governing equations. These alternatives represent two different ways of exploiting reduced physics: one retains an explicit numerical solver, whereas the other embeds the reduced equations inside a learned surrogate.

Both baselines are subsequently extended through residual learning. A dedicated correction network learns the discrepancy between the baseline prediction and the higher-fidelity reference data. This leads to two complementary hybrid strategies:

$$u_{\text{hybrid}} = u_{\text{baseline}} + \widehat{\delta u}, \quad (1)$$

where u_{baseline} may be either the simplified finite-volume solution or the frozen PINN prediction, and $\widehat{\delta u}$ denotes the learned residual correction.

The study therefore investigates not only whether residual learning improves accuracy, but also whether the correction preserves the physical properties of the baseline, remains valid under trajectory and parameter perturbations, and provides a computational advantage large enough to justify its development and training cost.

1.3 Research Questions

The central scientific question is decomposed into six research questions. These questions define the logical structure of the study and are answered quantitatively by the final operational benchmark.

- RQ1 – Reduction decision.** Under which accuracy, physical-reliability, and computational requirements is a reduced representation preferable to the higher-fidelity reference simulator?
- RQ2 – Baseline selection.** When a reduced representation is appropriate, should it be constructed as an explicit simplified physical model or as a physics-informed learned surrogate?
- RQ3 – Reliability.** How accurately does each model reproduce the reference solution, which physical properties are satisfied, and which constraints are exact rather than empirical?
- RQ4 – Computational performance.** What offline cost is required to generate data and train each model, how fast is the resulting online prediction, and after how many simulations is the initial computational investment recovered?
- RQ5 – Value of residual correction.** When does a separately trained residual model provide a meaningful improvement over the simplified baseline, and how should the correction be constructed for physical and physics-informed baselines?
- RQ6 – Operational validity.** How sensitive are the learned models to stochastic initialization and to changes in the operating trajectory or simplified-model parameters, and when are retraining, fallback, or additional validation required?

These questions deliberately separate predictive accuracy from physical reliability and computational convenience. A model can be fast but physically inadmissible, accurate but expensive to train, or reliable only inside a narrow operating domain. The final recommendation must therefore depend on the requirements of the specific use case.

1.4 Case Study and Modelling Scope

The methodology is investigated through lithium diffusion inside a spherical electrode particle. The problem is derived from the Single Particle Model and captures a fundamental transport mechanism occurring in lithium-ion battery electrodes while remaining sufficiently controlled for systematic numerical analysis.

The case study is methodological rather than intended as a complete electrochemical-cell representation. The higher-fidelity simulator is used as a numerical reference within the scope of this work; it should not be interpreted as experimental ground truth or as a fully validated battery model.

A conservative finite-volume formulation is first developed for solid-phase diffusion with a constant diffusion coefficient. The numerical implementation is assessed through comparison with the Guo–White approximate analytical solution, mesh-convergence analysis, mass-conservation checks, and numerical-stability diagnostics.

The model is subsequently extended by introducing two mechanisms that are omitted from the reduced description:

1. a solid-state diffusion coefficient depending on local lithium concentration and particle temperature;
2. a lumped thermal balance accounting for current-induced heating and heat exchange with the surrounding environment.

The resulting thermo-diffusive formulation constitutes the higher-fidelity reference model used to generate the synthetic concentration and temperature datasets. The constant-diffusivity formulation is retained as the simplified physical baseline. This construction creates a controlled discrepancy between two models that share the same dominant diffusion mechanism but differ in constitutive and thermal complexity.

1.5 Two Reduced-Modelling Streams

1.5.1 Simplified Physical Model

The first reduced-modelling stream retains an explicit finite-volume solver while adopting a constant solid-state diffusion coefficient and omitting the thermal dependence included in the reference model.

This strategy preserves several desirable properties:

- the concentration field is generated through a conservative discretization;
- physical parameters remain explicit and interpretable;
- the model can be executed without training data;
- the solver can provide a robust fallback when a learned model is evaluated outside its validated domain.

Its main limitation is model-form error. Even when mass conservation is satisfied, the simplified constitutive description may not reproduce the radial lithium redistribution generated by the thermo-diffusive reference.

The simplified FVM is therefore evaluated both as an independent candidate and as the physical baseline of an FVM–FNN hybrid model.

1.5.2 Physics-Informed Neural Network

The second stream adopts a PINN trained using observations generated by the higher-fidelity simulator while enforcing a deliberately simplified diffusion equation. PINNs normally combine observational information with differential constraints and have been proposed as surrogate solvers in several engineering domains [Cuomo et al., 2022; Khanra et al., 2026; Niraimathi et al., 2026].

In the present work, the PINN is not supplied with the complete equations of the reference simulator. It is constrained by reduced physics, including a constant-diffusivity equation and a simplified surface-flux representation. The experiment consequently investigates whether observational data can compensate implicitly for physical mechanisms that are absent or misspecified in the embedded equations.

This setting differs from the conventional use of a PINN as a direct surrogate for the same equations used to generate the reference solution. It reflects an engineering situation in which partial physical knowledge is available, while the complete process can only be observed through data or an expensive simulator. Model misspecification can introduce a persistent bias even when the training loss is successfully minimized [Zou et al., 2024].

The standalone PINN is retained as an independent baseline. Its accuracy, physical consistency, training cost, and inference speed are measured before any residual correction is introduced.

1.6 Hybrid Correction of the Two Baselines

Hybrid modelling combines mechanistic structure with data-driven flexibility [Jia et al., 2020; Tu et al., 2024; Garg et al., 2022]. In this work, the same high-level residual-learning principle is applied to the two baseline streams, while preserving their distinct numerical and physical structures.

1.6.1 FVM-FNN Hybrid Model

In the first hybrid branch, a feedforward neural network learns the discrepancy between the thermo-diffusive reference field and the independently executable simplified FVM.

The network receives reduced physical states extracted from the simplified trajectory and predicts a complete radial correction. A hard affine projection is applied to the predicted correction so that its volume average is consistent with the lithium-inventory relationship identified from the training data.

This branch investigates whether a low-cost physical model can supply the main dynamics while a compact neural network corrects its systematic radial error. It also investigates the locality of the learned correction by applying a frozen nominal network to perturbed values of the simplified initial concentration and diffusion coefficient.

The correction is considered operationally useful only if it improves the baseline prediction without introducing unacceptable physical violations and if its offline cost can be recovered through repeated online simulations.

1.6.2 Post-Training Hybrid-PINN

In the second hybrid branch, the trained PINN is frozen and treated as a reusable physics-informed baseline. A coordinate-based feedforward network is then trained to predict the remaining pointwise residual:

$$C_{\text{Hybrid-PINN}}(\rho, \tau) = C_{\text{PINN}}(\rho, \tau) + \widehat{\Delta C}(\rho, \tau). \quad (2)$$

Freezing the PINN separates the expensive physics-informed optimization from the residual-fitting stage. The correction can therefore be studied without altering the original PINN or repeating its complete training procedure.

The residual architecture preserves the PINN initial condition and centre symmetry by construction. Other properties, including global mass balance, surface-flux consistency, and full-physics PDE consistency, are evaluated through separate diagnostics. They are not assumed to be inherited automatically from the baseline.

This distinction is central to the study: improving agreement with reference data does not necessarily imply that the corrected model satisfies the governing physics.

1.7 Data-Driven Reconstruction of the Thermal Dynamics

The thermal mechanism omitted from the simplified concentration model is also examined independently. Two data-driven approaches are compared:

- Sparse Identification of Nonlinear Dynamics with control (SINDyC), which searches for a compact governing equation within a predefined function library;
- a feedforward neural network, which learns the thermal derivative without requiring the governing terms to be specified in advance.

This comparison introduces a complementary form of discovery. Residual learning reconstructs the discrepancy left by a baseline model, whereas SINDyC attempts to identify an explicit mathematical structure Brunton et al. [2016b] for the omitted thermal dynamics.

The thermal models are trained using multiple current trajectories and are tested under both interpolation and extrapolation conditions. Their comparison addresses the trade-off between interpretability and representational flexibility. A sparse equation is attractive when the governing terms are contained in the candidate library, while a neural model may be preferable when the missing dynamics are more complex or structurally unknown.

The thermal models are evaluated separately from the concentration surrogates. Their rollout times and prediction errors refer to a different output and should not be interpreted as a direct ranking against the diffusion models.

1.8 Overview of the Investigated Strategies

Table 2 summarizes the role assigned to each modelling strategy.

The models do not receive identical information and are not intended to constitute a perfectly symmetric algorithmic competition. The simplified FVM and its neural correction operate on an explicit physical trajectory, whereas the PINN embeds a deliberately simplified boundary representation. Similarly, the thermal models address temperature dynamics rather than the complete concentration field.

The final comparison therefore evaluates the implemented strategies according to their intended roles. It does not establish the universal superiority of one model class over another.

Table 2: Overview of the modelling strategies investigated in this work.

Strategy	Physical baseline	Learned component	Role in the study
Reference FVM	Thermo-diffusive model	None	Higher-fidelity numerical reference, dataset generator, and validation standard.
Simplified FVM	Constant-diffusivity model	None	Training-free reduced physical model and robust operational fallback.
Standalone PINN	Simplified PDE embedded in the loss and architecture	Concentration surrogate	Physics-informed reduced representation and study of bias caused by model misspecification.
FVM-FNN	Simplified FVM	Radial residual correction	Hybrid strategy designed to improve the accuracy of the inexpensive physical solver.
Hybrid-PINN	Frozen standalone PINN	Coordinate-based residual correction	Post-training strategy designed to reduce PINN bias without repeating the complete PINN optimization.
SINDyC	Thermal state definition	Sparse controlled equation	Interpretable discovery of the omitted thermal dynamics.
Thermal FFN	Thermal state definition	Neural derivative model	Flexible black-box reconstruction used as a comparison with sparse equation discovery.

1.9 Methodological Contributions

The principal contributions of the work are:

1. the development and numerical validation of a conservative finite-volume solver for spherical solid-state diffusion;
2. the construction of a higher-fidelity thermo-diffusive numerical reference and the generation of aligned synthetic datasets;
3. the systematic comparison of two reduced-modelling streams: an explicit simplified physical solver and a PINN constrained by simplified physics;
4. the implementation of two modular residual-learning strategies for correcting the simplified FVM and the frozen PINN;
5. the distinction between exact architectural constraints, inherited physical properties, and physical consistency assessed only a posteriori;
6. the reconstruction of the omitted thermal dynamics through both sparse equation discovery and neural learning;
7. the evaluation of nominal accuracy, physical consistency, out-of-domain behaviour, stochastic stability, computational cost, and engineering effort within a common reproducible benchmark;
8. the translation of the numerical results into scenario-dependent model-selection, fallback, and retraining recommendations.

1.10 Methodological Roadmap

The study is organized into eight successive stages.

1. Numerical formulation and validation.

The spherical diffusion equation is non-dimensionalized and discretized using a conservative finite-volume formulation. The implementation is assessed through analytical comparison, mesh convergence, conservation checks, and numerical diagnostics.

2. Construction of the higher-fidelity reference.

Concentration- and temperature-dependent diffusion and a lumped thermal balance are introduced. The resulting thermo-diffusive simulator generates the reference concentration and temperature datasets.

3. Construction of the reduced baselines.

A constant-diffusivity FVM is retained as the simplified physical model. In parallel, a PINN is trained using reference observations and deliberately simplified physical constraints. Both models are evaluated independently before correction.

4. Residual correction of the simplified FVM.

An FNN learns the radial discrepancy between the simplified and reference concentration fields. A generalized lithium-inventory constraint is imposed on every corrected profile. Nominal accuracy and robustness to changes in the simplified-model parameters are evaluated.

5. Comparison and simplified-model calibration.

The reference, simplified FVM, standalone PINN, and FVM-FNN predictions are compared on a common structured grid. The analysis investigates how inexpensive candidate simplified models can be pre-screened before a correction network is trained.

6. Thermal-dynamics reconstruction.

Multiple thermal trajectories are generated. SINDyC and a feedforward neural network are trained and evaluated under interpolation and extrapolation conditions to compare structural interpretability with black-box flexibility.

7. Post-training correction of the PINN.

The standalone PINN is frozen and a separate residual network is trained. The Hybrid-PINN is evaluated in terms of accuracy, inherited hard constraints, mass conservation, boundary consistency, PDE residuals, regularity, and computational cost.

8. Operational benchmark and decision guidance.

All implemented strategies are executed using controlled seeds, repeats, timing repetitions, and recorded software configurations. The benchmark compares accuracy, physical reliability, training cost, online latency, robustness, stochastic stability, complexity, and engineering effort. The resulting evidence is translated into application-dependent model-selection and retraining recommendations.

2 Physical Model of Lithium Diffusion in a Spherical Particle

The spherical solid-diffusion formulation and its principal validation benchmark follow the model presented by Guo and White [Guo and White, 2012].

2.1 Physical Domain and State Variable

The physical system consists of a single spherical active material particle belonging to one of the electrodes of a lithium-ion cell. The particle is assumed to have radius R_p . This representative spherical-particle description is also central to Single Particle Model formulations and to the inference of solid-state diffusivity [Gümrükçüoğlu et al., 2024].

Due to spherical symmetry, the lithium concentration depends only on the radial coordinate r and on time t . The concentration field is therefore represented as

$$c_s = c_s(r, t),$$

where

$$r \in [0, R_p], \quad t \geq 0.$$

The subscript s indicates that the concentration refers to lithium in the solid phase of the active material particle.

The concentration $c_s(r, t)$ is measured in

$$\text{mol m}^{-3}.$$

2.2 Modelling Assumptions

The following assumptions are introduced:

1. the active material particle is perfectly spherical;
2. the lithium concentration is spherically symmetric, so that no angular dependence is considered;
3. lithium transport inside the particle occurs exclusively through solid-state diffusion;
4. the solid-phase diffusion coefficient D_s is spatially uniform and constant in time;
5. the particle radius R_p remains constant;
6. electrochemical intercalation and deintercalation occur only at the external surface of the particle;
7. the molar flux at the particle surface may vary with time and is denoted by $j(t)$.

Under these assumptions, the three-dimensional diffusion problem is reduced to a one-dimensional problem along the radial coordinate.

2.3 Mass Conservation

Consider a spherical shell bounded by the radial coordinates r and $r + \Delta r$.

The volume of the shell is approximately

$$\Delta V = 4\pi r^2 \Delta r.$$

Let $N_r(r, t)$ denote the radial molar flux of lithium through the solid particle. The molar flow rate crossing a spherical surface of radius r is

$$\dot{n}(r, t) = 4\pi r^2 N_r(r, t).$$

The accumulation of lithium inside the spherical shell is equal to the difference between the incoming and outgoing molar flow rates:

$$\frac{\partial c_s}{\partial t} \Delta V = \dot{n}(r, t) - \dot{n}(r + \Delta r, t).$$

Substituting the expressions for the shell volume and molar flow rate gives

$$\frac{\partial c_s}{\partial t} 4\pi r^2 \Delta r = 4\pi r^2 N_r(r, t) - 4\pi (r + \Delta r)^2 N_r(r + \Delta r, t).$$

Dividing by $4\pi r^2 \Delta r$ and taking the limit $\Delta r \rightarrow 0$ yields the local mass conservation equation

$$\frac{\partial c_s}{\partial t} = -\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 N_r).$$

2.4 Fick's Law

The radial molar flux inside the solid particle is described by Fick's first law:

$$N_r(r, t) = -D_s \frac{\partial c_s}{\partial r},$$

where D_s is the solid-phase diffusion coefficient, measured in

$$\text{m}^2 \text{s}^{-1}.$$

The negative sign indicates that diffusion occurs from regions of higher concentration toward regions of lower concentration.

Substituting Fick's law into the mass conservation equation gives

$$\frac{\partial c_s}{\partial t} = -\frac{1}{r^2} \frac{\partial}{\partial r} \left(-r^2 D_s \frac{\partial c_s}{\partial r} \right).$$

Assuming that D_s is constant, it can be taken outside the spatial derivative:

$$\boxed{\frac{\partial c_s}{\partial t} = D_s \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_s}{\partial r} \right)}$$

for

$$0 < r < R_p, \quad t > 0.$$

This is the solid-phase diffusion equation in spherical coordinates.

2.5 Expanded Form of the Diffusion Equation

The spatial differential operator can be expanded using the product rule:

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial c_s}{\partial r} \right) = 2r \frac{\partial c_s}{\partial r} + r^2 \frac{\partial^2 c_s}{\partial r^2}.$$

Dividing by r^2 gives

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_s}{\partial r} \right) = \frac{\partial^2 c_s}{\partial r^2} + \frac{2}{r} \frac{\partial c_s}{\partial r}.$$

The governing equation can therefore also be written as

$$\boxed{\frac{\partial c_s}{\partial t} = D_s \left(\frac{\partial^2 c_s}{\partial r^2} + \frac{2}{r} \frac{\partial c_s}{\partial r} \right)}$$

for $0 < r < R_p$.

The additional term

$$\frac{2}{r} \frac{\partial c_s}{\partial r}$$

is a consequence of the spherical geometry and would not appear in a one-dimensional Cartesian diffusion problem.

2.6 Initial Condition

The initial lithium concentration inside the particle is prescribed as

$$c_s(r, 0) = c_{s,0}(r).$$

A common simplifying assumption is a spatially uniform initial concentration:

$$\boxed{c_s(r, 0) = c_{s,0}} \quad 0 \leq r \leq R_p.$$

This means that, before the application of the external flux, lithium is uniformly distributed inside the particle.

2.7 Boundary Condition at the Particle Centre

At the centre of the spherical particle, no preferential radial direction exists. Spherical symmetry therefore requires the concentration gradient to vanish:

$$\boxed{\left. \frac{\partial c_s}{\partial r} \right|_{r=0} = 0}$$

for $t > 0$.

This is a homogeneous Neumann boundary condition.

Physically, it implies that there is no net molar flux through the centre of the particle:

$$N_r(0, t) = 0.$$

2.8 Boundary Condition at the Particle Surface

At the external surface of the particle, the diffusive flux is related to the intercalation or deintercalation process.

The boundary condition is written as

$$\boxed{-D_s \left. \frac{\partial c_s}{\partial r} \right|_{r=R_p} = j(t)}$$

where $j(t)$ is the prescribed molar flux at the particle surface, measured in

$$\text{mol m}^{-2} \text{s}^{-1}.$$

The sign convention must be defined consistently. According to the formulation adopted above:

- $j(t) > 0$ corresponds to an outward lithium flux;
- $j(t) < 0$ corresponds to an inward lithium flux.

An inward flux increases the average lithium concentration inside the particle, whereas an outward flux decreases it.

2.9 Complete Initial-Boundary Value Problem

The complete mathematical problem is therefore

$$\frac{\partial c_s}{\partial t} = D_s \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_s}{\partial r} \right), \quad 0 < r < R_p, \quad t > 0,$$

subject to the initial condition

$$c_s(r, 0) = c_{s,0}(r), \quad 0 \leq r \leq R_p,$$

and the boundary conditions

$$\begin{aligned} \left. \frac{\partial c_s}{\partial r} \right|_{r=0} &= 0, \\ -D_s \left. \frac{\partial c_s}{\partial r} \right|_{r=R_p} &= j(t). \end{aligned}$$

The solution of this problem provides the spatial and temporal evolution of the lithium concentration inside the active material particle.

2.10 Quantities of Interest

Although the complete solution is the concentration field $c_s(r, t)$, two additional quantities are particularly relevant.

The surface concentration is defined as

$$c_{s,\text{surf}}(t) = c_s(R_p, t).$$

The volume-averaged concentration inside the particle is

$$\bar{c}_s(t) = \frac{1}{V_p} \int_{V_p} c_s dV,$$

where the particle volume is

$$V_p = \frac{4}{3} \pi R_p^3.$$

Using spherical symmetry,

$$\bar{c}_s(t) = \frac{3}{R_p^3} \int_0^{R_p} c_s(r, t) r^2 dr.$$

Therefore,

$$\bar{c}_s(t) = \frac{3}{R_p^3} \int_0^{R_p} c_s(r, t) r^2 dr$$

The surface concentration is important because the electrochemical reaction takes place at the particle surface, while the average concentration measures the total amount of lithium stored inside the particle.

3 Non-Dimensional Formulation

3.1 Motivation

The dimensional diffusion model involves quantities characterized by very different orders of magnitude. For example, the particle radius is typically of the order of micrometres, the solid-state diffusion coefficient may be of the order of $10^{-14} \text{ m}^2 \text{ s}^{-1}$, and the lithium concentration may be of the order of 10^4 mol m^{-3} .

Working directly with these dimensional variables may lead to poorly scaled numerical equations. This issue is particularly relevant for Physics-Informed Neural Networks, since the different terms entering the loss function may have considerably different magnitudes.

For this reason, the governing equations are rewritten in terms of dimensionless variables.

3.2 Characteristic Scales

The characteristic length scale of the problem is the particle radius

$$r_{\text{ref}} = R_p.$$

The characteristic concentration scale is chosen as the maximum lithium concentration in the solid particle,

$$c_{\text{ref}} = c_{s,\text{max}}.$$

The characteristic diffusion time is obtained by balancing the temporal and spatial terms of the diffusion equation.

Starting from

$$\frac{\partial c_s}{\partial t} = D_s \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_s}{\partial r} \right),$$

the temporal term scales as

$$\frac{\partial c_s}{\partial t} \sim \frac{c_{\text{ref}}}{t_{\text{ref}}},$$

while the diffusion term scales as

$$D_s \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_s}{\partial r} \right) \sim D_s \frac{c_{\text{ref}}}{R_p^2}.$$

Balancing the two terms gives

$$\frac{c_{\text{ref}}}{t_{\text{ref}}} = D_s \frac{c_{\text{ref}}}{R_p^2}.$$

Therefore,

$$\boxed{t_{\text{ref}} = \frac{R_p^2}{D_s}}$$

is the characteristic diffusion time of the particle.

This quantity represents the typical time required for lithium to diffuse over a distance comparable to the particle radius.

3.3 Dimensionless Variables

The following dimensionless variables are introduced:

$$\rho = \frac{r}{R_p}$$

$$\tau = \frac{D_s t}{R_p^2}$$

$$C(\rho, \tau) = \frac{c_s(r, t)}{c_{s, \max}}$$

where

$$\rho \in [0, 1], \quad \tau \geq 0, \quad 0 \leq C \leq 1.$$

The dimensional variables can consequently be written as

$$r = R_p \rho,$$

$$t = \frac{R_p^2}{D_s} \tau,$$

$$c_s(r, t) = c_{s, \max} C(\rho, \tau).$$

3.4 Transformation of the Time Derivative

Since

$$\tau = \frac{D_s t}{R_p^2},$$

the chain rule gives

$$\frac{\partial \tau}{\partial t} = \frac{D_s}{R_p^2}.$$

The dimensional concentration is

$$c_s = c_{s, \max} C.$$

Therefore,

$$\frac{\partial c_s}{\partial t} = c_{s, \max} \frac{\partial C}{\partial \tau} \frac{\partial \tau}{\partial t}.$$

Substituting the derivative of τ gives

$$\boxed{\frac{\partial c_s}{\partial t} = c_{s,\max} \frac{D_s}{R_p^2} \frac{\partial C}{\partial \tau}}$$

3.5 Transformation of the Radial Derivatives

Since

$$\rho = \frac{r}{R_p},$$

the chain rule gives

$$\frac{\partial \rho}{\partial r} = \frac{1}{R_p}.$$

The first radial derivative becomes

$$\frac{\partial c_s}{\partial r} = c_{s,\max} \frac{\partial C}{\partial \rho} \frac{\partial \rho}{\partial r}.$$

Hence,

$$\boxed{\frac{\partial c_s}{\partial r} = \frac{c_{s,\max}}{R_p} \frac{\partial C}{\partial \rho}}$$

The quantity inside the spherical diffusion operator is therefore

$$r^2 \frac{\partial c_s}{\partial r} = (R_p \rho)^2 \frac{c_{s,\max}}{R_p} \frac{\partial C}{\partial \rho}.$$

Simplifying,

$$r^2 \frac{\partial c_s}{\partial r} = R_p c_{s,\max} \rho^2 \frac{\partial C}{\partial \rho}.$$

Taking the derivative with respect to r ,

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial c_s}{\partial r} \right) = \frac{1}{R_p} \frac{\partial}{\partial \rho} \left(R_p c_{s,\max} \rho^2 \frac{\partial C}{\partial \rho} \right).$$

Since R_p and $c_{s,\max}$ are constant,

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial c_s}{\partial r} \right) = c_{s,\max} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial C}{\partial \rho} \right).$$

Finally, since

$$r^2 = R_p^2 \rho^2,$$

we obtain

$$\boxed{\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_s}{\partial r} \right) = \frac{c_{s,\max}}{R_p^2} \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial C}{\partial \rho} \right)}$$

3.6 Dimensionless Governing Equation

Substituting the transformed derivatives into the dimensional equation

$$\frac{\partial c_s}{\partial t} = D_s \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_s}{\partial r} \right)$$

gives

$$c_{s,\max} \frac{D_s}{R_p^2} \frac{\partial C}{\partial \tau} = D_s \frac{c_{s,\max}}{R_p^2} \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial C}{\partial \rho} \right).$$

The common factor

$$\frac{D_s c_{s,\max}}{R_p^2}$$

can be cancelled from both sides, resulting in

$$\boxed{\frac{\partial C}{\partial \tau} = \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial C}{\partial \rho} \right)}$$

for

$$0 < \rho < 1, \quad \tau > 0.$$

The diffusion coefficient and the particle radius no longer appear explicitly in the dimensionless PDE. Their influence is incorporated into the definition of the dimensionless time τ .

3.7 Expanded Dimensionless Equation

Using the product rule,

$$\frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial C}{\partial \rho} \right) = 2\rho \frac{\partial C}{\partial \rho} + \rho^2 \frac{\partial^2 C}{\partial \rho^2}.$$

Therefore,

$$\boxed{\frac{\partial C}{\partial \tau} = \frac{\partial^2 C}{\partial \rho^2} + \frac{2}{\rho} \frac{\partial C}{\partial \rho}}$$

for $0 < \rho < 1$.

As in the dimensional formulation, the expanded equation appears singular at $\rho = 0$. The centre of the particle must consequently be treated using spherical symmetry and an appropriate limiting expression.

3.8 Dimensionless Initial Condition

The dimensional initial condition is

$$c_s(r, 0) = c_{s,0}(r).$$

Dividing by $c_{s,\max}$ gives

$$C(\rho, 0) = \frac{c_{s,0}(R_p \rho)}{c_{s,\max}}.$$

Defining

$$C_0(\rho) = \frac{c_{s,0}(R_p \rho)}{c_{s,\max}},$$

the dimensionless initial condition becomes

$$\boxed{C(\rho, 0) = C_0(\rho)}$$

If the initial concentration is spatially uniform,

$$c_{s,0}(r) = c_{s,0},$$

then

$$\boxed{C(\rho, 0) = C_0 = \frac{c_{s,0}}{c_{s,\max}}}$$

3.9 Dimensionless Boundary Condition at the Centre

The dimensional symmetry condition is

$$\left. \frac{\partial c_s}{\partial r} \right|_{r=0} = 0.$$

Using

$$\frac{\partial c_s}{\partial r} = \frac{c_{s,\max}}{R_p} \frac{\partial C}{\partial \rho},$$

we obtain

$$\frac{c_{s,\max}}{R_p} \left. \frac{\partial C}{\partial \rho} \right|_{\rho=0} = 0.$$

Therefore,

$$\boxed{\left. \frac{\partial C}{\partial \rho} \right|_{\rho=0} = 0}$$

3.10 Dimensionless Surface Flux

The dimensional surface condition is

$$-D_s \left. \frac{\partial c_s}{\partial r} \right|_{r=R_p} = j(t).$$

Substituting the transformed radial derivative gives

$$-D_s \frac{c_{s,\max}}{R_p} \left. \frac{\partial C}{\partial \rho} \right|_{\rho=1} = j(t).$$

Rearranging,

$$-\left. \frac{\partial C}{\partial \rho} \right|_{\rho=1} = \frac{j(t)R_p}{D_s c_{s,\max}}.$$

The dimensionless surface flux is therefore defined as

$$\boxed{\delta(\tau) = \frac{j(t)R_p}{D_s c_{s,\max}}}$$

where dimensional and dimensionless time are related by

$$t = \frac{R_p^2}{D_s} \tau.$$

The surface boundary condition becomes

$$\boxed{-\left. \frac{\partial C}{\partial \rho} \right|_{\rho=1} = \delta(\tau)}$$

or, equivalently,

$$\boxed{\left. \frac{\partial C}{\partial \rho} \right|_{\rho=1} = -\delta(\tau)}$$

3.11 Complete Dimensionless Problem

The complete dimensionless initial-boundary value problem is

$$\frac{\partial C}{\partial \tau} = \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial C}{\partial \rho} \right), \quad 0 < \rho < 1, \quad \tau > 0,$$

subject to

$$C(\rho, 0) = C_0(\rho), \quad 0 \leq \rho \leq 1,$$

$$\left. \frac{\partial C}{\partial \rho} \right|_{\rho=0} = 0,$$

and

$$-\left.\frac{\partial C}{\partial \rho}\right|_{\rho=1} = \delta(\tau).$$

This formulation is equivalent to the dimensional model but is more suitable for numerical implementation and Physics-Informed Neural Network training.

3.12 Dimensionless Quantities of Interest

The dimensionless surface concentration is

$$C_{\text{surf}}(\tau) = C(1, \tau)$$

The dimensionless volume-averaged concentration is

$$\bar{C}(\tau) = 3 \int_0^1 C(\rho, \tau) \rho^2 d\rho.$$

Therefore,

$$\bar{C}(\tau) = 3 \int_0^1 C(\rho, \tau) \rho^2 d\rho$$

The corresponding dimensional quantities can be recovered through

$$c_{s,\text{surf}}(t) = c_{s,\text{max}} C_{\text{surf}}(\tau),$$

and

$$\bar{c}_s(t) = c_{s,\text{max}} \bar{C}(\tau).$$

3.13 Interpretation of the Dimensionless Time

The dimensionless time is defined as

$$\tau = \frac{D_s t}{R_p^2}.$$

Consequently:

- $\tau \ll 1$ corresponds to times much shorter than the characteristic diffusion time;
- $\tau \approx 1$ corresponds to times comparable to the diffusion time across the particle;
- $\tau \gg 1$ corresponds to times much longer than the characteristic diffusion time.

The dimensional time can always be recovered as

$$t = \frac{R_p^2}{D_s} \tau$$

Thus, the same dimensionless solution can represent particles with different radii and diffusivities, provided that dimensional time is rescaled accordingly.

3.14 Operational Summary

The previous sections established the complete mathematical formulation of the lithium diffusion problem inside a spherical particle. Although the derivation is analytical, its ultimate purpose is the numerical implementation of a reference solver that will be used to generate the reference data.

More specifically, the mathematical formulation provides all the ingredients required to construct the numerical model:

- the governing partial differential equation (PDE), describing the evolution of the lithium concentration inside the particle;
- the initial condition, defining the concentration profile at the beginning of the simulation;
- the symmetry boundary condition at the particle centre;
- the flux boundary condition at the particle surface.

Rather than solving the PDE analytically, the objective is to solve it numerically. The implementation will therefore follow the pipeline

$$\boxed{\text{PDE} \rightarrow \text{Spatial Discretization} \rightarrow \text{System of ODEs} \rightarrow \text{Numerical Integration}}$$

The role of the dimensionless formulation is equally important from a computational perspective. By introducing the variables

$$\rho = \frac{r}{R_p}, \quad \tau = \frac{D_s t}{R_p^2}, \quad C = \frac{c_s}{c_{s,\max}},$$

the computational domain becomes fixed,

$$\rho \in [0, 1],$$

while the governing equation no longer explicitly depends on the physical parameters R_p , D_s and $c_{s,\max}$.

Consequently, the numerical implementation will operate entirely on the dimensionless problem

$$\frac{\partial C}{\partial \tau} = \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial C}{\partial \rho} \right),$$

subject to the corresponding initial and boundary conditions.

In Python, this mathematical model will not be implemented directly as a continuous PDE. Instead, the spatial domain will first be discretized into a finite number of radial nodes,

$$\rho_0, \rho_1, \dots, \rho_N,$$

so that the unknown concentration field becomes the finite-dimensional vector

$$\mathbf{C}(\tau) = [C_0(\tau), C_1(\tau), \dots, C_N(\tau)]^T.$$

The spatial derivatives appearing in the diffusion equation will then be approximated through finite-difference formulas, transforming the original PDE into a coupled system of ordinary differential equations (ODEs). This ODE system will finally be integrated in time using SciPy's `solve_ivp` routine, producing the numerical evolution of the concentration field.

The resulting solver will constitute the reference physical model of the project. Its numerical solutions will subsequently be employed to generate the synthetic datasets.

4 Spatial Discretization

The governing equation is defined over the continuous spatial domain

$$0 \leq \rho \leq 1.$$

Since no analytical solution is sought, the continuous domain is replaced by a finite set of discrete spatial locations.

This procedure, known as spatial discretization, converts the original partial differential equation into a finite-dimensional dynamical system, which can subsequently be integrated in time.

4.1 Uniform Radial Grid

The interval

$$[0, 1]$$

is divided into N equal subintervals, generating $N + 1$ computational nodes. The spacing between two consecutive nodes is

$$\Delta\rho = \frac{1}{N}$$

The coordinates of the nodes are therefore

$$\rho_i = i\Delta\rho, \quad i = 0, \dots, N.$$

Consequently,

$$\rho_0 = 0,$$

corresponds to the particle centre, while

$$\rho_N = 1$$

corresponds to the particle surface.

4.2 Discrete Unknowns

The continuous concentration field

$$C(\rho, \tau)$$

is no longer represented at every point of the spatial domain. Instead, only its values at the computational nodes are retained. The unknown becomes

$$C_i(\tau) = C(\rho_i, \tau), \quad i = 0, \dots, N.$$

The complete concentration profile is therefore represented by the vector

$$\mathbf{C}(\tau) = \begin{bmatrix} C_0(\tau) \\ C_1(\tau) \\ \vdots \\ C_N(\tau) \end{bmatrix}$$

Instead of computing an unknown function over a continuous domain, the numerical solver computes the evolution of this finite-dimensional vector.

4.3 Connection with the Python Implementation

This discretization directly determines the data structure used in the numerical solver.

The computational grid will be generated as

$$\rho_i = i\Delta\rho,$$

which in Python corresponds to

$$\text{rho} = \text{np.linspace}(0.0, 1.0, N+1)$$

The unknown concentration field will be stored as

$$\mathbf{C} = \text{np.zeros}(N+1)$$

where

$$C[i] = C_i(\tau).$$

Each entry of the vector represents the lithium concentration at one specific radial position inside the particle.

4.4 From a PDE to a Finite-Dimensional System

At this stage, the governing equation

$$\frac{\partial C}{\partial \tau} = \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial C}{\partial \rho} \right)$$

is still continuous.

The only modification introduced so far is that the solution will be evaluated only at the discrete locations

$$\rho_0, \rho_1, \dots, \rho_N.$$

The spatial derivatives appearing in the governing equation have not yet been approximated.

The next step consists of replacing these derivatives with finite difference formulas, thereby transforming the partial differential equation into a coupled system of ordinary differential equations.

5 Conservative Finite-Volume Discretization

5.1 Motivation

The dimensionless diffusion equation is written in conservative form as

$$\frac{\partial C}{\partial \tau} = \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial C}{\partial \rho} \right).$$

Instead of expanding the spherical diffusion operator and approximating its derivatives separately, the finite-volume method integrates the conservation equation over a finite control volume surrounding each computational node.

This approach directly preserves the physical balance

$$\boxed{\text{accumulation} = \text{incoming contribution} - \text{outgoing contribution}}$$

and is therefore particularly suitable for diffusion problems.

5.2 Computational Nodes and Control-Volume Interfaces

The computational nodes remain those previously introduced:

$$\rho_i = i\Delta\rho, \quad i = 0, \dots, N,$$

with

$$\Delta\rho = \frac{1}{N}.$$

Each node ρ_i is associated with a spherical control volume bounded by two interfaces:

$$\rho_{i-\frac{1}{2}}, \quad \rho_{i+\frac{1}{2}}.$$

For internal nodes, the interfaces are positioned halfway between consecutive nodes:

$$\boxed{\rho_{i+\frac{1}{2}} = \frac{\rho_i + \rho_{i+1}}{2} = \left(i + \frac{1}{2}\right) \Delta\rho}$$

and

$$\boxed{\rho_{i-\frac{1}{2}} = \frac{\rho_{i-1} + \rho_i}{2} = \left(i - \frac{1}{2}\right) \Delta\rho.}$$

At the physical boundaries,

$$\rho_{-\frac{1}{2}} = 0, \quad \rho_{N+\frac{1}{2}} = 1.$$

Therefore, the control volumes associated with the centre and surface nodes are half control volumes.

5.3 Dimensionless Control-Volume Measure

The dimensional volume of a spherical shell is

$$\Delta V_i = \frac{4\pi}{3} \left(r_{i+\frac{1}{2}}^3 - r_{i-\frac{1}{2}}^3 \right).$$

In the dimensionless formulation, the common geometrical factor 4π cancels from the conservation equation. It is therefore sufficient to introduce the dimensionless volume weight

$$\mathcal{V}_i = \int_{\rho_{i-\frac{1}{2}}}^{\rho_{i+\frac{1}{2}}} \rho^2 d\rho$$

which gives

$$\mathcal{V}_i = \frac{\rho_{i+\frac{1}{2}}^3 - \rho_{i-\frac{1}{2}}^3}{3}.$$

This quantity accounts for the fact that spherical control volumes increase in size as the radial coordinate increases.

Consequently, equally spaced radial nodes do not represent equal physical volumes.

5.4 Integration of the Governing Equation

The governing equation is integrated over the control volume associated with node i :

$$\int_{\rho_{i-\frac{1}{2}}}^{\rho_{i+\frac{1}{2}}} \rho^2 \frac{\partial C}{\partial \tau} d\rho = \int_{\rho_{i-\frac{1}{2}}}^{\rho_{i+\frac{1}{2}}} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial C}{\partial \rho} \right) d\rho.$$

Assuming that the nodal value $C_i(\tau)$ represents the average concentration inside the corresponding control volume, the left-hand side is approximated as

$$\int_{\rho_{i-\frac{1}{2}}}^{\rho_{i+\frac{1}{2}}} \rho^2 \frac{\partial C}{\partial \tau} d\rho \approx \mathcal{V}_i \frac{dC_i}{d\tau}.$$

Applying the fundamental theorem of calculus to the right-hand side gives

$$\int_{\rho_{i-\frac{1}{2}}}^{\rho_{i+\frac{1}{2}}} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial C}{\partial \rho} \right) d\rho = \rho^2 \frac{\partial C}{\partial \rho} \Big|_{\rho_{i+\frac{1}{2}}} - \rho^2 \frac{\partial C}{\partial \rho} \Big|_{\rho_{i-\frac{1}{2}}}.$$

The semi-discrete conservation equation is therefore

$$\mathcal{V}_i \frac{dC_i}{d\tau} = \rho^2 \frac{\partial C}{\partial \rho} \Big|_{\rho_{i+\frac{1}{2}}} - \rho^2 \frac{\partial C}{\partial \rho} \Big|_{\rho_{i-\frac{1}{2}}}$$

This equation states that the rate of concentration accumulation inside a control volume is determined by the difference between the diffusive contributions at its outer and inner interfaces.

5.5 Interface Approximation

At an internal interface $\rho_{i+\frac{1}{2}}$, the radial concentration gradient is approximated using the two neighbouring nodal values:

$$\left. \frac{\partial C}{\partial \rho} \right|_{\rho_{i+\frac{1}{2}}} \approx \frac{C_{i+1} - C_i}{\Delta \rho}.$$

Similarly,

$$\left. \frac{\partial C}{\partial \rho} \right|_{\rho_{i-\frac{1}{2}}} \approx \frac{C_i - C_{i-1}}{\Delta \rho}.$$

The corresponding geometrically weighted interface contributions are

$$\rho^2 \left. \frac{\partial C}{\partial \rho} \right|_{\rho_{i+\frac{1}{2}}} \approx \rho_{i+\frac{1}{2}}^2 \frac{C_{i+1} - C_i}{\Delta \rho},$$

and

$$\rho^2 \left. \frac{\partial C}{\partial \rho} \right|_{\rho_{i-\frac{1}{2}}} \approx \rho_{i-\frac{1}{2}}^2 \frac{C_i - C_{i-1}}{\Delta \rho}.$$

5.6 Finite-Volume Equation at an Internal Node

Substituting the interface approximations into the conservation equation gives

$$\mathcal{V}_i \frac{dC_i}{d\tau} = \rho_{i+\frac{1}{2}}^2 \frac{C_{i+1} - C_i}{\Delta \rho} - \rho_{i-\frac{1}{2}}^2 \frac{C_i - C_{i-1}}{\Delta \rho}.$$

Therefore,

$$\frac{dC_i}{d\tau} = \frac{1}{\mathcal{V}_i \Delta \rho} \left[\rho_{i+\frac{1}{2}}^2 (C_{i+1} - C_i) - \rho_{i-\frac{1}{2}}^2 (C_i - C_{i-1}) \right]$$

for

$$i = 1, \dots, N-1.$$

This is the conservative finite-volume equation for the internal nodes.

The time derivative at node i depends only on

$$C_{i-1}, \quad C_i, \quad C_{i+1},$$

but the two contributions are weighted by the areas of the corresponding spherical interfaces.

6 Treatment of the Particle Centre

6.1 Centre Control Volume

Unlike the internal control volumes, the control volume associated with the particle centre extends only from

$$\rho = 0$$

to the first interface,

$$\rho_{\frac{1}{2}} = \frac{\Delta\rho}{2}.$$

Consequently, the centre control volume occupies only half of the width of an internal control volume.

Its dimensionless volume measure is

$$\mathcal{V}_0 = \int_0^{\rho_{\frac{1}{2}}} \rho^2 d\rho.$$

Evaluating the integral gives

$$\boxed{\mathcal{V}_0 = \frac{\rho_{\frac{1}{2}}^3}{3} = \frac{\Delta\rho^3}{24}.}$$

6.2 Symmetry Boundary Condition

The governing conservation equation for the centre control volume is

$$\mathcal{V}_0 \frac{dC_0}{d\tau} = \rho^2 \frac{\partial C}{\partial \rho} \Big|_{\rho_{\frac{1}{2}}} - \rho^2 \frac{\partial C}{\partial \rho} \Big|_{\rho=0}.$$

The second contribution vanishes.

Indeed, spherical symmetry requires

$$\frac{\partial C}{\partial \rho} \Big|_{\rho=0} = 0,$$

while the surface area of a sphere also approaches zero as

$$\rho \rightarrow 0.$$

Therefore,

$$\boxed{\rho^2 \frac{\partial C}{\partial \rho} \Big|_{\rho=0} = 0.}$$

Only the flux through the outer interface contributes to the mass balance.

6.3 Semi-Discrete Equation at the Centre

The concentration gradient at the first interface is approximated using the two neighbouring nodal values,

$$\left. \frac{\partial C}{\partial \rho} \right|_{\rho_{\frac{1}{2}}} \approx \frac{C_1 - C_0}{\Delta \rho}.$$

Since

$$\rho_{\frac{1}{2}} = \frac{\Delta \rho}{2},$$

the interface contribution becomes

$$\rho^2 \left. \frac{\partial C}{\partial \rho} \right|_{\rho_{\frac{1}{2}}} = \left(\frac{\Delta \rho}{2} \right)^2 \frac{C_1 - C_0}{\Delta \rho}.$$

Substituting this approximation into the conservation equation gives

$$\mathcal{V}_0 \frac{dC_0}{d\tau} = \frac{\Delta \rho}{4} (C_1 - C_0).$$

Finally, substituting

$$\mathcal{V}_0 = \frac{\Delta \rho^3}{24}$$

yields

$$\boxed{\frac{dC_0}{d\tau} = \frac{6(C_1 - C_0)}{\Delta \rho^2}}.$$

This equation replaces the internal-node equation at the particle centre.

6.4 Physical Interpretation

The concentration inside the centre control volume changes only because of diffusion through its external interface.

If

$$C_1 > C_0,$$

lithium diffuses towards the particle centre and the concentration at the centre increases.
Conversely, if

$$C_1 < C_0,$$

lithium diffuses away from the particle centre and the concentration decreases.
When

$$C_1 = C_0,$$

no concentration gradient exists across the interface and therefore

$$\frac{dC_0}{d\tau} = 0.$$

The resulting discrete equation therefore preserves the expected diffusive behaviour while remaining fully consistent with the finite-volume conservation principle.

7 Treatment of the Particle Surface

7.1 Surface Control Volume

The control volume associated with the particle surface is bounded by the last internal interface,

$$\rho_{N-\frac{1}{2}} = 1 - \frac{\Delta\rho}{2},$$

and by the physical particle surface,

$$\rho_{N+\frac{1}{2}} = 1.$$

Therefore, similarly to the centre control volume, the surface control volume occupies only half of the width of a standard internal control volume.

Its dimensionless volume measure is

$$\mathcal{V}_N = \int_{\rho_{N-\frac{1}{2}}}^1 \rho^2 d\rho.$$

Evaluating the integral gives

$$\mathcal{V}_N = \frac{1 - \rho_{N-\frac{1}{2}}^3}{3}.$$

Using

$$\rho_{N-\frac{1}{2}} = 1 - \frac{\Delta\rho}{2},$$

the volume measure can also be written as

$$\mathcal{V}_N = \frac{1 - \left(1 - \frac{\Delta\rho}{2}\right)^3}{3}.$$

Unlike the centre control volume, the surface control volume is located at the largest radial coordinate and therefore represents a comparatively large physical volume.

7.2 Finite-Volume Balance at the Surface

The conservative diffusion equation integrated over the surface control volume gives

$$\mathcal{V}_N \frac{dC_N}{d\tau} = \rho^2 \frac{\partial C}{\partial \rho} \Big|_{\rho=1} - \rho^2 \frac{\partial C}{\partial \rho} \Big|_{\rho_{N-\frac{1}{2}}}.$$

The first term represents the contribution through the physical surface of the particle, while the second represents diffusion through the last internal interface.

The concentration gradient at the internal interface is approximated by

$$\frac{\partial C}{\partial \rho} \Big|_{\rho_{N-\frac{1}{2}}} \approx \frac{C_N - C_{N-1}}{\Delta \rho}.$$

Therefore,

$$\rho^2 \frac{\partial C}{\partial \rho} \Big|_{\rho_{N-\frac{1}{2}}} \approx \rho_{N-\frac{1}{2}}^2 \frac{C_N - C_{N-1}}{\Delta \rho}.$$

7.3 Incorporation of the Surface Boundary Condition

The dimensionless boundary condition at the particle surface is

$$-\frac{\partial C}{\partial \rho} \Big|_{\rho=1} = \delta(\tau),$$

where $\delta(\tau)$ is the dimensionless intercalation flux.
Equivalently,

$$\frac{\partial C}{\partial \rho} \Big|_{\rho=1} = -\delta(\tau).$$

Since

$$\rho = 1$$

at the particle surface,

$$\rho^2 \frac{\partial C}{\partial \rho} \Big|_{\rho=1} = -\delta(\tau).$$

Substituting the boundary condition and the internal-interface approximation into the finite-volume balance gives

$$\mathcal{V}_N \frac{dC_N}{d\tau} = -\delta(\tau) - \rho_{N-\frac{1}{2}}^2 \frac{C_N - C_{N-1}}{\Delta \rho}.$$

7.4 Semi-Discrete Equation at the Surface

Dividing by the volume measure of the surface control volume yields

$$\frac{dC_N}{d\tau} = \frac{1}{\mathcal{V}_N} \left[-\delta(\tau) - \rho_{N-\frac{1}{2}}^2 \frac{C_N - C_{N-1}}{\Delta\rho} \right].$$

Equivalently, the internal diffusion contribution can be written with the opposite concentration difference:

$$\frac{dC_N}{d\tau} = \frac{1}{\mathcal{V}_N} \left[\rho_{N-\frac{1}{2}}^2 \frac{C_{N-1} - C_N}{\Delta\rho} - \delta(\tau) \right].$$

This equation replaces the internal-node equation at $i = N$.

It depends on

$$C_{N-1}, \quad C_N, \quad \delta(\tau).$$

Unlike the finite-difference ghost-node formulation, the prescribed surface flux is incorporated directly into the conservation balance and no computational node outside the particle is required.

7.5 Physical Interpretation

The evolution of the surface concentration is controlled by two contributions.

The term

$$\rho_{N-\frac{1}{2}}^2 \frac{C_{N-1} - C_N}{\Delta\rho}$$

represents diffusion between the last internal control volume and the particle surface.

The term

$$-\delta(\tau)$$

represents the prescribed molar exchange through the external particle surface.

Under the adopted sign convention,

$$\delta(\tau) > 0$$

corresponds to a positive outward flux and therefore tends to decrease the concentration inside the surface control volume.

Conversely,

$$\delta(\tau) < 0$$

corresponds to an inward flux and tends to increase the surface concentration.

If

$$\delta(\tau) = 0,$$

the particle surface is impermeable and its concentration evolves only because of diffusion between the surface and the particle interior.

8 Complete Semi-Discrete System

After discretizing the spatial domain using the finite-volume method, the original diffusion partial differential equation is transformed into a finite-dimensional system of ordinary differential equations.

The concentration field is represented by the vector

$$\mathbf{C}(\tau) = [C_0(\tau) \ C_1(\tau) \ \cdots \ C_N(\tau)]^T,$$

whose components evolve according to different equations depending on their position within the computational domain.

8.1 Centre Node

The centre node is governed by

$$\boxed{\frac{dC_0}{d\tau} = \frac{6(C_1 - C_0)}{\Delta\rho^2}}.$$

This equation results from the finite-volume balance over the half control volume adjacent to the centre and naturally incorporates the symmetry boundary condition.

8.2 Internal Nodes

For the internal control volumes,

$$i = 1, \dots, N-1,$$

the governing equations are

$$\boxed{\frac{dC_i}{d\tau} = \frac{1}{\mathcal{V}_i \Delta\rho} \left[\rho_{i+\frac{1}{2}}^2 (C_{i+1} - C_i) - \rho_{i-\frac{1}{2}}^2 (C_i - C_{i-1}) \right]}.$$

Each internal control volume exchanges mass only with its two adjacent neighbours through the corresponding spherical interfaces.

8.3 Surface Node

The surface control volume satisfies

$$\boxed{\frac{dC_N}{d\tau} = \frac{1}{\mathcal{V}_N} \left[\rho_{N-\frac{1}{2}}^2 \frac{C_{N-1} - C_N}{\Delta\rho} - \delta(\tau) \right]}.$$

Unlike the internal nodes, the surface node is directly affected by the prescribed intercalation flux imposed through the boundary condition.

8.4 Compact Vector Form

Collecting all equations together, the semi-discrete model can be written compactly as

$$\boxed{\frac{d\mathbf{C}}{d\tau} = \mathbf{f}(\tau, \mathbf{C}).}$$

Here,

$$\mathbf{C}(\tau) = [C_0 \quad C_1 \quad \cdots \quad C_N]^T$$

contains the nodal concentrations, while the nonlinear operator

$$\mathbf{f}(\tau, \mathbf{C})$$

represents the finite-volume discretization of the spherical diffusion equation together with the imposed boundary conditions.

The original diffusion partial differential equation has therefore been transformed into a conservative system of coupled ordinary differential equations, which can be integrated numerically in time.

9 Numerical Time Integration

The finite-volume discretization transforms the original diffusion partial differential equation into the semi-discrete system

$$\frac{d\mathbf{C}}{d\tau} = \mathbf{f}(\tau, \mathbf{C}),$$

where the spatial derivatives have been replaced by algebraic expressions, while time remains a continuous variable.

This system generally does not admit a closed-form analytical solution. Consequently, the nodal concentrations are obtained through numerical time integration.

9.1 Initial Value Problem

The semi-discrete system is solved together with the dimensionless initial condition

$$\mathbf{C}(0) = \mathbf{C}_0,$$

where

$$\mathbf{C}_0 = [C_0(0) \quad C_1(0) \quad \cdots \quad C_N(0)]^T$$

contains the initial concentration at every computational node.

The complete problem therefore consists of the following initial value problem:

$$\boxed{\begin{cases} \frac{d\mathbf{C}}{d\tau} = \mathbf{f}(\tau, \mathbf{C}), \\ \mathbf{C}(0) = \mathbf{C}_0. \end{cases}}$$

9.2 Numerical Integration

The resulting system is integrated using a standard numerical solver for ordinary differential equations.

In the present work, the integration is performed using the `solve_ivp` routine provided by the SciPy scientific computing library.

At each integration step, the solver performs the following operations:

1. evaluates the current concentration vector

$$\mathbf{C}(\tau),$$

2. computes the corresponding time derivative

$$\mathbf{f}(\tau, \mathbf{C}),$$

by evaluating the finite-volume equations derived in the previous sections,

3. estimates the solution at the next time instant,
4. repeats the procedure until the prescribed final simulation time is reached.

The integration algorithm automatically selects the intermediate time steps required to satisfy the prescribed numerical tolerances, thus providing an accurate approximation of the transient diffusion process.

9.3 Simulation Output

The numerical integration produces the complete evolution of the dimensionless concentration field

$$C(\rho, \tau),$$

at every computational node and at every sampled time instant.

In particular, the surface concentration

$$C(1, \tau) = C_N(\tau)$$

is obtained directly from the last component of the numerical solution vector.

Depending on the intended application, either the complete concentration field or only the surface concentration can be stored for subsequent analysis.

9.4 Role in the Present Work

The numerical solution generated in this part constitutes the constant-diffusivity baseline used to verify the finite-volume implementation. It supports the analytical comparison, mesh-convergence analysis, conservation checks, and numerical diagnostics presented in the following sections.

This baseline should not be confused with the higher-fidelity reference simulator adopted in the learning experiments. The latter is introduced in Section 13 by adding state-dependent diffusivity and a lumped thermal balance. The constant-diffusivity solver is subsequently retained as the simplified physical model used in the reduced-modelling and FVM-FNN investigations.

10 Python Implementation

The finite-volume formulation developed in the previous sections is implemented in Python to generate the constant-diffusivity baseline solution of the diffusion problem. The implementation follows directly the mathematical derivation and can be divided into four main stages: generation of the computational mesh, evaluation of the geometrical quantities, assembly of the semi-discrete system, and numerical time integration.

10.1 Mesh Generation

The first step consists in discretizing the dimensionless radial domain

$$0 \leq \rho \leq 1$$

using a uniform grid composed of $N + 1$ computational nodes,

$$\rho_i = i\Delta\rho, \quad \Delta\rho = \frac{1}{N}.$$

The interface positions are then computed as

$$\rho_{i+\frac{1}{2}} = \frac{\rho_i + \rho_{i+1}}{2},$$

thereby defining the boundaries of every control volume.

10.2 Evaluation of the Control-Volume Geometry

Once the mesh has been generated, the geometrical quantities required by the finite-volume formulation are evaluated.

For each control volume, the dimensionless volume measure is computed as

$$\mathcal{V}_i = \frac{\rho_{i+\frac{1}{2}}^3 - \rho_{i-\frac{1}{2}}^3}{3},$$

while the interface locations are stored for the subsequent evaluation of the diffusive fluxes.

Since the computational mesh remains fixed throughout the simulation, these quantities are calculated only once before the numerical integration begins.

10.3 Assembly of the Semi-Discrete System

The core of the implementation consists of assembling the right-hand side of the ordinary differential equation,

$$\frac{d\mathbf{C}}{d\tau} = \mathbf{f}(\tau, \mathbf{C}),$$

where

$$\mathbf{C}(\tau) = [C_0 \quad C_1 \quad \cdots \quad C_N]^T.$$

Given the current concentration vector, the algorithm evaluates the time derivative of every nodal concentration according to the equations derived in the previous sections.

The centre node is updated using the symmetry condition, the internal nodes are computed from the diffusive fluxes across the neighbouring interfaces, whereas the surface node additionally incorporates the prescribed intercalation flux through the particle boundary.

As a result, the function implementing $\mathbf{f}(\tau, \mathbf{C})$ returns the complete vector of concentration derivatives required by the ODE solver.

10.4 Numerical Integration

The assembled right-hand side is supplied to the `solve_ivp` routine of the SciPy library together with

- the initial concentration vector;
- the integration interval;
- the desired sampling instants.

The solver repeatedly evaluates the right-hand side and advances the solution in time until the final simulation time is reached.

The numerical output consists of the concentration vector

$$\mathbf{C}(\tau)$$

evaluated at every requested time instant.

11 Numerical Setup and Baseline Simulations

The numerical framework presented in the previous sections was implemented in Python to validate the conservative finite-volume formulation under a constant-diffusivity assumption. The objective of these simulations is not to reproduce a complete battery cell model or to define the final high-fidelity reference, but to establish a physically consistent and numerically verified baseline for lithium diffusion inside a single spherical active-material particle. The same solver is later retained as the simplified physical model, whereas the thermo-diffusive extension introduced in Section 13 generates the higher-fidelity reference data used in the learning experiments.

The physical parameters adopted in the simulations are taken from the positive-electrode model reported by Guo and White [Guo and White, 2012], ensuring that the baseline configuration remains representative of a realistic lithium-ion battery particle. The parameters are summarized in Table 3.

Using these values, the characteristic diffusion time becomes

$$t_D = \frac{R^2}{D_s} \approx 5.96 \times 10^3 \text{ s},$$

which defines the dimensionless time variable

$$\tau = \frac{D_s t}{R^2}.$$

Consequently, a simulation over one diffusion time corresponds approximately to

$$0 \leq \tau \leq 1,$$

or equivalently

$$0 \leq t \leq 6000 \text{ s.}$$

The governing diffusion equation is driven by the intercalation flux imposed at the particle surface,

$$-D_s \left. \frac{\partial c}{\partial r} \right|_{r=R} = j(t),$$

which physically represents the molar flux of lithium ions exchanged between the electrolyte and the active material during electrochemical reactions.

In a complete Single Particle Model, this quantity depends on the applied current and electrode properties according to electrochemical relations involving Faraday's law and the specific surface area. Therefore, the intercalation flux is generally expressed as

$$j(t) = f(I(t), a_s, F, L, \dots),$$

where the exact expression depends on the adopted electrochemical model.

In the present implementation, the electrochemical submodel is not solved explicitly. Instead, the current-to-flux relation defined below provides the time-dependent boundary input shared with the higher-fidelity simulator. The exact behaviour is exhibit into simplified FVM version and consequently in its hybrid form. Instead, the PINN replaces this input with a constant approximation, thereby creating a controlled boundary-condition mismatch whose effect can be evaluated separately from the constitutive and thermal simplifications.

11.1 Time-Dependent Intercalation Flux

The lithium flux imposed at the particle surface is determined by the externally applied battery current. The dimensional surface boundary condition is written as

$$-D_s \left. \frac{\partial c}{\partial r} \right|_{r=R_p} = j(t),$$

where $j(t)$ is the molar intercalation flux at the particle surface.

For the positive-electrode particle considered in the present work, the molar flux is related to the applied current through Faraday's law,

$$j(t) = -\frac{I(t)}{FS_p},$$

where $I(t)$ is the applied current, F is Faraday's constant, and S_p is the total electroactive surface area.

The negative sign denotes lithium intercalation according to the sign convention adopted in this work. A positive applied current therefore produces a negative outward radial flux, corresponding to lithium entering the solid particle.

After nondimensionalization, the surface flux becomes

$$\delta(\tau) = \frac{R_p}{D_s c_{s,\max}} j(t),$$

and consequently

$$\delta(\tau) = -\frac{I(t)R_p}{D_s F S_p c_{s,\max}}.$$

The dimensionless and dimensional times are related by

$$t = \frac{R_p^2}{D_s} \tau.$$

Therefore, the complete expression of the dimensionless flux is

$$\delta(\tau) = -\frac{R_p}{D_s F S_p c_{s,\max}} I \left(\frac{R_p^2}{D_s} \tau \right).$$

11.2 Applied Current Profile

To obtain a time-dependent reference input while avoiding discontinuous changes at the beginning and end of the loading phase, the applied current is described by a smooth ramp–constant–ramp profile:

$$I(t) = \begin{cases} \frac{I_{1C}}{2} \left[1 - \cos \left(\frac{\pi t}{t_r} \right) \right], & 0 \leq t < t_r, \\ I_{1C}, & t_r \leq t < t_{\text{load}}, \\ \frac{I_{1C}}{2} \left[1 + \cos \left(\frac{\pi(t - t_{\text{load}})}{t_r} \right) \right], & t_{\text{load}} \leq t < t_{\text{load}} + t_r, \\ 0, & t_{\text{load}} + t_r \leq t \leq t_f. \end{cases}$$

Here, t_r denotes the current ramp duration, t_{load} defines the beginning of the ramp-down phase, and t_f is the final simulation time.

The profile satisfies

$$I(0) = 0, \quad I(t_f) = 0,$$

and reaches the nominal 1C current during the central loading interval.

The corresponding dimensionless flux is therefore

$$\delta(\tau) = -\delta_{1C} g(\tau),$$

where

$$\delta_{1C} = \frac{I_{1C} R_p}{D_s F S_p c_{s,\max}} = 0.200452,$$

and $g(\tau)$ is the dimensionless representation of the applied current profile.

11.3 Dimensionless Current Profile

Introducing

$$\tau_r = \frac{D_s t_r}{R_p^2}, \quad \tau_{\text{load}} = \frac{D_s t_{\text{load}}}{R_p^2}, \quad \tau_f = \frac{D_s t_f}{R_p^2},$$

the normalized current profile becomes

$$g(\tau) = \begin{cases} \frac{1}{2} \left[1 - \cos \left(\frac{\pi \tau}{\tau_r} \right) \right], & 0 \leq \tau < \tau_r, \\ 1, & \tau_r \leq \tau < \tau_{\text{load}}, \\ \frac{1}{2} \left[1 + \cos \left(\frac{\pi(\tau - \tau_{\text{load}})}{\tau_r} \right) \right], & \tau_{\text{load}} \leq \tau < \tau_{\text{load}} + \tau_r, \\ 0, & \tau_{\text{load}} + \tau_r \leq \tau \leq \tau_f. \end{cases}$$

Hence, the reference boundary condition is

$$-\left. \frac{\partial C}{\partial \rho} \right|_{\rho=1} = -\delta_{1C} g(\tau).$$

11.4 Simplified Flux Model

The time-dependent flux defined above constitutes the reference boundary input. In the simplified PINN formulation of Section 15.1, this input is replaced by a constant approximation,

$$\delta_{\text{simp}}(\tau) = -\delta_{1C},$$

throughout the represented time interval.

The discrepancy introduced by this approximation is therefore

$$\Delta \delta(\tau) = \delta(\tau) - \delta_{\text{simp}}(\tau).$$

This simplification removes the transient current dynamics while preserving the nominal 1C operating level. It therefore provides a controlled modification of the boundary physics whose effect on the predicted concentration field can be quantified.

11.5 Generation of the Baseline Simulation Data

The computed concentration profiles constitute the verified baseline solution of the constant-diffusivity problem.

For each simulation, the solver provides the complete concentration field

$$C(\rho, \tau),$$

from which quantities of interest can be directly extracted. In particular, the surface concentration

$$C(1, \tau) = C_N(\tau)$$

is obtained from the last component of the solution vector.

These numerical solutions are used to validate the implementation and to characterize the constant-diffusivity baseline. They are distinct from the higher-fidelity thermo-diffusive dataset introduced later and used for the training and evaluation of the learned models.

The adopted parameter values and boundary-input definitions are summarized in the following tables.

Table 3: Parameters adopted in the constant-diffusivity baseline simulations.

Category	Parameter	Symbol	Value
Physical	Particle radius	R_p	$8.5 \times 10^{-6} \text{ m}$
	Solid diffusion coefficient	D_s	$1.213 \times 10^{-14} \text{ m}^2/\text{s}$
	Maximum lithium concentration	$c_{s,\max}$	51410 mol/m^3
	Faraday constant	F	96487 C/mol
	Particle surface area	S_p	1.167 m^2
Simulation	Initial normalized concentration	C_0	0.5
	Nominal current (1C)	I_{1C}	1.656 A
	Current ramp duration	t_r	300 s
	Constant-current duration	t_{load}	3600 s
	Final simulation time	t_f	6000 s
	Characteristic diffusion time	$t_D = \frac{R_p^2}{D_s}$	5956 s
Numerical	Number of radial intervals	N	100
	Number of radial nodes	$N + 1$	101
	Radial spacing	$\Delta\rho$	0.01
	Number of stored time instants	N_τ	201
	Absolute tolerance	atol	10^{-8}
	Relative tolerance	rtol	10^{-6}
	ODE solver		<code>solve_ivp</code> (RK45)

Table 4: Definition of the boundary input.

Quantity	Expression	Description
Applied current	$I(t)$	Ramp-constant-ramp profile
Molar intercalation flux	$j(t) = -\frac{I(t)}{FS_p}$	Faraday's law
Dimensionless flux	$\delta(\tau) = \frac{R_p}{D_s c_{s,\max}} j(t)$	Surface boundary condition
Dimensionless PDE BC	$-\left. \frac{\partial C}{\partial \rho} \right _{\rho=1} = \delta(\tau)$	Neumann boundary condition

12 Baseline Solver Validation Results

The proposed finite-volume solver was evaluated using the numerical setup described in the previous section. The computational domain was discretized using 101 uniformly distributed radial nodes and 201 stored time instants over a physical simulation horizon of 6000 s, corresponding to a final dimensionless time of $\tau_f = 1.007$. The resulting concentration field contains $101 \times 201 = 20\,301$ samples and is used for solver verification and baseline characterization. It is not the final thermo-diffusive reference dataset used in the subsequent experiments.

Initially, the particle is assumed to be uniformly lithiated with a dimensionless concentration equal to $C_0 = 0.5$. Consequently, the concentration field is spatially uniform at $\tau = 0$, satisfying the prescribed initial condition. During the charging process, lithium enters the particle through the external surface, generating a concentration gradient directed from the surface towards the particle centre.

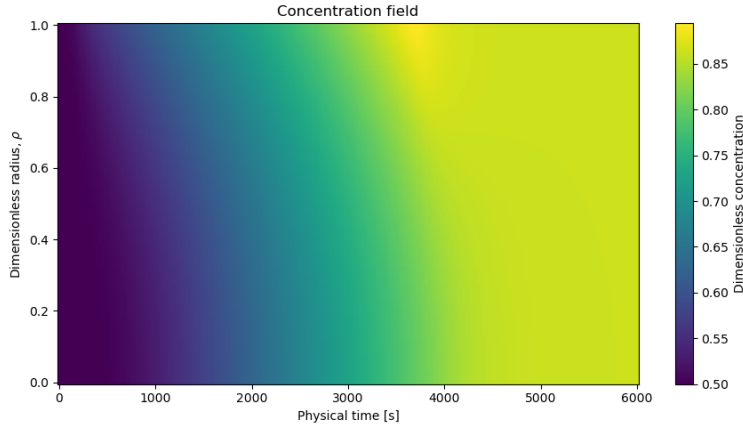


Figure 1: Concentration Map of the Particle

Figure 1 illustrates the spatio-temporal evolution of the concentration field. As expected, the concentration increases progressively throughout the particle while remaining higher near the surface during the charging stage. Once the applied current is removed, diffusion gradually redistributes lithium inside the particle until an almost uniform concentration field is recovered.

Figure 2 reports the temporal evolution of the concentrations at the particle centre and at the particle surface. The surface concentration reacts immediately to the imposed intercalation flux, reaching a maximum value of approximately $C_s = 0.894$ at $t \approx 3690$ s. During the subsequent relaxation stage, the surface concentration decreases slightly, whereas the concentration at the particle centre continues to increase as diffusion redistributes lithium inside the particle. At the end of the simulation both quantities converge towards the same equilibrium value of approximately 0.863, indicating that the internal concentration gradient has almost completely vanished.

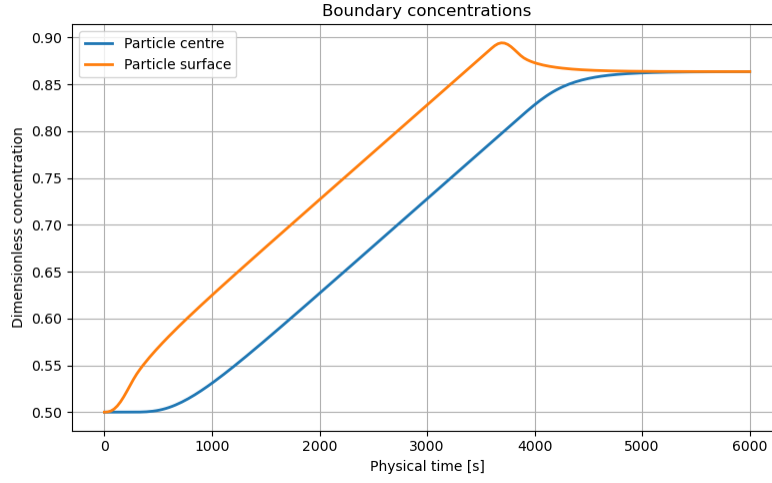


Figure 2: Concentration on Boundary

The corresponding evolution of the volume-averaged concentration is presented in Figure 3. During the charging stage, the average concentration increases almost linearly, consistently with the nearly constant lithium insertion rate imposed by the operating current profile. After the current is progressively reduced and eventually removed, the average concentration reaches a constant value close to 0.8635, reflecting the conservation of the total lithium content.

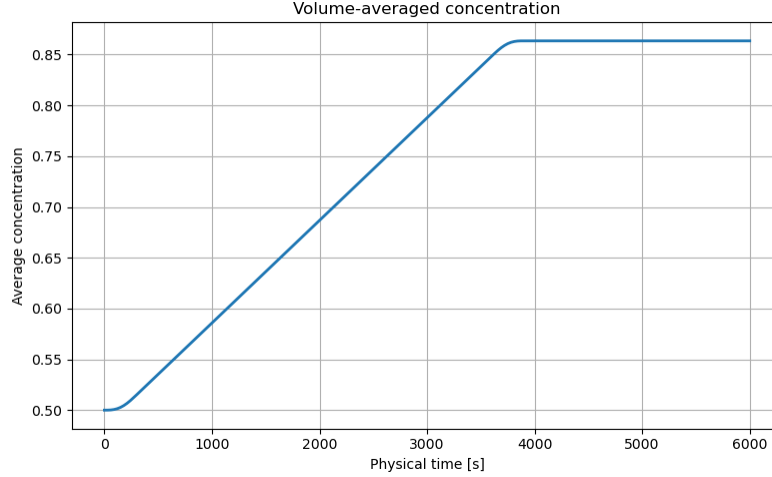


Figure 3: Average Concentration in the Particle

The radial concentration profiles at different time instants are shown in Figure 4. At intermediate times the highest concentration is always observed at the particle surface, while the particle centre exhibits a delayed response due to finite diffusion kinetics. As time progresses, the concentration

gradients progressively decrease until an almost uniform profile is recovered at the end of the simulation.

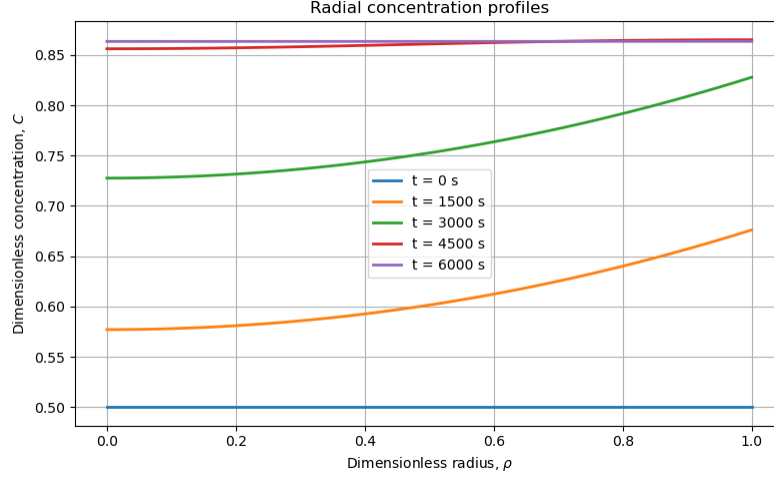


Figure 4: Radial Profiles

From a numerical perspective, the proposed implementation remained stable throughout the entire simulation. No **NaN** or infinite values were generated, all concentration values remained within the physically admissible interval $0 \leq C \leq 1$, and the maximum mass-balance residual was equal to 4.52×10^{-5} , as reported in Figure 5. These observations indicate that the finite-volume discretization preserves both numerical stability and physical consistency under the selected operating conditions.

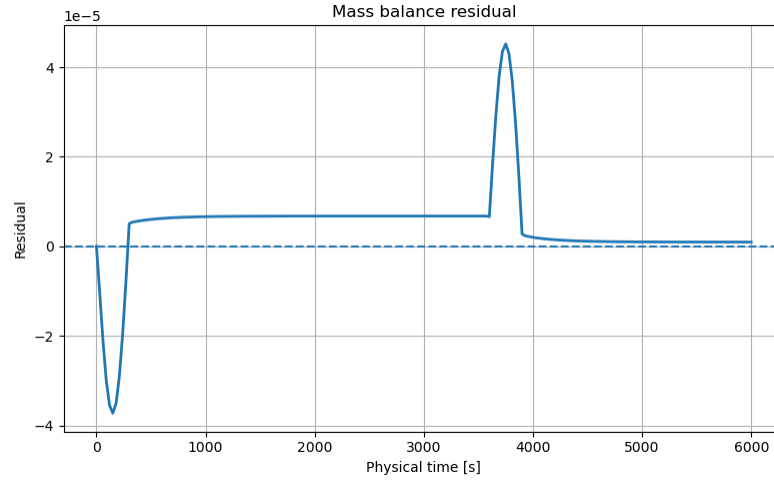


Figure 5: Mass Balance Residual

12.1 Validation

The accuracy of the proposed finite-volume implementation was assessed through two complementary validation procedures. First, the numerical solution was compared against the approximate analytical solution proposed by Guo and White [Guo and White, 2012]. Second, a mesh-convergence study was carried out to verify that the numerical solution is independent of the spatial discretization.

12.1.1 Comparison with the Guo–White Approximate Solution

The finite-volume solution was compared with the approximate surface-concentration solution proposed by Guo and White [Guo and White, 2012] for a constant dimensionless intercalation flux,

$$\delta(\tau) = -0.2,$$

with initial concentration $C_0 = 0.5$.

The reference curve was not obtained by digitizing the published figure. Instead, it was reconstructed directly from the analytical approximation given in Eq. (46) of Guo and White [Guo and White, 2012].

The eigenvalues λ_n were computed numerically as the positive roots of

$$\lambda_n - \tan(\lambda_n) = 0.$$

The first five eigenvalues were retained, consistently with the $N = 5$ approximation adopted in the paper. The following eigenvalue, λ_6 , was additionally used in the truncation-error correction term proposed by the authors.

For a constant flux, the resulting reference surface concentration was evaluated as

$$C_s^{\text{GW}}(\tau) = C_0 - 3\delta\tau - 2\delta \sum_{n=1}^N \frac{1 - \exp(-\lambda_n^2 \tau)}{\lambda_n^2} - 2\delta e_N^{\text{apx}}(\tau),$$

where $e_N^{\text{apx}}(\tau)$ denotes the approximation of the neglected eigenfunction contributions.

The Guo–White solution was evaluated over 300 uniformly distributed dimensionless time instants in the interval $0 \leq \tau \leq 0.68$. Since the finite-volume solution and the reference solution were defined on different temporal grids, the finite-volume surface concentration was linearly interpolated at the reference time instants before computing the error.

Figure 6 shows that the two solutions are nearly indistinguishable over the complete time interval. The resulting root-mean-square error is

$$\text{RMSE} = 6.33 \times 10^{-4},$$

confirming the accuracy of the finite-volume implementation for the constant-flux benchmark.

12.1.2 Mesh Convergence Study

A mesh-convergence analysis was performed using five spatial discretizations,

$$N = \{25, 50, 100, 200, 400\},$$

while keeping all the remaining simulation parameters unchanged.

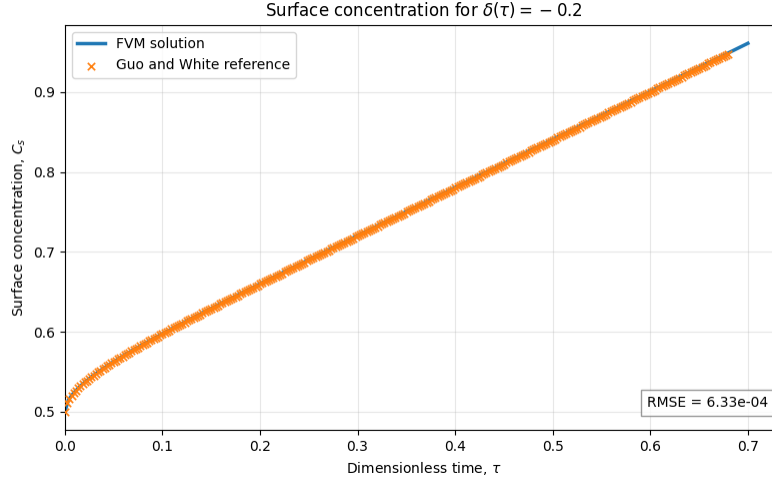


Figure 6: Comparison between the finite-volume and Guo–White surface-concentration solutions.

Figure 7 reports the evolution of the surface concentration, the volume-averaged concentration and the final radial concentration profile obtained with the different meshes. All the solutions rapidly converge towards the same response as the mesh is refined. In particular, the curves corresponding to $N = 100$, $N = 200$ and $N = 400$ are almost perfectly superimposed for all the monitored quantities.

The final radial concentration profile exhibits only negligible differences between the finest discretizations, indicating that the numerical error associated with the spatial discretization is already very small for $N = 100$. Consequently, this mesh resolution was selected throughout the remainder of this work as a suitable compromise between numerical accuracy and computational efficiency.

Table 5 reports the Euclidean differences in surface and volume-averaged concentration relative to the $N = 400$ numerical solution. The errors decrease systematically as the mesh is refined. The $N = 100$ discretization already yields differences below 6×10^{-5} for both quantities and is therefore retained as an appropriate compromise between accuracy and computational cost.

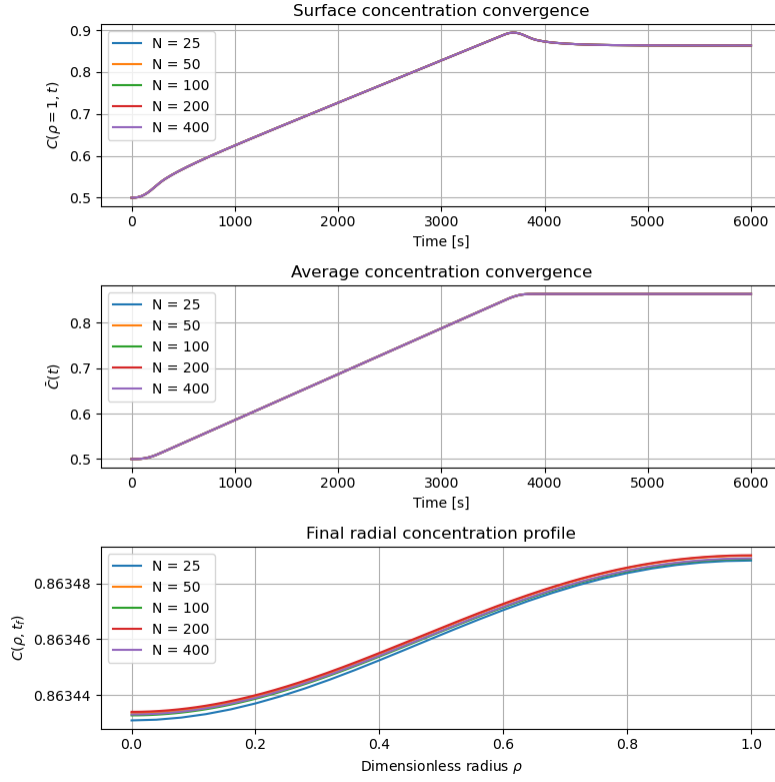


Figure 7: Effect of mesh refinement on the monitored concentration quantities.

Table 5: Mesh convergence analysis using the $N = 400$ solution as reference.

Radial nodes (N)	$\ \mathbf{C}_s - \mathbf{C}_{s,\text{ref}}\ _2$	$\ \bar{\mathbf{C}} - \bar{\mathbf{C}}_{\text{ref}}\ _2$
25	9.3471×10^{-4}	6.1095×10^{-4}
50	2.3872×10^{-4}	1.4729×10^{-4}
100	5.6692×10^{-5}	3.3493×10^{-5}
200	1.9012×10^{-5}	1.1247×10^{-5}
400	0	0

13 Model Extension

13.1 Motivation

The baseline model introduced in the previous sections assumes a constant solid-phase diffusion coefficient and neglects thermal effects. Under these assumptions, the dimensionless governing equation is linear, and the concentration evolution is controlled only by the prescribed intercalation flux and by diffusion within the spherical particle.

Although this formulation provides a suitable benchmark for validating the finite-volume implementation, data generated by the same constant-diffusivity model would define a relatively simple learning problem for a Physics-Informed Neural Network, as a PINN constrained by the identical governing equation would already be supplied with an almost complete description of the data-generating process. Moreover actual formulation provides non-sufficient complexity to remove when generating a simplified FVM solver.

The objective of the present extension is therefore to construct a more realistic and physically richer reference model. The extended reference model introduces two additional mechanisms:

1. a lumped thermal balance accounting for irreversible heating and convective heat exchange with the environment;
2. a solid-phase diffusion coefficient depending on both the local lithium concentration and the particle temperature.

The resulting formulation establishes a one-way thermo-diffusive coupling. The applied current determines the temperature evolution, while temperature and concentration modify the solid diffusion coefficient. The modified diffusivity then affects the spatial and temporal evolution of the lithium concentration.

The coupling structure can be summarized as

$$\boxed{I(t) \longrightarrow T(t) \longrightarrow D_s(C, T) \longrightarrow C(\rho, \tau)}$$

with an additional direct dependence of D_s on the local concentration field.

It is important to clarify that the thermal extension does not introduce a spatial heat-conduction PDE inside the particle. Instead, the temperature is represented through a lumped thermal state $T(t)$, assumed uniform over the entire particle domain. Convective heat transfer is introduced in the thermal energy balance, whereas the lithium transport problem remains spatially distributed.

This extended thermo-diffusive model is adopted as the high-fidelity reference model used for synthetic data generation. In the subsequent PINN formulation, part of this additional physics will deliberately be removed by imposing a constant diffusion coefficient and neglecting the thermal state.

13.2 Thermally Coupled Diffusion Equation

When the solid diffusion coefficient depends on the system state, it can no longer be extracted from the spatial derivative. The dimensional lithium diffusion equation must therefore be written as

$$\boxed{\frac{\partial c_s}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 D_s(c_s, T) \frac{\partial c_s}{\partial r} \right]}$$

for

$$0 < r < R_p, \quad t > 0.$$

The corresponding symmetry condition at the particle centre remains

$$\left. \frac{\partial c_s}{\partial r} \right|_{r=0} = 0.$$

At the particle surface, the dimensional intercalation-flux condition becomes

$$-D_s(c_s(R_p, t), T(t)) \left. \frac{\partial c_s}{\partial r} \right|_{r=R_p} = j(t).$$

Consequently, the diffusion coefficient now influences both the internal transport rate and the concentration gradient required to satisfy the prescribed surface flux.

13.3 State-Dependent Solid Diffusion

The constant-diffusivity formulation adopted in the preliminary model provides a linear description of lithium transport inside the active material particle. In real electrode materials, however, the solid-state diffusion coefficient may vary with both the local lithium concentration and the operating temperature. The extended reference model therefore replaces the constant coefficient D_s with a state-dependent diffusivity. Concentration-dependent solid diffusivity is also a central quantity in experimental parameter-inference studies [Gümrükçüoğlu et al., 2024]:

$$D_s = D_s(C, T),$$

where $C = C(\rho, \tau)$ is the local dimensionless lithium concentration and $T = T(\tau)$ is the particle temperature.

The purpose of this extension is to construct a more realistic and more demanding reference model. The resulting concentration dynamics are no longer governed by a linear diffusion equation with constant coefficients. Instead, the local transport rate changes during the simulation according to the evolving concentration field and thermal state. This introduces a controlled physical mismatch with respect to the simplified version.

13.3.1 Concentration and Temperature Dependence

In the implemented reference model, the solid diffusion coefficient is defined as

$$D_s(C, T) = D_{s,\text{ref}} \exp[\alpha (C - C_0)] \exp\left[-\frac{E_a}{R_g} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}}\right)\right],$$

where:

- $D_{s,\text{ref}}$ is the reference solid-state diffusion coefficient;
- C_0 is the initial dimensionless concentration;

- α controls the sensitivity of diffusivity to lithium concentration;
- E_a is the activation energy associated with solid-state diffusion;
- R_g is the universal gas constant;
- T_{ref} is the reference temperature.

The two exponential factors describe distinct physical effects. The first one introduces a concentration dependence,

$$f_C(C) = \exp[\alpha(C - C_0)],$$

while the second corresponds to an Arrhenius-type temperature correction,

$$f_T(T) = \exp\left[-\frac{E_a}{R_g} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}}\right)\right].$$

The complete diffusivity can therefore be written compactly as

$$D_s(C, T) = D_{s,\text{ref}} f_C(C) f_T(T).$$

At the initial reference state,

$$C = C_0, \quad T = T_{\text{ref}},$$

both correction factors are equal to one. Consequently,

$$\boxed{D_s(C_0, T_{\text{ref}}) = D_{s,\text{ref}}}.$$

Thus, $D_{s,\text{ref}}$ represents the diffusivity of the particle at the initial concentration and reference temperature.

13.3.2 Effect of Lithium Concentration

The concentration-dependent factor is

$$f_C(C) = \exp[\alpha(C - C_0)].$$

For the value adopted in the present simulations,

$$\alpha = 1,$$

an increase in lithium concentration with respect to the initial state produces an exponential increase in the local diffusion coefficient. In particular,

$$C > C_0 \implies f_C(C) > 1,$$

whereas

$$C < C_0 \implies f_C(C) < 1.$$

Since the concentration is spatially distributed, the resulting diffusivity also varies along the radial coordinate. Therefore, during the charging phase,

$$D_s = D_s(\rho, \tau),$$

even though the temperature is assumed spatially uniform.

Regions with different lithium concentrations are consequently characterized by different local transport rates. In the present charging simulation, the concentration initially increases more rapidly near the particle surface. The corresponding local diffusivity therefore becomes larger near the surface than in the particle interior, modifying the transient redistribution of lithium.

13.3.3 Effect of Temperature

The temperature correction follows the Arrhenius-type relation

$$f_T(T) = \exp \left[-\frac{E_a}{R_g} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}} \right) \right].$$

When the particle temperature exceeds the reference temperature,

$$T > T_{\text{ref}},$$

the reciprocal-temperature difference satisfies

$$\frac{1}{T} - \frac{1}{T_{\text{ref}}} < 0.$$

Hence, the exponent becomes positive and

$$f_T(T) > 1.$$

The diffusion coefficient therefore increases as the particle temperature rises. Conversely, temperatures below the reference value reduce the diffusivity.

The activation energy adopted in the reference model is

$$E_a = 3.5 \times 10^4 \text{ J mol}^{-1},$$

while the universal gas constant is

$$R_g = 8.314462618 \text{ J mol}^{-1} \text{ K}^{-1}.$$

Even a moderate temperature variation can therefore produce a non-negligible modification of the diffusion coefficient.

13.3.4 Dimensional Governing Equation

When the diffusion coefficient depends on the evolving state, it must remain inside the divergence operator. The dimensional governing equation becomes

$$\boxed{\frac{\partial c_s}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left[r^2 D_s(c_s, T) \frac{\partial c_s}{\partial r} \right]}.$$

This equation differs from the constant-diffusivity formulation

$$\frac{\partial c_s}{\partial t} = D_s \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial c_s}{\partial r} \right),$$

because D_s can no longer be extracted from the spatial derivative.

Expanding the variable-coefficient operator gives

$$\frac{\partial c_s}{\partial t} = D_s \left(\frac{\partial^2 c_s}{\partial r^2} + \frac{2}{r} \frac{\partial c_s}{\partial r} \right) + \frac{\partial D_s}{\partial r} \frac{\partial c_s}{\partial r}.$$

The additional contribution

$$\frac{\partial D_s}{\partial r} \frac{\partial c_s}{\partial r}$$

arises exclusively from the spatial variation of the diffusion coefficient. Since D_s depends on concentration, its radial derivative is indirectly determined by the concentration gradient.

For this reason, the conservative form is retained in the numerical implementation, avoiding the explicit evaluation of $\partial D_s / \partial r$.

13.3.5 Dimensionless Diffusion Coefficient

The dimensionless time variable continues to be defined using the constant reference coefficient,

$$\tau = \frac{D_{s,\text{ref}} t}{R_p^2}.$$

A normalized diffusion coefficient is consequently introduced as

$$\hat{D}(C, T) = \frac{D_s(C, T)}{D_{s,\text{ref}}}.$$

Using the adopted state-dependent relation gives

$$\hat{D}(C, T) = \exp[\alpha(C - C_0)] \exp \left[-\frac{E_a}{R_g} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}} \right) \right].$$

At the initial reference state,

$$\hat{D}(C_0, T_{\text{ref}}) = 1.$$

The variable $\hat{D}$ is dimensionless and measures the local diffusivity relative to its reference value.

13.3.6 Dimensionless Governing Equation

After introducing the normalized diffusivity, the extended dimensionless diffusion equation becomes

$$\frac{\partial C}{\partial \tau} = \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left[\rho^2 \hat{D}(C, T) \frac{\partial C}{\partial \rho} \right].$$

The initial condition remains

$$C(\rho, 0) = C_0.$$

At the particle centre, spherical symmetry gives

$$\left. \frac{\partial C}{\partial \rho} \right|_{\rho=0} = 0.$$

The surface condition is still scaled using $D_{s,\text{ref}}$. Therefore,

$$-\hat{D}(C, T) \left. \frac{\partial C}{\partial \rho} \right|_{\rho=1} = \delta(\tau),$$

where

$$\delta(\tau) = \frac{R_p j(t)}{D_{s,\text{ref}} c_{s,\text{max}}}.$$

The complete extended problem is thus

$$\left\{ \begin{array}{ll} \frac{\partial C}{\partial \tau} = \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left[\rho^2 \hat{D}(C, T) \frac{\partial C}{\partial \rho} \right], & 0 < \rho < 1, \quad \tau > 0, \\ C(\rho, 0) = C_0, & 0 \leq \rho \leq 1, \\ \left. \frac{\partial C}{\partial \rho} \right|_{\rho=0} = 0, & \tau > 0, \\ -\hat{D}(C, T) \left. \frac{\partial C}{\partial \rho} \right|_{\rho=1} = \delta(\tau), & \tau > 0. \end{array} \right.$$

Because $\hat{D}$ depends on the unknown concentration and on the temperature state, the resulting diffusion problem is nonlinear and thermally coupled.

13.3.7 Finite-Volume Discretization with Variable Diffusivity

The conservative finite-volume formulation is retained for the extended model. At each numerical evaluation, the normalized diffusivity is first computed at every radial node:

$$\hat{D}_i = \hat{D}(C_i, T), \quad i = 0, \dots, N.$$

Since the diffusive transport terms are evaluated at the control-volume interfaces, the nodal diffusivities are interpolated using the arithmetic mean,

$$\hat{D}_{i+\frac{1}{2}} = \frac{\hat{D}_i + \hat{D}_{i+1}}{2}.$$

The geometrically weighted diffusive contribution at an internal interface becomes

$$\rho^2 \hat{D} \frac{\partial C}{\partial \rho} \bigg|_{\rho_{i+\frac{1}{2}}} \approx \rho_{i+\frac{1}{2}}^2 \hat{D}_{i+\frac{1}{2}} \frac{C_{i+1} - C_i}{\Delta \rho}.$$

Consequently, the equation for an internal control volume is

$$\frac{dC_i}{d\tau} = \frac{1}{\mathcal{V}_i \Delta \rho} \left[\rho_{i+\frac{1}{2}}^2 \hat{D}_{i+\frac{1}{2}} (C_{i+1} - C_i) - \rho_{i-\frac{1}{2}}^2 \hat{D}_{i-\frac{1}{2}} (C_i - C_{i-1}) \right],$$

for

$$i = 1, \dots, N-1.$$

At the particle centre, the corresponding equation becomes

$$\frac{dC_0}{d\tau} = \frac{6 \hat{D}_{\frac{1}{2}} (C_1 - C_0)}{\Delta \rho^2}.$$

At the particle surface, the prescribed boundary flux is incorporated directly into the finite-volume balance:

$$\frac{dC_N}{d\tau} = \frac{1}{\mathcal{V}_N} \left[\rho_{N-\frac{1}{2}}^2 \hat{D}_{N-\frac{1}{2}} \frac{C_{N-1} - C_N}{\Delta \rho} - \delta(\tau) \right].$$

The variable diffusivity is therefore evaluated dynamically throughout the integration. At every ODE-solver evaluation, the current concentration field and temperature are used to update $\hat{D}_i$, after which the interface coefficients and the finite-volume derivatives are recomputed.

13.3.8 Coupled State Representation

The temperature affects lithium diffusion through $\hat{D}(C, T)$, while the temperature itself evolves according to the applied current and the thermal balance introduced in the following subsection.

The complete numerical state is therefore extended from the concentration vector

$$\mathbf{C}(\tau) = [C_0(\tau) \quad \dots \quad C_N(\tau)]^T$$

to

$$\mathbf{y}(\tau) = [C_0(\tau) \quad C_1(\tau) \quad \dots \quad C_N(\tau) \quad T(\tau)]^T.$$

The coupled semi-discrete problem can then be written as

$$\boxed{\frac{d\mathbf{y}}{d\tau} = \begin{bmatrix} \mathbf{f}_C(\tau, \mathbf{C}, T) \\ f_T(\tau, T) \end{bmatrix}},$$

where $\mathbf{f}_C$ denotes the finite-volume diffusion operator with state-dependent diffusivity and f_T denotes the lumped thermal balance.

This formulation allows the concentration field and temperature to be integrated simultaneously within the same implicit ODE solver.

13.4 Lumped Thermal Model

The previous formulation assumes that the particle temperature remains constant throughout the entire simulation. Although this assumption is reasonable under mild operating conditions, it neglects the thermal effects generated during battery operation. Since the diffusion coefficient is strongly temperature dependent, even relatively small temperature variations may alter the lithium transport dynamics.

Rather than introducing a full heat-conduction equation inside the particle, the present work adopts a lumped thermal model. The particle is therefore assumed to be thermally homogeneous, so that its temperature depends only on time,

$$\boxed{T = T(\tau)}.$$

This approximation is justified by the small dimensions of the active material particle, for which internal thermal gradients are typically much smaller than concentration gradients. Consequently, the thermal state can be represented by a single ordinary differential equation, while the spatial dependence remains confined to the diffusion problem.

13.4.1 Energy Balance

The particle temperature evolves according to the balance between heat generation and heat dissipation,

$$\boxed{\frac{dT}{dt} = \beta_Q I^2(t) - \beta_H (T - T_{\text{amb}})},$$

where

- β_Q is the heat-generation coefficient, which converts the electrical current into thermal power;
- β_H is the effective convective heat-transfer coefficient;
- T_{amb} denotes the ambient temperature;
- $I(t)$ is the externally applied current.

The first term models irreversible heat generation caused by Joule losses, while the second term represents heat exchange with the surrounding environment according to Newton's cooling law.

13.4.2 Dimensionless Thermal Equation

Since the diffusion problem is solved using the dimensionless time

$$\tau = \frac{D_{s,\text{ref}} t}{R_p^2},$$

the thermal model is rewritten using the same temporal variable. Applying the chain rule,

$$\frac{dT}{d\tau} = \frac{R_p^2}{D_{s,\text{ref}}} \frac{dT}{dt},$$

which yields

$$\frac{dT}{d\tau} = \frac{R_p^2}{D_{s,\text{ref}}} [\beta_Q I^2(t) - \beta_H (T - T_{\text{amb}})].$$

The factor

$$\frac{R_p^2}{D_{s,\text{ref}}}$$

corresponds to the characteristic diffusion time previously introduced. Consequently, both the concentration field and the thermal state evolve using the same dimensionless temporal coordinate.

13.4.3 Coupling with the Diffusion Model

Although the temperature is assumed spatially uniform, it directly affects the diffusion process through the state-dependent diffusion coefficient introduced in the previous subsection.

At each integration step, the numerical solver first evaluates the current particle temperature,

$$T(\tau),$$

which is subsequently used to compute the Arrhenius correction of the solid-state diffusion coefficient,

$$D_s = D_s(C, T).$$

The updated diffusion coefficient is then employed to assemble the finite-volume diffusion operator. Consequently, concentration and temperature are solved simultaneously, producing a one-way thermal coupled thermo-diffusion model. One-way refers to the fact that concentration does not influences thermal balance, while temperature acts on concentration through diffusive coefficient.

The complete numerical state therefore becomes

$$\mathbf{y}(\tau) = [C_0(\tau) \quad C_1(\tau) \quad \cdots \quad C_N(\tau) \quad T(\tau)]^T,$$

whose evolution is governed by the coupled system

$$\frac{d\mathbf{y}}{d\tau} = \begin{bmatrix} \mathbf{f}_C(\mathbf{C}, T) \\ f_T(T, I) \end{bmatrix}.$$

This formulation preserves the conservative finite-volume treatment of lithium diffusion while extending the reference model with a simplified yet physically meaningful thermal dynamics.

13.5 Thermal and Transport Parameters

The introduction of a state-dependent diffusion coefficient requires a number of additional physical parameters describing both the thermal dynamics of the particle and the dependence of lithium diffusivity on its operating state.

The thermal model is initialized at the ambient temperature,

$$T(0) = T_{\text{amb}} = T_{\text{ref}},$$

so that the diffusion coefficient initially coincides with its reference value,

$$D_s(C_0, T_{\text{ref}}) = D_{s,\text{ref}}.$$

As the simulation progresses, both the concentration field and the particle temperature modify the effective diffusion coefficient through the exponential relations introduced in the previous subsection.

Table 6 summarizes all the additional parameters employed in the extended reference model.

Table 6: Additional parameters introduced in the thermo-diffusion model.

Category	Parameter	Symbol	Value
Thermal	Reference temperature	T_{ref}	298.15 K
	Ambient temperature	T_{amb}	298.15 K
	Activation energy	E_a	$3.5 \times 10^4 \text{ J mol}^{-1}$
	Universal gas constant	R_g	$8.314462618 \text{ J mol}^{-1} \text{ K}^{-1}$
	Equivalent cell mass	m	0.045 kg
	Specific heat capacity	c_p	$900 \text{ J kg}^{-1} \text{ K}^{-1}$
	Equivalent electrical resistance	R_{el}	0.10 Ω
	Convective heat-transfer coefficient	h	$10 \text{ W m}^{-2} \text{ K}^{-1}$
	Equivalent heat-transfer area	A_{th}	0.01 m^2
Transport	Reference diffusion coefficient	$D_{s,\text{ref}}$	$1.213 \times 10^{-14} \text{ m}^2 \text{ s}^{-1}$
	Concentration sensitivity	α	1.0

The activation energy determines the sensitivity of lithium diffusion to temperature through the Arrhenius correction, while the parameter α controls the influence of the local lithium concentration on the diffusion coefficient.

The coefficients β_Q and β_H govern the thermal balance. The first determines the amount of heat generated by the applied current, whereas the second regulates the rate at which the particle releases heat to the surrounding environment.

The adopted values were selected so that the simulated particle experiences a moderate temperature excursion during the charging process. Consequently, the temperature remains within the typical operating range of lithium-ion batteries while still producing a measurable variation of the solid-state diffusion coefficient.

14 Results of the Extended Reference Model

The thermo-diffusive model presented in the previous section was solved using the same numerical configuration adopted for the constant diffusivity reference model. The spatial domain was discretized with $N = 100$ finite-volume cells (101 computational nodes), while the solution was stored at 201 uniformly distributed time instants over the interval

$$0 \leq t \leq 6000 \text{ s.}$$

Unlike the previous simulations, however, the diffusion coefficient is continuously updated according to both the local lithium concentration and the particle temperature.

The objective of this section is twofold. First, the numerical stability of the extended model is verified. Second, the influence of the adaptive diffusion coefficient on the concentration dynamics is analysed.

14.1 Temperature Evolution

Figure 8 reports the evolution of the lumped particle temperature during the complete charging and relaxation cycle.

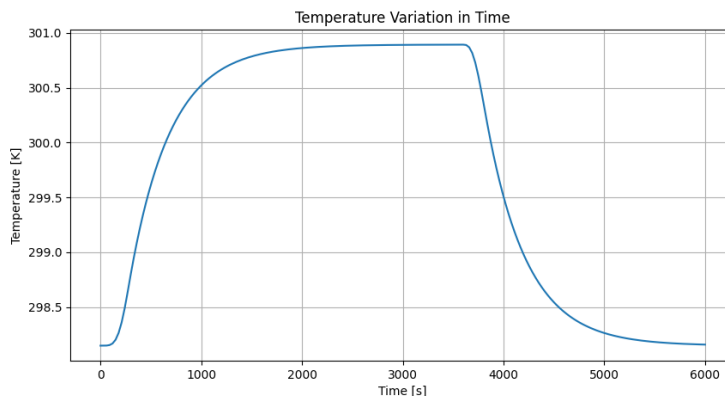


Figure 8: Temporal evolution of the particle temperature.

Initially, the particle is in thermal equilibrium with the surrounding environment,

$$T(0) = T_{\text{amb}} = 298.15 \text{ K.}$$

As the charging current increases, irreversible Joule heating becomes the dominant contribution in the thermal balance, producing a gradual temperature rise. The maximum temperature reached during the simulation is

$$T_{\max} = 300.892 \text{ K},$$

corresponding to an increase of approximately

$$\Delta T_{\max} = 2.74 \text{ K}$$

with respect to the ambient temperature.

After the charging current decreases, the heat-generation term rapidly vanishes, whereas convective cooling continues to remove heat from the particle. Consequently, the temperature progressively returns towards the ambient value, reaching

$$T_{\text{final}} = 298.160 \text{ K}$$

at the end of the simulation.

The limited temperature excursion confirms that the adopted thermal parameters produce physically realistic operating conditions while still introducing a measurable variation of the diffusion coefficient through the Arrhenius correction.

14.2 Temperature Distribution

Since the thermal model is lumped, the particle temperature is assumed to be spatially uniform at every instant. The temperature map shown in Figure 9 therefore exhibits no radial gradients.

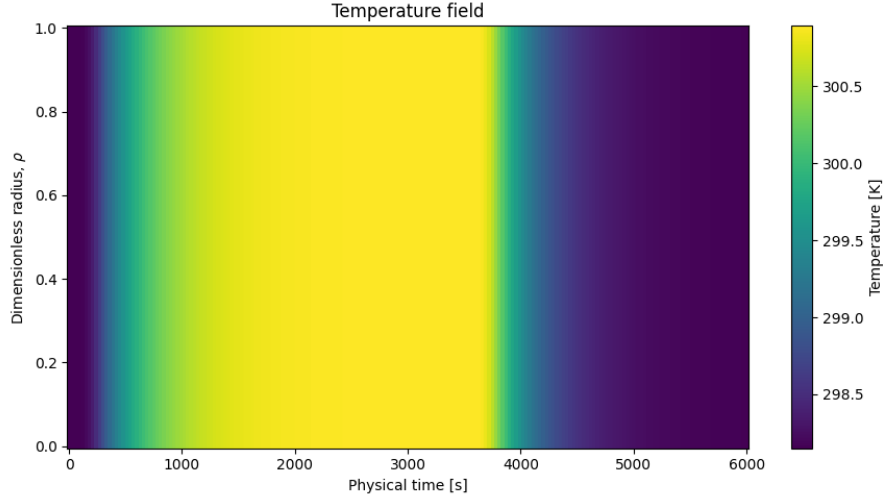


Figure 9: Spatio-temporal representation of the lumped temperature state.

Although represented over the entire radial domain for visualization purposes,

$$T(\rho, t) = T(t)$$

holds for every radial position.

The figure should therefore not be interpreted as the solution of a heat-conduction equation, but rather as a graphical representation of the single thermal state replicated along the spatial coordinate.

14.3 Adaptive Diffusion Behaviour

The increase in temperature produces a corresponding increase in the solid-state diffusion coefficient through the Arrhenius correction. Simultaneously, the progressive lithiation of the particle modifies the concentration-dependent contribution.

Consequently, the effective diffusion coefficient evolves continuously throughout the simulation according to

$$D_s = D_s(C, T).$$

Unlike the constant-diffusivity model, different regions of the particle may therefore exhibit different local transport rates.

Near the particle surface, where lithium concentration initially increases more rapidly, the concentration-dependent correction produces a larger diffusion coefficient than in the particle interior. At the same time, the global temperature increase amplifies diffusion over the entire domain.

The combined action of these two mechanisms accelerates the redistribution of lithium from the surface towards the particle centre, reducing concentration gradients during the charging phase.

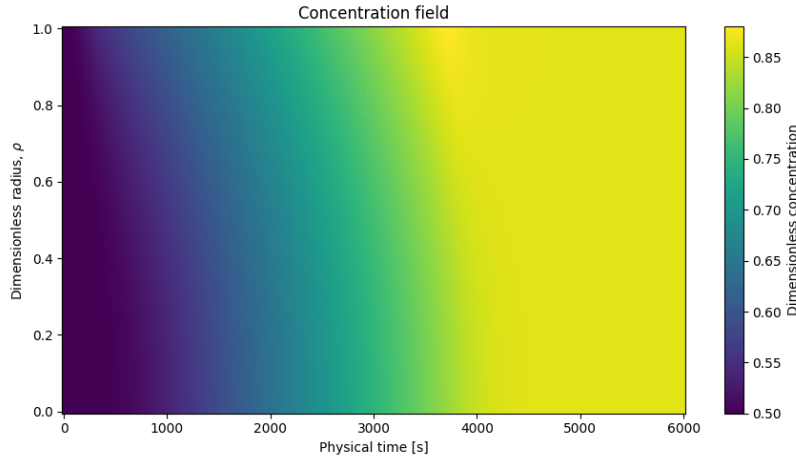


Figure 10: Concentration map in the thermo-diffusive model

14.4 Concentration Evolution

Despite the introduction of variable diffusivity, the qualitative behaviour of the concentration field remains consistent with the expected physical evolution.

Lithium first accumulates near the particle surface because of the prescribed intercalation flux. As diffusion progresses, the concentration front gradually propagates towards the particle centre until the entire particle approaches equilibrium during the relaxation phase.

The maximum concentration observed during the simulation is

$$C_{\max} = 0.879933,$$

and occurs at the particle surface at approximately

$$t = 3720 \text{ s.}$$

The final concentrations become

$$C_{\text{centre}} = 0.863478,$$

and

$$C_{\text{surface}} = 0.863479,$$

yielding a difference smaller than

$$10^{-6},$$

which indicates that the particle has essentially reached a uniform lithium distribution by the end of the simulation.

14.5 Mass Conservation and Numerical Stability

The conservative finite-volume discretization remains valid after the introduction of state-dependent diffusivity.

Throughout the simulation, the concentration remained bounded within the physically admissible interval,

$$0.5 \leq C \leq 0.879933,$$

while no numerical instabilities were observed.

In particular,

$$\text{NaN} = 0, \quad \text{Inf} = 0,$$

for every stored solution.

The maximum absolute mass-balance residual was

$$4.47 \times 10^{-5},$$

confirming that the conservative formulation preserves lithium mass despite the nonlinear diffusion coefficient.

These results demonstrate that the proposed thermo-diffusion model is numerically stable over the complete simulation interval and can therefore be employed as a reliable high-fidelity generator of synthetic training data.

14.6 Reference Data for Model Development and Evaluation

The extended thermo-diffusive finite-volume model serves as the numerical reference throughout the present work. For each simulated operating trajectory, it provides the concentration field $C_{\text{ref}}(\rho, \tau)$ and the temperature evolution $T_{\text{ref}}(\tau)$, together with the prescribed current profile and the simulation parameters. These outputs support the construction of training datasets and the evaluation of the investigated modelling strategies.

The reference data are used in different ways according to the quantity predicted by each model. For the standalone PINN, the network receives only the spatial and temporal coordinates,

$$(\rho, \tau),$$

and predicts the corresponding lithium concentration. The temperature is not supplied as an input, although its influence is present in the reference concentration data through the state-dependent diffusivity.

For the FVM–FNN hybrid model and the post-training Hybrid-PINN, the reference concentration field is used to construct residual targets relative to the respective baseline predictions,

$$\Delta C(\rho, \tau) = C_{\text{ref}}(\rho, \tau) - C_{\text{base}}(\rho, \tau).$$

These residuals define the corrections to be learned by the associated neural networks. The temperature trajectories generated by the reference model also support the identification and evaluation of the thermal models based on SINDyC and feedforward neural networks.

The extended FVM additionally provides the numerical benchmark against which concentration predictions and derived quantities, including surface concentration and volume-averaged concentration, are compared. The inputs, targets, and data partitions adopted for each modelling strategy are specified in the corresponding sections.

Within this framework, the standalone PINN is trained against concentration data generated by the extended thermo-diffusive model while enforcing a simplified physical formulation with constant diffusivity. Thermal effects are therefore represented implicitly in the observational targets but are absent from the governing equation embedded in the network. This controlled discrepancy allows the study to assess how observational data and subsequent residual correction compensate for the physical effects omitted from the simplified formulation.

The extended FVM is thus used as a common numerical reference for data generation and comparative assessment. Agreement with this reference measures the ability to reproduce the simulated dynamics; experimental validation against a physical cell remains a separate requirement.

15 Physics-Informed Neural Network with Simplified Physics

The high-fidelity thermo-diffusive finite-volume model introduced in Section 13 provides a physically consistent reference concentration field. The Physics-Informed Neural Network (PINN) developed in this work is not supplied with the complete physics used by that solver. Instead, the network is deliberately constrained by a reduced constant-diffusivity model and is simultaneously trained on concentration samples generated by the full-physics simulator.

This configuration differs from the conventional use of a PINN as a direct surrogate of a known differential equation. The objective is not to reproduce the finite-volume solution using an identical mathematical model, but to investigate whether observational data can compensate, at least partially, for deliberately missing physical information. The conventional PINN formulation combines observational, initial, boundary and differential-equation residuals in a common optimization objective [Cuomo et al., 2022; Khanra et al., 2026]. Here, the resulting prediction is therefore the outcome of a competition between two sources of information:

- the full-physics concentration data, which contain the effects of the operating current, temperature evolution and state-dependent diffusivity;
- the simplified differential equation and boundary condition, which impose constant normalized diffusivity and a constant surface flux.

This tension is intentional and constitutes the central methodological experiment of the present work.

15.1 Simplified Physical Model Enforced by the PINN

The PINN uses the dimensionless radial coordinate and time as inputs,

$$\mathbf{x} = [\rho \quad \tau]^T, \quad 0 \leq \rho \leq 1, \quad 0 \leq \tau \leq \tau_f,$$

and predicts the dimensionless solid concentration

$$\hat{C} = \hat{C}_\theta(\rho, \tau),$$

where θ denotes the trainable neural-network parameters.

The governing equation embedded in the PINN is the constant-coefficient spherical diffusion equation in expanded form,

$$\boxed{\frac{\partial \hat{C}}{\partial \tau} - \frac{\partial^2 \hat{C}}{\partial \rho^2} - \frac{2}{\rho} \frac{\partial \hat{C}}{\partial \rho} = 0.} \quad (3)$$

The initial concentration is

$$\hat{C}(\rho, 0) = C_0, \quad C_0 = 0.5, \quad (4)$$

and spherical symmetry requires

$$\left. \frac{\partial \hat{C}}{\partial \rho} \right|_{\rho=0} = 0. \quad (5)$$

At the particle surface, the PINN enforces the constant Neumann condition

$$\boxed{-\left.\frac{\partial \hat{C}}{\partial \rho}\right|_{\rho=1} = \delta_0, \quad \delta_0 = -0.2004616.} \quad (6)$$

The value δ_0 corresponds to the nominal 1C dimensionless flux evaluated using the reference diffusivity. It is held constant over the complete time interval, including the ramp-up, ramp-down and relaxation phases of the full-physics simulation.

Table 7 summarizes the deliberate mismatch between the model used to generate the data and the physics enforced during PINN training.

Table 7: Deliberate physical mismatch between the reference model and the PINN.

Modelling component	Full-physics FVM reference	Physics enforced by the PINN
Diffusion coefficient	$\hat{D} = \hat{D}(C, T)$	$\hat{D} = 1$
Diffusion equation	$C_\tau = \rho^{-2} \partial_\rho (\rho^2 \hat{D} C_\rho)$	$C_\tau = C_{\rho\rho} + 2C_\rho/\rho$
Temperature	Computed through a lumped thermal balance	Not supplied to the network
Surface flux	Ramp-constant-ramp-rest, $\delta = \delta(\tau)$	Constant, $\delta = \delta_0$
Initial condition	Uniform concentration $C_0 = 0.5$	Imposed exactly by construction
Centre symmetry	Zero radial derivative	Imposed exactly by construction

The thermal state and the adaptive diffusivity are therefore available only indirectly through the target concentration values. This choice prevents the network from simply reproducing the complete reference equations and makes it possible to evaluate the compensating role of data.

15.2 Exact Enforcement of Initial and Centre Conditions

The initial condition and the centre symmetry condition are not included as penalty terms in the optimization objective. They are embedded analytically in the network representation and are therefore satisfied for every value of the trainable parameters.

Let N_θ denote the unconstrained output of the multilayer perceptron. The final concentration prediction is constructed as

$$\boxed{\hat{C}_\theta(\rho, \tau) = C_0 + \tau N_\theta(\rho^2, \tau).} \quad (7)$$

At $\tau = 0$, Equation (7) immediately gives

$$\hat{C}_\theta(\rho, 0) = C_0,$$

independently of N_θ . The initial condition is consequently exact rather than approximately recovered through training.

The use of ρ^2 as the spatial network input produces

$$\frac{\partial \hat{C}_\theta}{\partial \rho} = 2\rho\tau \frac{\partial N_\theta}{\partial(\rho^2)},$$

and therefore

$$\left. \frac{\partial \widehat{C}_\theta}{\partial \rho} \right|_{\rho=0} = 0.$$

This transformation removes the need for an additional centre-boundary loss and guarantees regular behaviour at the origin. Only the outer Neumann boundary condition contributes to the optimization loss. This distinction is important: no initial-condition loss and no centre symmetry loss are introduced anywhere in the training procedure.

15.3 Neural-Network Architecture

The approximation N_θ is represented by a fully connected multilayer perceptron. After the analytical input transformation, the network receives the two-dimensional vector

$$\widetilde{\mathbf{x}} = [\rho^2 \quad \tau]^T.$$

Four hidden layers are used, each containing 100 neurons with hyperbolic tangent activation. The output layer is linear and returns one scalar. The architecture is therefore

$$2 \longrightarrow 100 \longrightarrow 100 \longrightarrow 100 \longrightarrow 100 \longrightarrow 1.$$

All linear-layer weights are initialized using Xavier normal initialization with the gain associated with the tanh activation, while every bias is initialized to zero. The complete model contains 30,701 trainable parameters.

Table 8: PINN architecture and initialization.

Component	Configuration	Purpose
Raw inputs	(ρ, τ)	Space-time coordinates
Transformed inputs	(ρ^2, τ)	Exact centre symmetry
Hidden layers	4 layers, 100 neurons each	Nonlinear approximation
Activation	tanh	Smooth automatic derivatives
Output layer	1 linear neuron	Unconstrained scalar N_θ
Output transform	$C_0 + \tau N_\theta$	Exact initial condition
Weight initialization	Xavier normal	Stable initial signal scale
Bias initialization	Zero	Deterministic neutral offset
Trainable parameters	30,701	Complete parameter count

The tanh activation is particularly suitable for this problem because the PDE residual requires first-order temporal derivatives and first- and second-order radial derivatives Khanra et al. [2026]. A smooth activation avoids the derivative discontinuities that would arise with piecewise-linear functions.

15.4 Training, Validation and Test Datasets

Three datasets are used independently during optimization.

1. **Reference-data points.** The complete full-physics finite-volume field provides tuples $(\rho, \tau, C_{\text{FVM}})$. These samples directly communicate the target high-fidelity behaviour to the network.

2. **Interior collocation points.** The spatial and temporal coordinates are extracted from the reference grid after excluding $\rho = 0$, $\rho = 1$ and $\tau = 0$. Removing the origin avoids the explicit evaluation of the singular coefficient $2/\rho$, while the centre condition remains exactly satisfied through the output transformation.
3. **Surface-boundary points.** Independent time coordinates are generated with a one-dimensional Sobol sequence and combined with $\rho = 1$. These points are used exclusively to enforce the simplified outer Neumann condition.

Each dataset is independently divided into 70% training, 15% validation and 15% test subsets using a fixed random seed. The resulting sample counts are reported in Table 9.

Table 9: Independent data, collocation and boundary dataset splits.

Dataset	Total	Training	Validation	Test
Full-physics data	20,301	14,210	3,045	3,046
Interior collocation	19,800	13,860	2,970	2,970
Surface boundary	20,301	14,210	3,045	3,046

Mini-batches of 512 samples are used during the Adam stage. The three loaders are evaluated jointly, but each loss contribution is computed from the dataset appropriate to that contribution. Thus, the same network is evaluated at data, collocation and boundary coordinates; no separate neural networks are introduced for the different terms.

15.5 Physics-Informed Objective Function

The data loss measures the discrepancy between the PINN prediction and the full-physics finite-volume concentration,

$$\mathcal{L}_{\text{data}} = \frac{1}{N_d} \sum_{i=1}^{N_d} \left| \hat{C}_\theta(\rho_i, \tau_i) - C_{\text{FVM},i} \right|^2. \quad (8)$$

At the interior collocation points, automatic differentiation is used to evaluate

$$\frac{\partial \hat{C}_\theta}{\partial \tau}, \quad \frac{\partial \hat{C}_\theta}{\partial \rho}, \quad \frac{\partial^2 \hat{C}_\theta}{\partial \rho^2}.$$

The simplified PDE residual is

$$R_f = \frac{\partial \hat{C}_\theta}{\partial \tau} - \frac{\partial^2 \hat{C}_\theta}{\partial \rho^2} - \frac{2}{\rho} \frac{\partial \hat{C}_\theta}{\partial \rho}, \quad (9)$$

This use of automatic differentiation to evaluate governing-equation residuals at collocation points is a defining element of PINN methodology [Cuomo et al., 2022; Khanra et al., 2026].

The associated loss is

$$\mathcal{L}_{\text{PDE}} = \frac{1}{N_f} \sum_{i=1}^{N_f} |R_{f,i}|^2. \quad (10)$$

At the independently generated surface points, the boundary residual is

$$R_b = - \left. \frac{\partial \hat{C}_\theta}{\partial \rho} \right|_{\rho=1} - \delta_0, \quad (11)$$

which yields

$$\mathcal{L}_{\text{BC}} = \frac{1}{N_b} \sum_{i=1}^{N_b} |R_{b,i}|^2. \quad (12)$$

The complete optimization objective is

$$\boxed{\mathcal{L} = \lambda_d \mathcal{L}_{\text{data}} + \lambda_f \mathcal{L}_{\text{PDE}} + \lambda_b \mathcal{L}_{\text{BC}},} \quad (13)$$

with

$$\lambda_d = \lambda_f = \lambda_b = 1.$$

Equal weighting provides a transparent baseline from which the natural competition between data fidelity and simplified-physics consistency can be observed. No adaptive loss weighting is used in the present model. Most importantly, Equation (13) contains neither an initial-condition term nor a centre-boundary term, because both conditions are hard constraints.

15.6 Learning and Model-Selection Procedure

The learning procedure combines a stochastic first-order stage with an optional deterministic full-batch refinement.

15.6.1 Adam stage

Adam is used with an initial learning rate of

$$\eta_0 = 10^{-3}$$

and a maximum of 10,000 epochs. At the end of every epoch, the three loss components are independently evaluated on the validation subsets. The reported epoch losses are sample-weighted averages rather than sums of mini-batch averages.

A `ReduceLROnPlateau` scheduler monitors the total validation loss. When progress stagnates, it multiplies the learning rate by 0.5, with a patience of 200 epochs, a relative threshold of 10^{-4} , a cooldown of 50 epochs and a minimum learning rate of 10^{-6} .

Model selection is intentionally based on the validation data loss. A copy of the model parameters is retained whenever $\mathcal{L}_{\text{data, val}}$ improves. This choice reflects the main objective of the experiment: approximating the full-physics reference field while still being regularized by simplified physical information.

If the data-validation loss fails to improve for 800 epochs, a plateau is declared and the Adam stage is terminated.

15.6.2 L-BFGS refinement

After a detected plateau, L-BFGS is applied to the complete training sets. The optimizer uses a maximum of 500 iterations, a history size of 100 and a strong-Wolfe line search. The full-batch closure recomputes the data, PDE and outer-boundary losses together with all required automatic derivatives.

The L-BFGS solution is not accepted unconditionally. Its validation data loss is compared with that of the best Adam checkpoint. If refinement worsens agreement with the full-physics observations, the best Adam parameters are restored. In the experiment reported here, this safeguard was activated and the final stored model corresponds to the best Adam state.

The final checkpoint records the model parameters, the validation losses, the physical constants, the three loss weights and the selected optimizer.

Table 10: Optimization and model-selection settings.

Item	Value	Role
Adam initial learning rate	10^{-3}	Mini-batch optimization
Maximum epochs	10,000	Upper training limit
Mini-batch size	512	Adam data processing
Scheduler factor	0.5	Learning-rate reduction
Scheduler patience	200 epochs	Plateau sensitivity
Early-stopping patience	800 epochs	Switch criterion
Minimum learning rate	10^{-6}	Scheduler lower bound
L-BFGS maximum iterations	500	Full-batch refinement
L-BFGS line search	Strong Wolfe	Step selection
Selection metric	Data-validation MSE	Full-physics fidelity
Final selected optimizer	Adam	Best retained checkpoint

15.7 Numerical Diagnostics

Three diagnostic checks are evaluated during learning to detect unstable optimization before it contaminates the stored model.

1. The concentration output is checked for non-finite values and its minimum, maximum, mean and standard deviation are monitored. A warning is emitted when predictions move significantly outside the normalized physical interval.
2. The root-mean-square and maximum absolute PDE residual are monitored to identify a sudden loss of physical consistency.
3. After backpropagation, the total gradient norm and the layer-wise gradient norms are inspected for non-finite, exploding or vanishing gradients.

After training, independent test subsets are used to evaluate the relative concentration error, relative surface-concentration error, PDE RMSE, boundary-condition RMSE and synchronized inference time. Separate post-processing checks verify the exact constraints, the physical concentration interval and the integral mass balance.

16 PINN Results and Comparison with the Full-Physics Model

The final PINN was evaluated in two complementary ways. First, the randomly held-out data, collocation and boundary test subsets were used to quantify generalization within the sampled domain. Second, the model was evaluated on the complete structured 101×201 finite-volume grid to obtain directly comparable fields, profiles and integral quantities.

This distinction is useful because the test split provides an unbiased estimate at unseen samples, while the structured-grid analysis exposes where and when the model discrepancy develops.

16.1 Validation and Held-Out Test Performance

The selected checkpoint achieved a data-validation mean-square error of

$$\mathcal{L}_{\text{data, val}} = 1.4707 \times 10^{-3},$$

and a total validation loss of

$$\mathcal{L}_{\text{val}} = 3.2028 \times 10^{-3}.$$

Table 11 summarizes the independent test-set results. The concentration error is measured against the full-physics finite-volume targets, whereas the PDE and boundary residuals are computed with respect to the simplified physics enforced by the PINN.

Table 11: PINN performance on the independent test subsets.

Metric	Value	Interpretation
Full-field relative L^2 error	4.9633×10^{-2} (4.9633%)	Agreement with held-out FVM data
Surface relative L^2 error	7.4243×10^{-2} (7.4243%)	Agreement at $\rho = 1$
Simplified-PDE RMSE	1.9074×10^{-2}	Interior physics residual
Constant-flux BC RMSE	3.6957×10^{-2}	Outer-boundary residual
Total inference time	4.5469 ms	3,046 held-out data samples on MPS
Inference time per sample	1.4927 μ s	Synchronized forward evaluation

A relative full-field error close to 5% is a meaningful result in the present setting. The network is not being tested against data generated by its own PDE: the targets contain a time-varying current, thermal dynamics and a concentration- and temperature-dependent diffusivity that are all absent from the PINN formulation. The result therefore shows that the observational term successfully transfers a substantial part of the missing behaviour into the learned approximation.

The larger surface error is physically interpretable. The surface is the location where the imposed constant PINN flux conflicts most directly with the ramped and subsequently vanishing reference flux. Errors are partially smoothed as lithium diffuses towards the particle interior, explaining why the centre and full-field measures are lower than the surface measure.

16.2 Standalone PINN Prediction

Figure 11 shows the concentration field predicted by the PINN over the complete space–time domain.

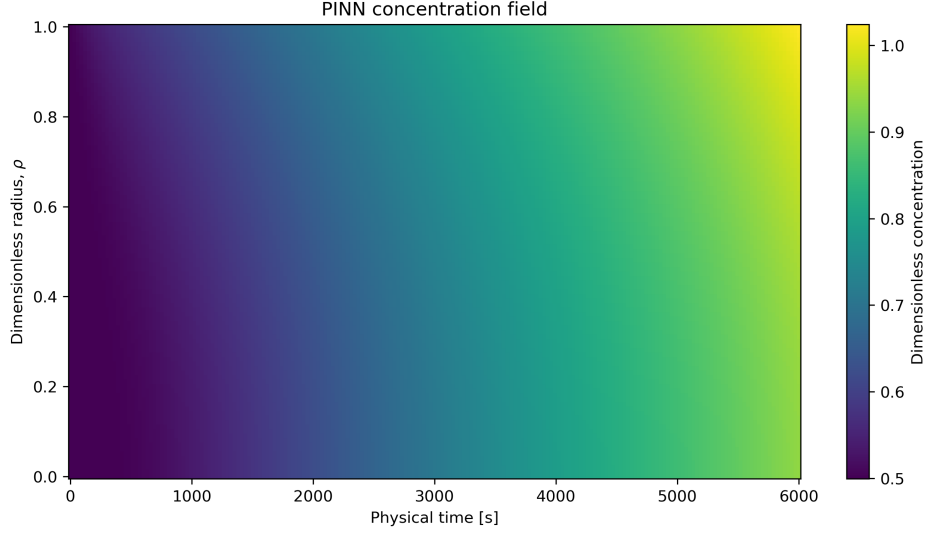
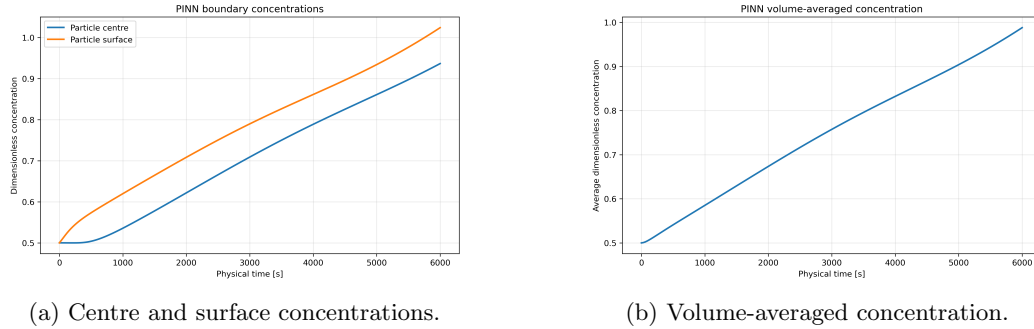


Figure 11: PINN concentration field.

The PINN correctly reproduces the initial uniform concentration and the development of a positive radial gradient, with larger concentration at the particle surface. Unlike the full-physics reference solution, however, the PINN concentration continues to increase during the final part of the simulation. This behaviour follows directly from the constant-flux condition: the simplified model continues to insert lithium even after the reference current has ramped down to zero.

The boundary and volume-averaged concentrations are reported in Figure 12.



(a) Centre and surface concentrations.

(b) Volume-averaged concentration.

Figure 12: Standalone temporal predictions of the PINN.

The surface concentration responds more rapidly than the centre concentration, as expected for diffusion driven from the external boundary. The volume average evolves smoothly and remains free from oscillations or discontinuities. The late-time monotonic increase is not a numerical instability; it is the deterministic consequence of the simplified boundary input.

Figure 13 further illustrates that the network produces smooth radial profiles and exactly satisfies the initial uniform state.

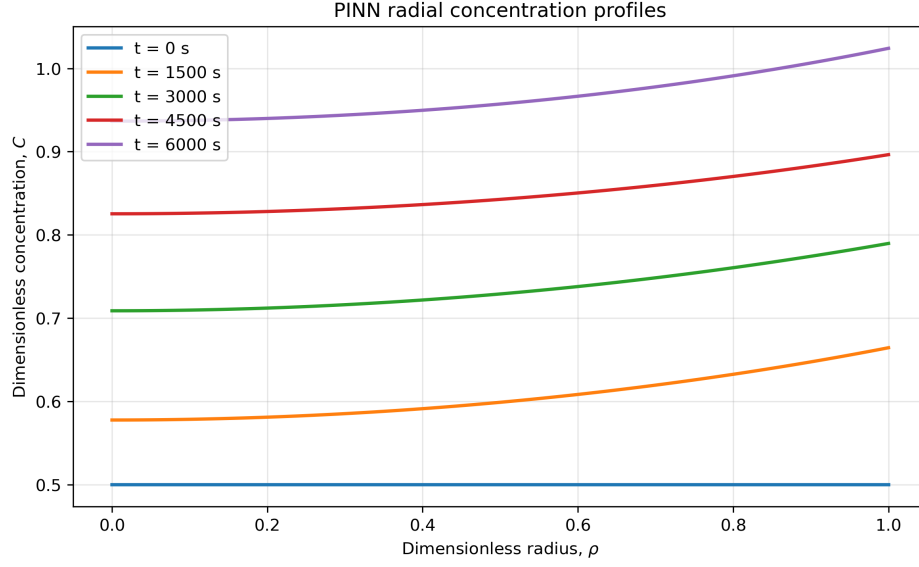


Figure 13: PINN radial concentration profiles at selected physical times.

At late times, the profiles retain a non-zero surface-to-centre difference, whereas the full-physics solution becomes almost spatially uniform after removal of the applied current. This discrepancy will be examined explicitly in the direct comparison.

The simplified-model mass-balance residual and the constant-flux Guo–White comparison are included in Figure 14.

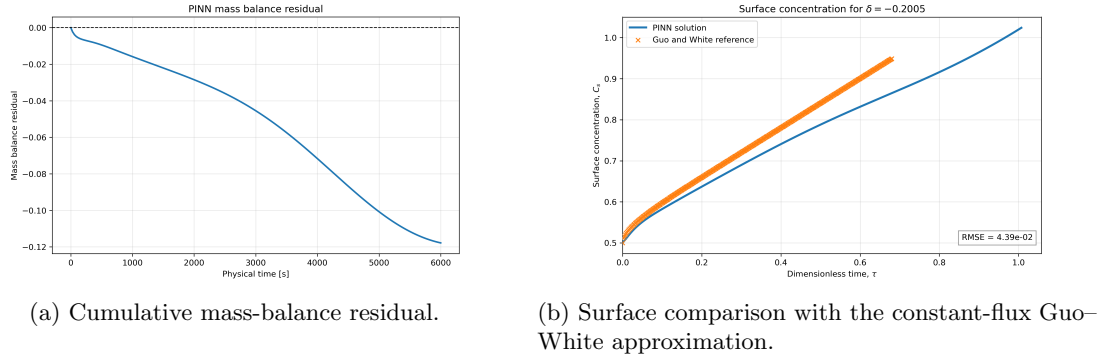


Figure 14: Physics-oriented diagnostics of the standalone PINN.

These diagnostics must be interpreted together with the supervised objective. The learned field is not a pure solution of the simplified constant-flux problem: the data loss pulls the prediction towards the full-physics response. A non-zero simplified-physics mass residual is therefore evidence of the trade-off made by the optimizer, rather than evidence that the finite-volume reference data violate conservation.

16.3 Direct Full-Field Comparison

For direct comparison, the PINN and FVM solutions were evaluated on the same 101×201 grid. Identical concentration limits were used for both colour maps. This shared colour normalization is essential: if each field were plotted using its own minimum and maximum, the two maps would appear artificially similar and the PINN overshoot would be visually hidden.

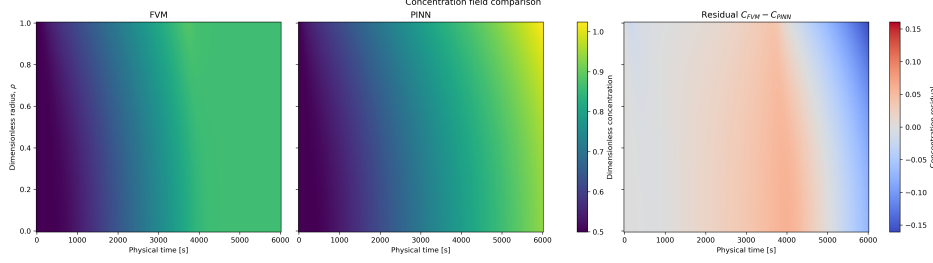


Figure 15: Full-physics FVM field, PINN field and pointwise residual $C_{\text{FVM}} - C_{\text{PINN}}$. A common colour scale is used for the two concentration maps.

The reference field reaches an almost uniform plateau after the current is removed. In contrast, the PINN continues to evolve because its constant surface flux remains active. The residual map is smooth and strongly structured in time. It remains small during the early loading stage, changes sign around the transition to relaxation (before on surface, after internal regions) and becomes largest close to the particle surface at late times. This pattern is a direct signature of model-form error rather than unstructured neural noise.

The quantitative dense-grid errors are reported in Table 12.

Table 12: PINN errors with respect to the full-physics FVM solution on the complete structured grid.

Quantity	Relative L^2 error	RMSE	MAE	Maximum absolute error
Complete field	0.050674	0.037757	0.027721	0.160606
Particle centre	0.045458	0.033251	0.026263	0.073184
Particle surface	0.068609	0.052938	0.034088	0.160606
Volume average	0.053791	0.040645	0.028642	0.124484

The 5.07% relative error computed on the complete grid is close to the 4.96% error obtained on the randomly held-out test samples. This agreement indicates that the random split does not conceal a substantial generalization failure. It also confirms that the comparison plots and the independent test routine describe the same overall model quality.

The surface remains the most demanding output, with a dense-grid relative error of 6.86% and a maximum absolute error of 0.1606. The largest discrepancy occurs at the end of the simulation, when the FVM has relaxed to approximately 0.8635 while the PINN surface prediction reaches approximately 1.0241. The centre error remains lower because the interior response is delayed and spatial diffusion attenuates the direct effect of the boundary mismatch.

16.4 Integral and Boundary Quantities

Figure 16 compares the quantities most directly related to lithium inventory and boundary behaviour.

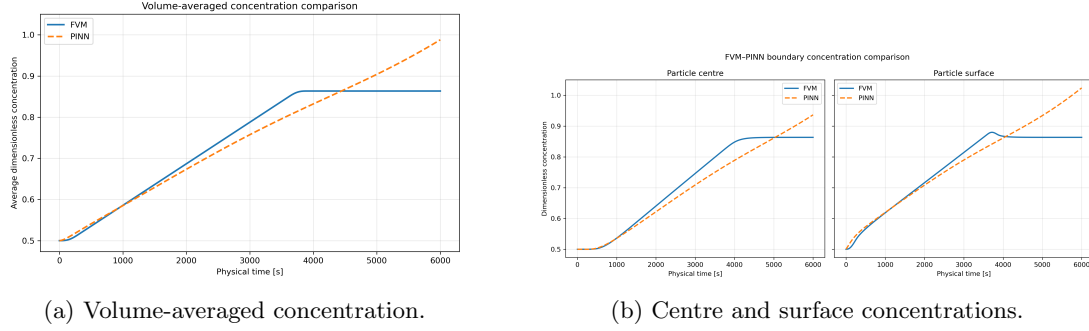


Figure 16: Direct comparison of integral and boundary concentration quantities.

The volume-average curves are close during the first part of the loading phase. The reference average then approaches a constant value as the current is removed, while the PINN average continues to increase. A similar trend is observed at both boundaries. The reference surface concentration displays a local maximum during ramp-down followed by relaxation, whereas the constant-flux PINN cannot reproduce this turning point exactly in such underlying physics configuration.

This observation identifies the time-dependent boundary input as the dominant source of late-time error. Adaptive diffusivity also affects the curvature and radial redistribution, but the persistent increase after the reference current reaches zero is specifically attributable to the simplified constant-flux assumption.

Radial profiles and the surface residual are reported in Figure 17.

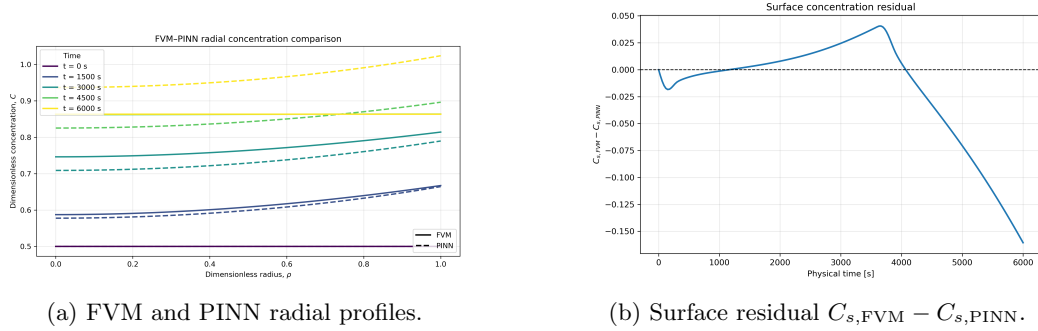


Figure 17: Spatial comparison and temporal development of the surface discrepancy.

The residual changes gradually rather than oscillating rapidly. This is important for the intended hybrid extension: a smooth, phase-dependent error is more amenable to a compact corrective model than high-frequency or apparently random errors.

16.5 Error Distribution over the Operating Phases

To connect prediction error with the imposed operating conditions, the full-field discrepancy was evaluated separately during ramp-up, constant-current operation, ramp-down and rest. The resulting metrics are shown in Table 13.

Table 13: Full-field PINN error over the different operating phases.

Operating phase	Relative L^2 error	RMSE	Physical interpretation
Ramp-up	0.009335	0.004683	Initial response captured accurately
Constant current	0.033957	0.022696	Moderate accumulated mismatch
Ramp-down	0.059622	0.050293	Boundary inputs begin to diverge
Rest	0.062203	0.053651	PINN constant flux remains active

The ramp-up relative error is below 1%, demonstrating that the network captures the initial thermo-diffusive response very accurately even though neither temperature nor adaptive diffusivity is provided as an input. The error rises to approximately 3.4% during the constant-current stage and to approximately 6% during ramp-down and rest.

This monotonic relation between operating-phase mismatch and prediction error is one of the most informative results of the experiment. The network performs best when the simplified constant flux is close to the reference operating condition and deteriorates when the two physical descriptions diverge. The behaviour therefore remains physically interpretable and supports the conclusion that the remaining error is primarily model-form error.

16.6 Physical Consistency Checks

The PINN and full-physics FVM fields were subjected to the same post-processing checks. The volume average was evaluated as

$$\overline{C}(\tau) = \frac{\int_0^1 C(\rho, \tau) \rho^2 d\rho}{\int_0^1 \rho^2 d\rho}, \quad (14)$$

and the cumulative mass-balance residual was defined consistently as

$$R_{\text{mass}}(\tau) = \overline{C}(\tau) - \left[C_0 - 3 \int_0^\tau \delta(s) ds \right]. \quad (15)$$

For the FVM, $\delta(\tau)$ is the ramp–constant–ramp–rest profile. For the PINN, $\delta(\tau) = \delta_0$ is the constant flux actually used in its loss function. Thus, both residuals use the same mathematical definition but are evaluated against the boundary law belonging to the corresponding model.

The numerical checks are summarized in Table 14.

The exact zero errors associated with the initial state and centre symmetry verify that the analytical output transformation behaves as intended. These conditions do not deteriorate during optimization and do not compete with the three soft loss terms.

The PINN maximum concentration of 1.0241 represents a limited but physically relevant overshoot. It occurs near the surface at late time, where the simplified model still applies an insertion

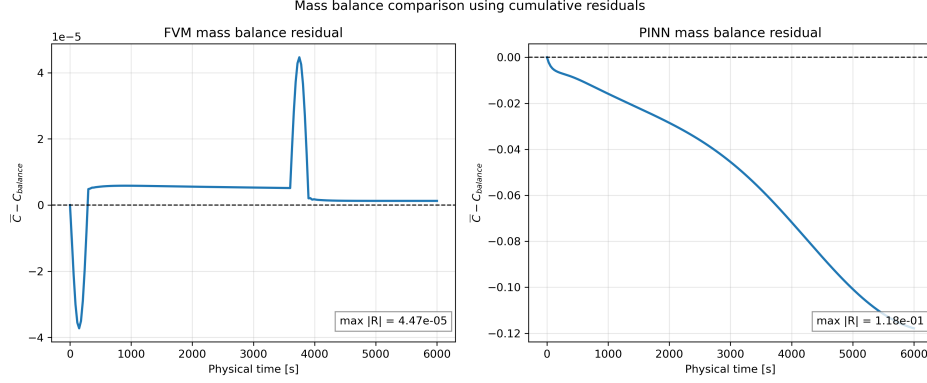


Figure 18: FVM and PINN cumulative mass-balance residuals. Each model is evaluated using its corresponding prescribed surface flux.

Table 14: Physical and hard-constraint checks on the structured grid.

Check	FVM	PINN	Comment
Minimum concentration	0.500000	0.499530	Both remain close to the initial lower level
Maximum concentration	0.879933	1.024085	PINN exceeds the normalized upper bound
Maximum $ R_{\text{mass}} $	4.47×10^{-5}	1.1783×10^{-1}	FVM conservative; PINN balances data and simplified physics
Initial-condition maximum error	—	0	Exact hard constraint
Centre-gradient maximum error	—	0	Exact hard constraint

flux after the full-physics model has entered relaxation. This result should not be hidden by rescaling the comparison plots. It identifies a clear regime in which a corrective strategy is needed.

The larger PINN mass residual also has a precise interpretation. A pure solution of the constant-flux PDE would follow its own integral balance, whereas the supervised full-physics data tend towards a plateau. The trained network adopts a compromise between these incompatible targets. The residual therefore quantifies the effect of combining deliberately inconsistent data and physics, while the much smaller FVM residual confirms that the reference dataset itself remains conservative.

16.7 Assessment of the Simplified-Physics Strategy

The results support four main conclusions.

1. **The dominant diffusion behaviour is learned.** The PINN reproduces the initial condition, the direction and smoothness of radial transport, the delayed centre response and the main increase in particle lithium content. A global relative error of approximately 5% is obtained despite the deliberately incomplete governing model.
2. **Data compensate for a significant part of the missing physics.** During ramp-up, the relative field error remains below 1%, even though the reference temperature and variable diffusivity are never provided to the network. The observational term therefore conveys information that cannot be generated by the simplified PDE alone.
3. **The remaining error is structured and physically explainable.** The largest discrepancy appears at the surface and after the reference current begins to decrease. This is exactly

where the constant-flux approximation becomes least representative. The residual is consequently linked to identifiable missing mechanisms rather than to numerical instability or random overfitting.

4. **The model retains useful physical structure.** The hard initial and symmetry conditions are satisfied to machine precision, while the PDE and boundary residuals remain finite and controlled. The model is therefore more than an unconstrained interpolator, even though data fidelity is deliberately prioritized during checkpoint selection.

Within the objective of this study, this constitutes a good result. A nearly exact PINN would leave little systematic discrepancy for the next stage and would make it difficult to study hybrid correction. Conversely, a poorly performing base model would force a subsequent network to learn the complete solution rather than a correction. The present PINN occupies a useful intermediate regime: it captures the dominant field with approximately 5% global error, while preserving a smooth and physically localized residual associated with the intentionally omitted physics.

The result should nevertheless be interpreted as a baseline rather than as a final high-fidelity surrogate. The late-time overshoot, surface error and mass-balance discrepancy prevent the standalone PINN from replacing the extended finite-volume model over the complete operating cycle. Their value lies in revealing where the simplified description ceases to be sufficient.

16.8 Transition to the Implemented Residual-Learning Extension

The structured discrepancy identified above is subsequently learned by a dedicated data-driven model. This separation between a misspecified PINN and a discrepancy network is consistent with corrective strategies developed for model-form error in PINNs [Zou et al., 2024]. For the PINN-based corrective branch, the supervised target is

$$r_C(\rho, \tau) = C_{\text{FVM}}(\rho, \tau) - \hat{C}_\theta(\rho, \tau). \quad (16)$$

and the corrected prediction is defined as

$$C_{\text{hybrid}}(\rho, \tau) = \hat{C}_\theta(\rho, \tau) + \hat{r}_\phi(\rho, \tau), \quad (17)$$

where $\hat{r}_\phi$ denotes the learned correction. The selected PINN checkpoint is frozen before the residual dataset is generated and is not retrained during this stage. The additional network therefore approximates only the part of the full-physics response that the reduced PINN does not represent.

The comparative analysis explains why residual learning is suitable in this case. The residual map is smooth, its magnitude changes systematically with the operating phase, and its largest values are localized near the surface and during relaxation. Consequently, the correction task is substantially more structured than learning the full concentration field from scratch. The simplified PINN can retain the dominant diffusion behaviour and exact analytical constraints, while the residual model focuses specifically on the effects omitted from the reduced physics.

The detailed architecture, data flow, numerical results and physical checks of this implemented post-training correction are reported in Section 21. This branch must not be confused with the standalone FVM-FNN hybrid model introduced next: the latter corrects a simplified numerical solver and returns a complete radial profile, whereas the Hybrid-PINN corrects a frozen coordinate-based PINN one point at a time.

17 Hybrid Physics–ML Modelling through Residual Learning

The simplified-physics PINN investigated in the previous section combines observational data and reduced governing equations within a single optimization problem. The present section develops a separate FVM–FNN hybrid strategy. An independently executable simplified finite-volume model first provides the dominant concentration dynamics. A feedforward neural network then learns the discrepancy between that baseline and the available high-fidelity concentration observations.

This formulation is distinct from the post-training Hybrid-PINN in Section 21. Here, the baseline is a simplified finite-volume solver and the network returns a complete 101-node radial correction. The Hybrid-PINN instead corrects a frozen coordinate-based PINN one point at a time.

The FVM–FNN prediction has the additive structure

$$\boxed{\mathbf{C}_{\text{hybrid}}(\tau) = \mathbf{C}_{\text{simpl}}(\tau) + \widehat{\Delta\mathbf{C}}_{\phi}(\tau)}, \quad (18)$$

where $\mathbf{C}_{\text{simpl}}$ is the radial concentration profile computed by the simplified solver and $\widehat{\Delta\mathbf{C}}_{\phi}$ is the constrained correction produced by the network with parameters ϕ . The simplified solver remains responsible for time integration, spherical geometry, the operating current and the dominant diffusion process. The FNN is assigned only the model-form discrepancy associated with the unresolved transport and thermal effects [Garg et al., 2022; Ghnatios and Chinesta, 2025].

17.1 Hybrid Modelling Objective

For the stored time instant τ_k , the reference and simplified profiles are

$$\mathbf{C}_{\text{ref}}(\tau_k) = [C_{\text{ref}}(\rho_0, \tau_k) \quad \cdots \quad C_{\text{ref}}(\rho_N, \tau_k)]^T,$$

$$\mathbf{C}_{\text{simpl}}(\tau_k) = [C_{\text{simpl}}(\rho_0, \tau_k) \quad \cdots \quad C_{\text{simpl}}(\rho_N, \tau_k)]^T.$$

The supervised residual target is

$$\boxed{\Delta\mathbf{C}_{\text{target}}(\tau_k) = \mathbf{C}_{\text{ref}}(\tau_k) - \mathbf{C}_{\text{simpl}}(\tau_k)}. \quad (19)$$

The word reference denotes the available concentration observations. In the synthetic experiment these observations are generated by the thermo-diffusive FVM, but the learning formulation does not require the true reference initial concentration to be supplied as an independent scalar. Only the observed concentration profiles are used to construct the target in Eq. (19).

The computational sequence is:

1. execute the simplified finite-volume model for a selected initial concentration and constant diffusivity;
2. construct reduced state variables from the simplified solution and combine them with the applied current;
3. predict a raw 101-node residual profile;
4. impose the constant-volume-average correction described in Section 17.5;

5. add the constrained correction to the simplified concentration profile.

Figure 19 summarizes the resulting data flow.

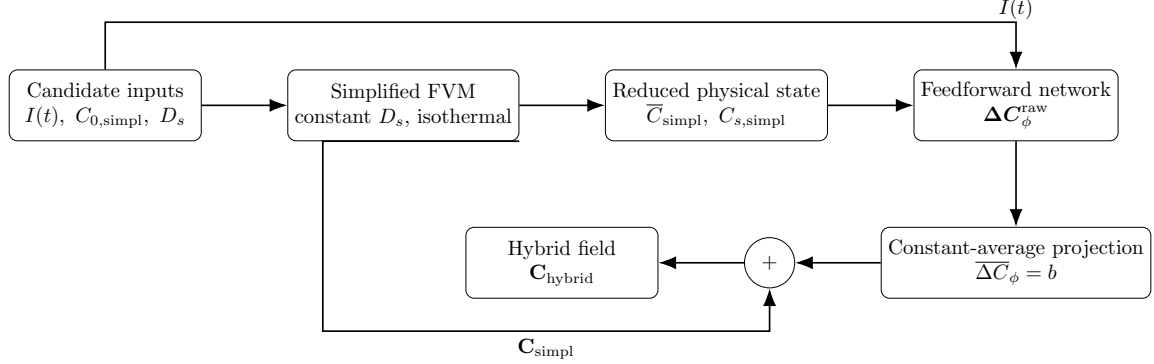


Figure 19: Architecture of the FVM-FNN residual model. The learned correction is projected onto the affine set of profiles whose spherical volume average is the constant offset b .

17.2 Simplified Physics-Based Baseline

The simplified component uses the conservative spherical finite-volume framework with a constant solid-state diffusion coefficient. In the dimensionless coordinates associated with the selected D_s , the governing equation is

$$\frac{\partial C_{\text{simpl}}}{\partial \tau} = \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial C_{\text{simpl}}}{\partial \rho} \right). \quad (20)$$

The initial and centre conditions are

$$C_{\text{simpl}}(\rho, 0) = C_{0,\text{simpl}}, \quad (21)$$

$$\left. \frac{\partial C_{\text{simpl}}}{\partial \rho} \right|_{\rho=0} = 0. \quad (22)$$

At the particle surface, the model retains the known time-dependent operating flux,

$$\left[- \frac{\partial C_{\text{simpl}}}{\partial \rho} \right]_{\rho=1} = \delta(\tau). \quad (23)$$

The simplified FVM therefore differs from the constant-flux model enforced in the standalone PINN. It receives the same ramp-constant-ramp-rest current history used by the reference data. Its principal constitutive approximation is the use of a constant diffusivity and an isothermal state,

$$T(\tau) = T_{\text{amb}}, \quad \hat{D}_{\text{simpl}} = 1. \quad (24)$$

Table 15: Physical content of the reference data and nominal simplified FVM.

Modelling component	Thermo-diffusive reference	Simplified FVM baseline
Spatial domain	Spherical particle	Spherical particle
Numerical method	Conservative FVM	Conservative FVM
Radial grid	101 nodes	101 nodes
Diffusivity	$\hat{D} = \hat{D}(C, T)$	Constant D_s
Temperature	Lumped evolving state	Constant ambient state
Applied current	Operating profile $I(t)$	Same profile $I(t)$
Surface flux	$\delta = \delta(\tau)$	Same physical flux history
Nominal initial concentration	$C_0 = 0.5$	$C_{0,\text{simpl}} = 0.5$
Centre symmetry	Zero radial derivative	Zero radial derivative

Table 15 summarizes the nominal comparison.

The simplified baseline remains an inexpensive, executable and conservative model. The initial concentration $C_{0,\text{simpl}}$ and the constant coefficient D_s can be varied without modifying the experimental or reference dataset. Their role in model calibration is discussed in Section 19.4.

17.3 State-Aware Residual Representation and Network Architecture

The FNN receives one input vector for each complete radial profile,

$$\mathbf{x}_k = \begin{bmatrix} \overline{C}_{\text{simpl}}(\tau_k) \\ C_{s,\text{simpl}}(\tau_k) \\ I(t_k) \end{bmatrix}, \quad (25)$$

where

$$\overline{C}_{\text{simpl}}(\tau) = \frac{\int_0^1 C_{\text{simpl}}(\rho, \tau) \rho^2 d\rho}{\int_0^1 \rho^2 d\rho} \quad (26)$$

and

$$C_{s,\text{simpl}}(\tau) = C_{\text{simpl}}(1, \tau). \quad (27)$$

The current separates loading, ramp-down and rest states that may otherwise have similar concentration summaries. The network returns one raw correction for each ordered radial node,

$$\Delta \mathbf{C}_\phi^{\text{raw}}(\mathbf{x}_k) \in \mathbb{R}^{101}. \quad (28)$$

The architecture is

$$3 \longrightarrow 32 \longrightarrow 32 \longrightarrow 32 \longrightarrow 101,$$

with ReLU hidden activations and a linear output layer. It contains 5,573 trainable parameters. The radial coordinate is encoded by the ordering of the 101 outputs; consequently, the present correction network is tied to the training mesh.

17.4 Dataset Construction, Splitting and Optimization

Reference and simplified fields are aligned on the same physical times and ordered radial grid. Each complete radial profile is treated as one learning sample. The input and target matrices have dimensions

$$\mathbf{X} \in \mathbb{R}^{201 \times 3}, \quad \mathbf{Y} \in \mathbb{R}^{201 \times 101}.$$

The 201 profiles are randomly divided with seed 26 into 140 training, 30 validation and 31 test profiles. Radial points belonging to the same time are never separated. This prevents spatial leakage, but all subsets still belong to one continuous operating trajectory. The test result therefore measures temporal interpolation within the nominal trajectory, not generalization to a new C_0 , D_s or current history.

The raw FNN output is transformed by the hard constraint in Section 17.5 before the loss is evaluated. The optimized objective is consequently

$$\mathcal{L}_{\Delta C}(\phi) = \frac{1}{N_s N_\rho} \sum_{k=1}^{N_s} \left\| \Delta \mathbf{C}_{\text{target}}(\tau_k) - \widehat{\Delta \mathbf{C}}_\phi(\tau_k) \right\|_2^2. \quad (29)$$

Thus, the constant-average condition is part of training and validation, not an inference-only correction. Adam is used with learning rate 10^{-3} , a batch size of 32 profiles, a maximum of 10,000 epochs and early-stopping patience of 800 epochs. The best validation checkpoint is retained.

17.5 Generalized Lithium-Inventory Constraint

17.5.1 Why a zero-average correction is not general

A natural starting point is to require the learned correction not to alter the lithium inventory already predicted by the conservative simplified model. Under this assumption, the neural network modifies only the radial distribution of lithium inside the particle, while leaving the spherical volume average of the concentration unchanged. Although physically meaningful, this assumption need not hold for every candidate simplified baseline. The corresponding zero-average constraint is

$$\overline{\Delta C}_\phi(\tau) = 0.$$

This is appropriate only when reference and simplified models have the same initial lithium inventory. If the simplified model starts from a different initial concentration, the correct residual may contain a non-zero uniform component. For models driven by the same physical surface flux, the volume averages satisfy

$$\frac{d\overline{C}_{\text{ref}}}{d\tau} = \frac{d\overline{C}_{\text{simpl}}}{d\tau}, \quad (30)$$

up to numerical integration error. Therefore,

$$\frac{d}{d\tau} (\overline{C}_{\text{ref}} - \overline{C}_{\text{simpl}}) = 0, \quad (31)$$

and the physically consistent average residual is a constant,

$$\boxed{\overline{\Delta C}(\tau) = b.} \quad (32)$$

If the observation window begins at the true initial instant, then

$$b = \overline{C}_{\text{ref}}(0) - \overline{C}_{\text{simpl}}(0).$$

The implementation does not require this difference to be supplied explicitly. The scalar is estimated from the residual targets in the training subset,

$$b = \frac{1}{N_{\text{train}}} \sum_{k \in \mathcal{I}_{\text{train}}} \overline{\Delta C}_{\text{target}}(\tau_k). \quad (33)$$

This definition is also meaningful when the first experimental sample is not the true electrochemical initial state. In that case, b represents an effective inventory offset over the observed time window, while $C_{0,\text{simpl}}$ is interpreted as an effective initialization of the simplified model.

17.5.2 Hard projection applied to every profile

Let w_i be normalized spherical trapezoidal weights,

$$w_i = \frac{\omega_i \rho_i^2}{\sum_{j=0}^N \omega_j \rho_j^2}, \quad \sum_{i=0}^N w_i = 1. \quad (34)$$

The volume average of the raw correction is

$$a_\phi(\tau_k) = \sum_{i=0}^N w_i \Delta C_{\phi,i}^{\text{raw}}(\tau_k). \quad (35)$$

The final correction used by the loss and by inference is

$$\widehat{\Delta C}_{\phi,i}(\tau_k) = \Delta C_{\phi,i}^{\text{raw}}(\tau_k) - a_\phi(\tau_k) + b. \quad (36)$$

Since the same scalar is subtracted from all radial nodes,

$$\sum_{i=0}^N w_i \widehat{\Delta C}_{\phi,i} = \sum_{i=0}^N w_i \Delta C_{\phi,i}^{\text{raw}} - a_\phi \sum_{i=0}^N w_i + b \sum_{i=0}^N w_i \quad (37)$$

$$= a_\phi - a_\phi + b = b. \quad (38)$$

The constraint is therefore satisfied by construction for every profile in training, validation and testing. Mini-batch shuffling does not affect it, because the projection is evaluated independently for every profile and the single value b is fixed before optimization. The checkpoint stores this offset together with the network parameters.

The projection removes only the average mode of the raw FNN output. It does not suppress radial variations, because subtracting a spatial constant does not change radial concentration gradients. Consequently, the network learns the redistribution error while the admissible average component is supplied by b .

17.5.3 Relation to the mass-balance residual

For any concentration field, define

$$R_{\text{mass}}(\tau) = \overline{C}(\tau) - \left[\overline{C}(0) - 3 \int_0^\tau \delta(s) ds \right]. \quad (39)$$

For the additive hybrid field,

$$R_{\text{hybrid}}(\tau) = R_{\text{simpl}}(\tau) + [\overline{\Delta C}_\phi(\tau) - \overline{\Delta C}_\phi(0)]. \quad (40)$$

The term in brackets is the temporal drift of the correction average. Since $\overline{\Delta C}_\phi = b$ at every instant,

$$\boxed{R_{\text{hybrid}}(\tau) = R_{\text{simpl}}(\tau)}. \quad (41)$$

This result holds whether b is zero or non-zero. The constraint therefore preserves the temporal lithium balance while allowing an initial inventory offset. It removes the need for an ϵ -dependent clipping rule: a bounded but time-varying average could remain inside $[-\epsilon, \epsilon]$ and still add drift to Eq. (40).

The constraint guarantees inventory consistency, but not concentration accuracy. A model may satisfy Eq. (41) and still predict an incorrect radial distribution. This distinction becomes central in the parameter-variation study of Section 19.3.

17.6 Reconstruction and Evaluation Metrics

The corrected profile is

$$\mathbf{C}_{\text{hybrid}}(\tau_k) = \mathbf{C}_{\text{simpl}}(\tau_k) + \widehat{\Delta \mathbf{C}}_\phi(\tau_k). \quad (42)$$

Surface and average concentrations are recomputed from this profile,

$$C_{s,\text{hybrid}}(\tau_k) = C_{\text{hybrid}}(1, \tau_k), \quad (43)$$

$$\overline{C}_{\text{hybrid}}(\tau_k) = \sum_{i=0}^N w_i C_{\text{hybrid},i}(\tau_k). \quad (44)$$

Outputted corrected results therefore contain mutually consistent field, surface and average concentrations. Errors are characterized through RMSE, MAE, maximum absolute error and relative L^2 error. The RMSE reduction relative to the simplified baseline is

$$\text{RER} = \frac{\text{RMSE}_{\text{simplified}} - \text{RMSE}_{\text{hybrid}}}{\text{RMSE}_{\text{simplified}}} \times 100\%. \quad (45)$$

Two additional diagnostics verify the new hard constraint:

$$d_{\text{temporal}} = \max_k |\overline{\Delta C}_\phi(\tau_k) - \overline{\Delta C}_\phi(\tau_0)|, \quad (46)$$

$$e_{\text{inventory}} = \max_k |\overline{\Delta C}_\phi(\tau_k) - b|. \quad (47)$$

18 Hybrid Model Results

The results below correspond to the nominal simplified model, $C_{0,\text{simpl}} = 0.5$ and $D_s = 1.213 \times 10^{-14} \text{ m}^2 \text{ s}^{-1}$.

18.1 Training Convergence and Computational Cost

The run stopped after 4,893 epochs. The best validation loss was

$$\mathcal{L}_{\Delta C, \text{val}} = 1.1121 \times 10^{-7}$$

at epoch 4,093. Training required 80.131 s on Apple MPS. The offset estimated from the training targets was

$$b = -5.2620 \times 10^{-7}. \quad (48)$$

The reported training and inference times refer to one representative development run on Apple MPS. They document the behaviour of this experiment but do not constitute the final cross-model timing evidence. Operational cost comparisons are based on the repeated, device-controlled measurements reported in Section 22.

Its magnitude confirms that the nominal reference and simplified trajectories have the same initial inventory up to numerical quadrature error. The value is not manually forced to zero.

Table 16: Training and nominal test summary for the constrained FVM–FNN model.

Quantity	Value
Maximum epochs	10,000
Completed epochs	4,893
Best validation epoch	4,093
Best validation MSE	1.1121×10^{-7}
Training time	80.131 s
Inventory offset b	-5.2620×10^{-7}
Test profiles	31
Corrective test inference time	19.146 ms
Inference time per profile	0.6176 ms
Execution device	Apple MPS

18.2 Held-Out Interpolation Performance

The test MSE was 9.8827×10^{-8} . Table 17 compares the simplified and hybrid predictions on the 31 held-out profiles.

The concentration correction substantially improves the radial field and the surface response. The average concentration changes by a much smaller absolute amount because the simplified FVM already predicts the lithium inventory with an error of order 10^{-6} .

The hard-constraint diagnostics are

$$d_{\text{temporal}} = 1.1593 \times 10^{-9}, \quad e_{\text{inventory}} = 1.4751 \times 10^{-9}.$$

Table 17: Errors on the nominal held-out profiles.

Quantity	Simplified RMSE	Hybrid RMSE	Reduction
Complete field	8.4231×10^{-3}	3.1437×10^{-4}	96.27%
Particle surface	8.7886×10^{-3}	3.7805×10^{-4}	95.70%
Volume average	1.0378×10^{-6}	8.0910×10^{-7}	22.04%

These values are at the level of floating-point and quadrature error. They demonstrate that the volume-average correction remains equal to b during inference.

18.3 Complete Structured-Grid Reconstruction

The checkpoint was subsequently evaluated over all 201 profiles. These values include training, validation and test times and therefore describe reconstruction of the sampled nominal trajectory, not an independent generalization experiment.

Table 18: Dense-grid errors on the complete nominal trajectory.

Quantity	Simplified RMSE	Hybrid RMSE	Hybrid MAE	RMSE reduction
Complete field	8.2082×10^{-3}	4.1431×10^{-4}	2.3846×10^{-4}	94.95%
Particle centre	1.2211×10^{-2}	7.3018×10^{-4}	4.7305×10^{-4}	94.02%
Particle surface	8.3272×10^{-3}	3.3805×10^{-4}	2.1539×10^{-4}	95.94%
Volume average	9.4944×10^{-7}	7.7267×10^{-7}	7.0450×10^{-7}	18.62%

The complete-field RMSE is approximately 19.8 times smaller than that of the simplified model. The principal benefit is the recovery of the radial distribution: improvements at the centre and surface are much larger than the already small change in the volume average.

The hybrid field remains smooth during loading and relaxation. The radial profiles retain the structure supplied by the conservative FVM while removing most of the diffusivity-related discrepancy.

18.4 Inventory Offset and Mass Balance

Figure 23 verifies the hard constraint by comparing the volume average of the final correction with b .

The small visible fluctuations are of order 10^{-9} , whereas the plotted offset is of order 10^{-7} . They are numerical representation errors and not a physical temporal drift.

The maximum mass-balance residual is

$$\max_{\tau} |R_{\text{hybrid}}(\tau)| = 4.5525 \times 10^{-5},$$

compared with 4.5522×10^{-5} for the simplified FVM. Their difference is approximately 2.5×10^{-9} , consistently with Eq. (41). The common peaks are associated with temporal integration, quadrature and the current transitions rather than with an inventory drift introduced by the FNN.

Over the complete corrected dataset,

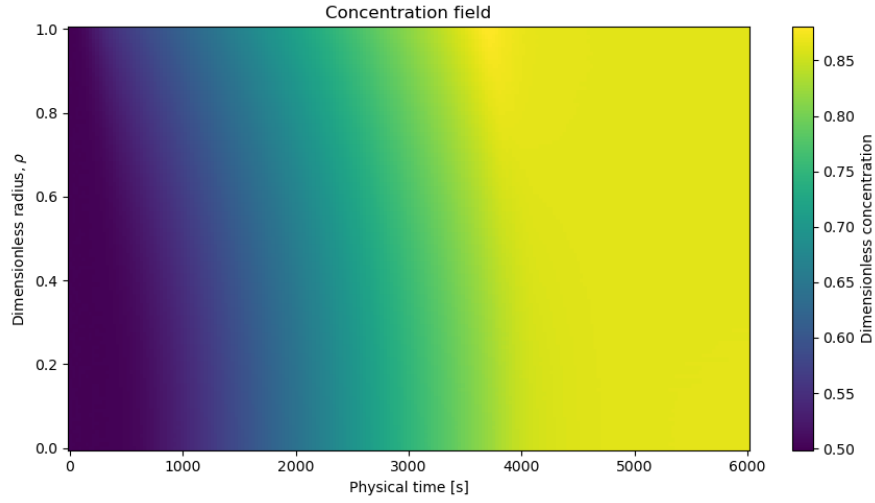


Figure 20: Concentration field reconstructed by the constrained FVM-FNN hybrid model on the nominal trajectory.

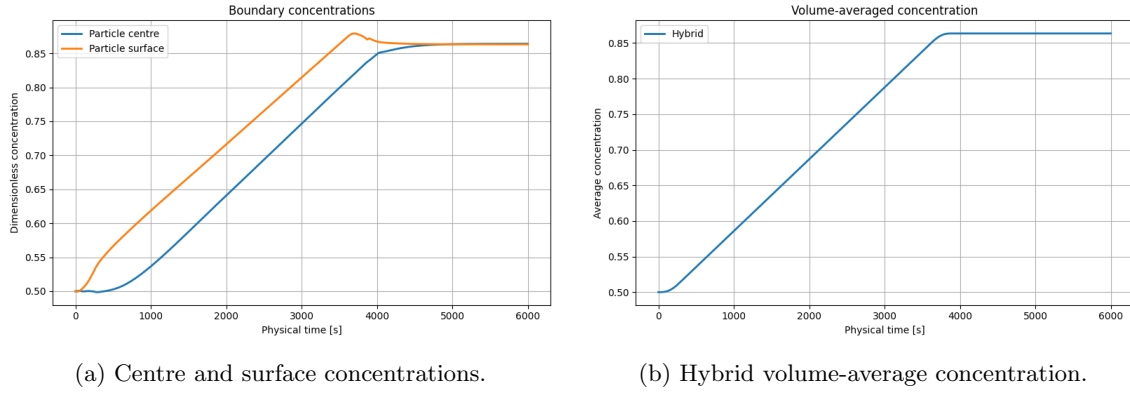


Figure 21: Boundary and integral quantities reconstructed consistently from the corrected radial field.

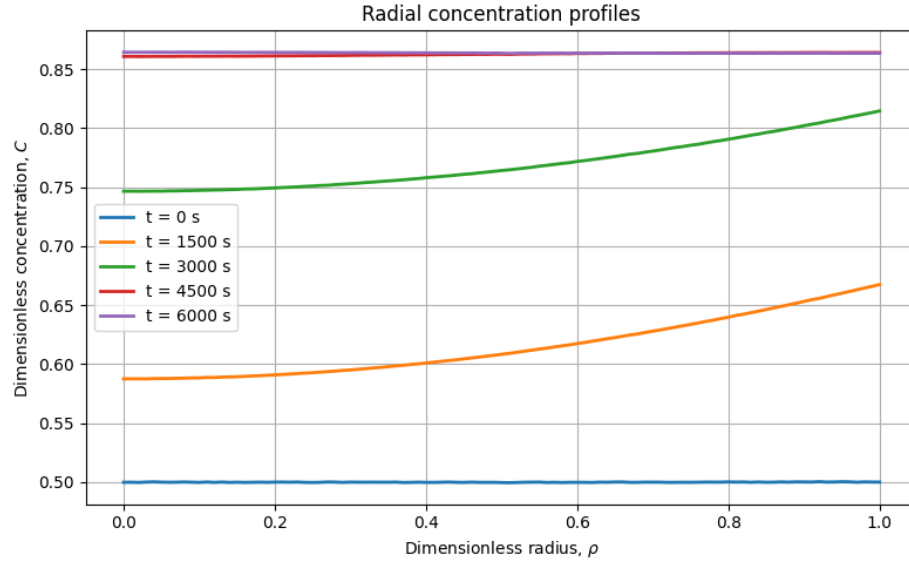


Figure 22: Hybrid radial concentration profiles at selected physical times.

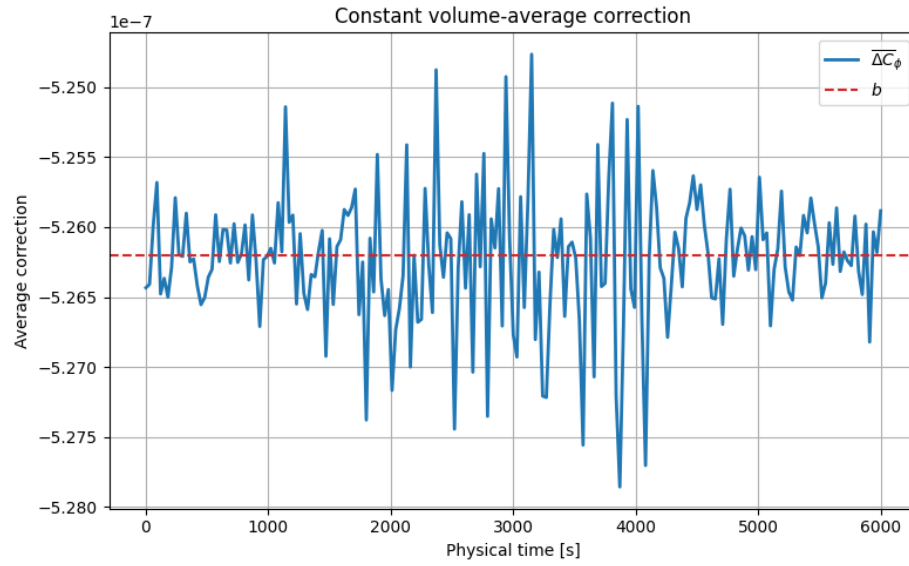


Figure 23: Volume-average correction and the constant inventory offset stored in the selected checkpoint.

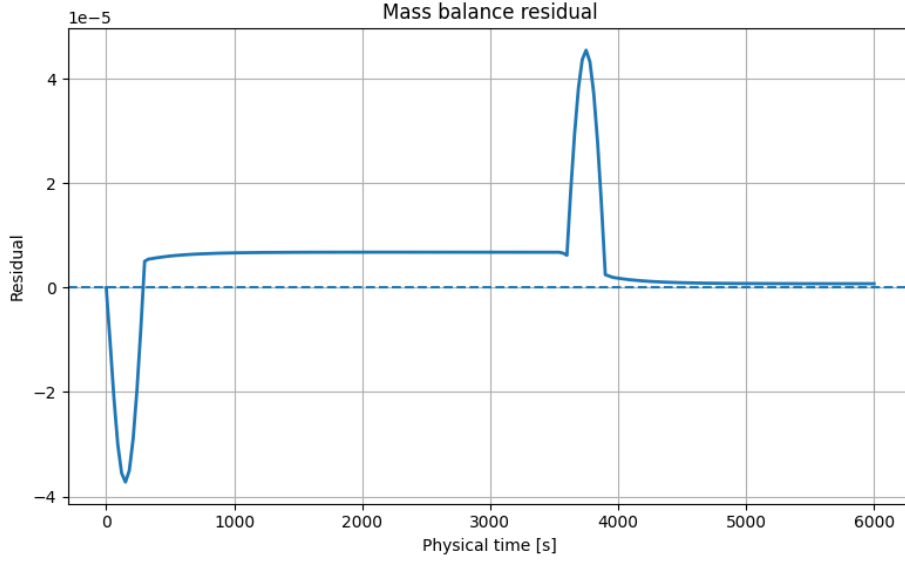


Figure 24: Cumulative mass-balance residual of the nominal hybrid field.

$$0.498489 \leq C_{\text{hybrid}} \leq 0.879384,$$

and no value violates the normalized interval $[0, 1]$.

18.5 Assessment of the Nominal Result

The nominal experiment supports three conclusions.

1. The residual network accurately corrects the radial redistribution omitted by the constant-diffusivity FVM, reducing the held-out field RMSE by 96.27%.
2. The hard affine projection preserves the temporal lithium balance without assuming that the average correction must be zero. In the nominal case, the estimated offset happens to be close to zero because the two trajectories start from the same concentration.
3. Accuracy and constraint satisfaction remain different properties. The inventory condition controls only one scalar mode of every radial profile; it cannot make a residual learned for one simplified model valid for a different C_0 or D_s .

19 Comparison of the Simplified, PINN and Hybrid Models

This section compares the simplified FVM, the standalone PINN and the FVM–FNN Hybrid model with the same thermo-diffusive reference field. The post-training Hybrid-PINN remains excluded because it is a different branch and is assessed directly against its frozen PINN baseline in Section 21.

19.1 Scope and Fairness of the Comparison

The three models do not receive identical information. The simplified FVM and the FVM–FNN Hybrid model follow the ramp–constant–ramp–rest operating current used by the reference simulation. The PINN checkpoint enforces the constant dimensionless flux

$$\delta(\tau) = \delta_0 \simeq -0.2005$$

over the complete time interval. Its late-time growth is therefore the expected response of a model evaluated outside the boundary-input history of the reference experiment.

The comparison documents the behaviour of the models as implemented, but it does not establish universal superiority of one class over another. Instead, it aims to provide a meaningful outcome explanation when different strategies are followed.

19.2 Nominal Dense-Grid Comparison

Table 19: Errors with respect to the thermo-diffusive reference on the common 101-by-201 grid.

Model	Quantity	Relative L^2	RMSE	MAE	Maximum error
Simplified	Field	0.011016	0.008208	0.005586	0.021888
Simplified	Centre	0.016695	0.012211	0.009360	0.021888
Simplified	Surface	0.010792	0.008327	0.006371	0.015136
Simplified	Average	1.2565×10^{-6}	9.4944×10^{-7}	8.1108×10^{-7}	1.6437×10^{-6}
PINN	Field	0.050674	0.037757	0.027721	0.160606
PINN	Centre	0.045458	0.033251	0.026263	0.073184
PINN	Surface	0.068609	0.052938	0.034088	0.160606
PINN	Average	0.053791	0.040645	0.028642	0.124484
Hybrid	Field	5.5605×10^{-4}	4.1431×10^{-4}	2.3846×10^{-4}	0.002694
Hybrid	Centre	9.9825×10^{-4}	7.3018×10^{-4}	4.7305×10^{-4}	0.002601
Hybrid	Surface	4.3812×10^{-4}	3.3805×10^{-4}	2.1539×10^{-4}	0.001962
Hybrid	Average	1.0226×10^{-6}	7.7267×10^{-7}	7.0450×10^{-7}	1.1165×10^{-6}

Relative to the simplified FVM, the Hybrid model main achievement is a more accurate radial distribution rather than a correction of the total lithium inventory.

The common colour scale includes the PINN values above one and therefore compresses the visible difference among the other three fields. The numerical metrics and radial profiles are required to resolve the smaller Hybrid error.

The Hybrid model nearly overlays the reference at the centre, surface and intermediate radii. The remaining maximum errors are localized and are larger than the global RMSE, which indicates small residual peaks rather than a systematic inventory bias.

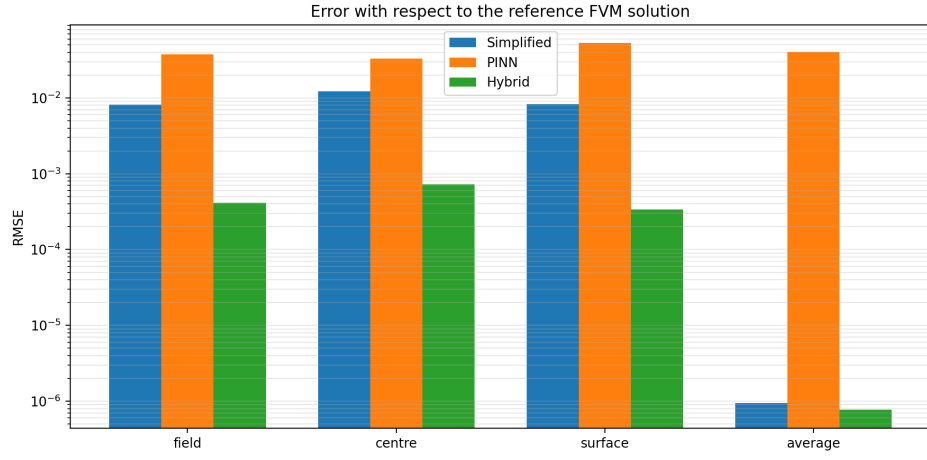


Figure 25: RMSE comparison with respect to the reference FVM solution.

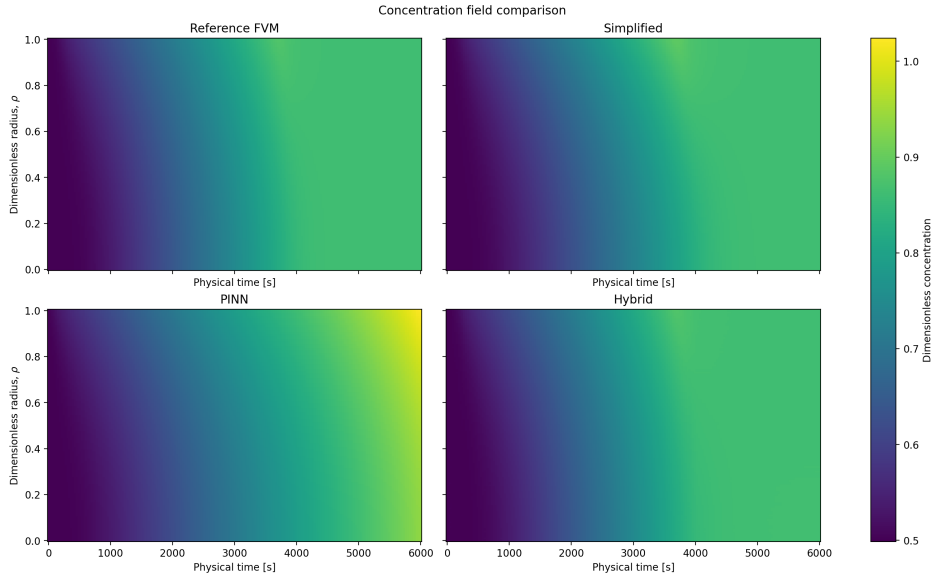


Figure 26: Reference, simplified, constant-flux PINN and FVM-FNN Hybrid concentration fields on a common colour scale.

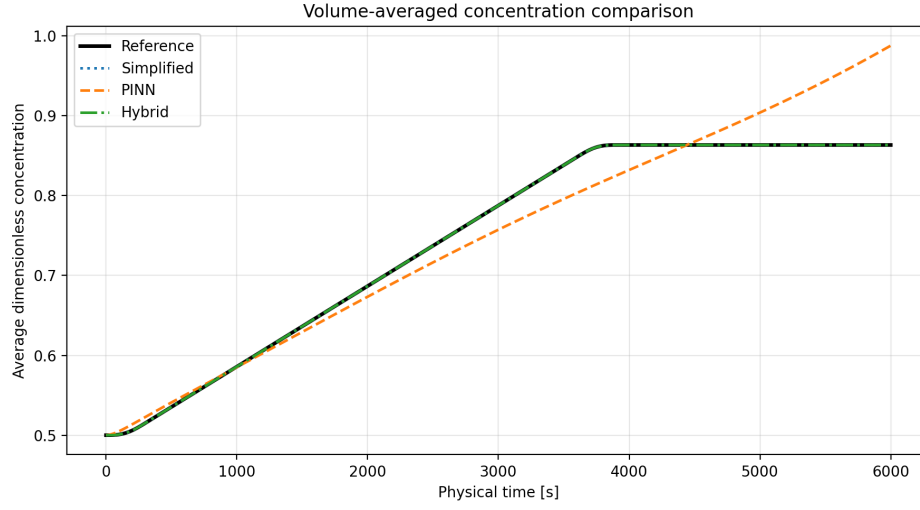


Figure 27: Volume-average concentration comparison. Reference, simplified and Hybrid curves nearly overlap because they share the physical operating flux; the PINN retains its constant-flux evolution.

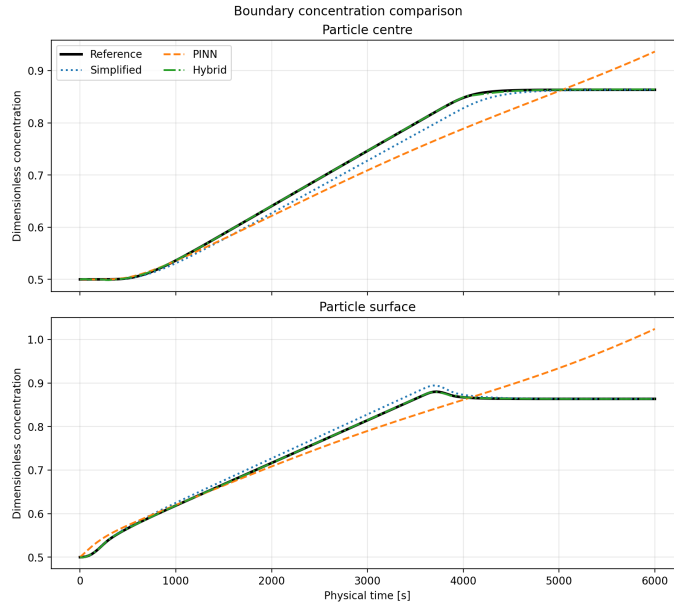


Figure 28: Centre and surface concentrations for all considered methods.

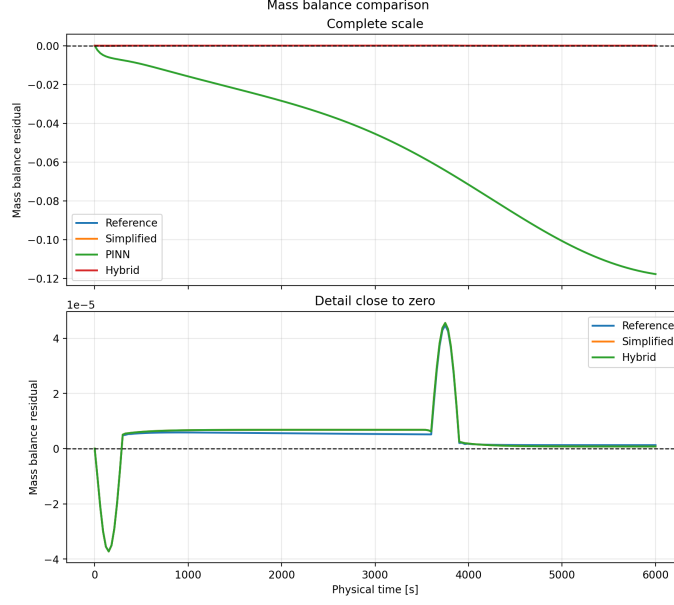


Figure 29: Mass-balance comparison. Each model is evaluated with the boundary flux belonging to its own formulation.

19.3 Robustness to Changes in C_0 and D_s

The frozen nominal Hybrid checkpoint was evaluated using five newly generated simplified trajectories. Both the simplified initial concentration and the constant diffusivity were varied, while the reference observations and FNN weights were kept fixed. Table 20 reports the held-out field errors.

Table 20: Frozen-checkpoint performance for simultaneous changes in $C_{0,\text{simpl}}$ and D_s .

Variant	$C_{0,\text{simpl}}$	D_s [$\text{m}^2 \text{s}^{-1}$]	Simplified RMSE	Hybrid RMSE	RER	Constraint error
0	0.48	1.35×10^{-14}	0.023040	0.022542	2.16%	1.80×10^{-9}
1	0.49	1.55×10^{-14}	0.011019	0.010405	5.57%	1.32×10^{-9}
2	0.50	1.75×10^{-14}	0.002618	0.008421	-221.64%	1.77×10^{-9}
3	0.51	1.87×10^{-14}	0.011847	0.018526	-56.38%	1.18×10^{-9}
4	0.52	2.01×10^{-14}	0.022501	0.030338	-34.83%	1.57×10^{-9}

All variants satisfy the inventory constraint to approximately 10^{-9} , but only the first two show a small positive field-error reduction. Variants 2–4 are degraded by the frozen correction. Variant 2 is especially informative: $C_{0,\text{simpl}} = 0.50$ matches the nominal initial concentration, yet changing D_s alone makes the learned radial correction inappropriate. The hard constraint therefore generalizes as a physical guardrail, whereas the learned radial residual does not.

For variants 0, 1, 3 and 4, the volume-average RMSE is approximately equal to the magnitude of the initial-concentration mismatch. This is expected because the frozen checkpoint still contains the nominal value $b = -5.2620 \times 10^{-7}$. The convergence test asks whether the nominal model can be transferred unchanged; it does not re-estimate b or retrain the radial residual for each candidate.

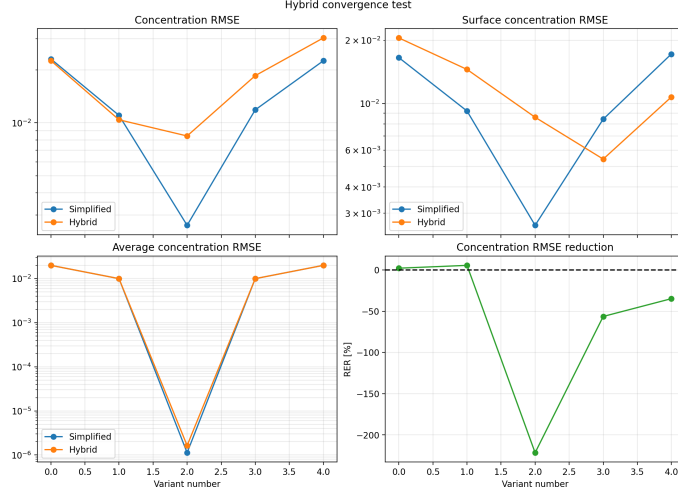


Figure 30: Convergence test for the frozen nominal correction under changes in simplified initial concentration and diffusivity.

The experiment establishes a local domain of validity. A residual network trained for

$$\Delta C_{\text{nominal}} = C_{\text{ref}} - C_{\text{simpl}}(C_{0,0}, D_{s,0})$$

cannot generally supply

$$\Delta C_{\text{candidate}} = C_{\text{ref}} - C_{\text{simpl}}(C_{0,\text{candidate}}, D_{s,\text{candidate}})$$

without retraining or explicit conditioning on the changed parameters.

19.4 Operational Candidate-Selection Strategy

In the intended application, the true reference initial concentration is not assumed to be available as a known model parameter. The experimental concentration dataset is fixed, whereas $C_{0,\text{simpl}}$ and D_s are candidate calibration parameters of the inexpensive simplified model. A new full-reference simulation is not required for every candidate.

The practical workflow is:

1. generate several simplified trajectories over a physically admissible grid of $C_{0,\text{simpl}}$ and D_s ;
2. align every candidate with the same experimental observation times;
3. pre-screen candidates using the concentration quantities that are experimentally available, without training one FNN per grid point;
4. retain one candidate, or a small set of finalists;
5. construct the corresponding residual targets, estimate b from the training subset using Eq. (33), and retrain the residual network;

6. validate both prediction accuracy and the hard-constraint metrics on held-out data.

If the first observation occurs after the true physical initialization, the selected $C_{0,\text{simpl}}$ should be described as the effective initial state of the observation window. The procedure does not infer an inaccessible historical concentration; it selects the simplified trajectory that best aligns with the available data and learns its corresponding residual.

The measured computational costs support this strategy. Generating one constant-diffusivity simplified dataset with 101 radial nodes and 201 time instants required approximately 0.59 s after software initialization. The complete nominal Hybrid training required 80.131 s on MPS, approximately 136 times longer. Candidate generation is therefore inexpensive, whereas training one network for every candidate would be wasteful. Fast physical pre-screening followed by training only the finalists provides the appropriate compromise.

The factor of 136 is specific to the development run and hardware configuration reported above. It is used here to motivate candidate pre-screening rather than as a general performance ratio; the aggregated benchmark provides the values used for the final operational comparison.

19.5 Overall Comparative Assessment

The updated comparison leads to the following conclusions.

1. **The simplified FVM already provides the correct lithium balance.** Its principal nominal error concerns radial redistribution, not the total inventory.
2. **The nominal FVM–FNN Hybrid model provides a substantial radial correction.** It reduces the complete-trajectory field RMSE by 94.95% and the held-out interpolation RMSE by 96.27%.
3. **The generalized hard constraint preserves mass evolution without forcing equal initial inventories.** The constant offset b is estimated from training residuals and is enforced during training, validation and testing.
4. **Constraint satisfaction is not a robustness certificate.** The convergence variants satisfy the inventory condition at 10^{-9} while several field predictions deteriorate. The radial residual remains local to the simplified configuration used during training.
5. **The PINN comparison is informative but asymmetric.** The reported PINN uses a constant flux, whereas the simplified and Hybrid FVMs receive the operating current profile. These results must not be presented as a universal ranking of PINNs and residual hybrid models.
6. **Unknown reference initialization can be handled through calibration rather than explicit identification.** The fixed experimental dataset is compared with rapidly generated simplified candidates. The selected candidate is then used to estimate b and retrain the residual model.

The present model is consequently a high-accuracy local correction around a selected simplified baseline. Its main limitations remain the fixed 101-node output, interpolation-based random time split, absence of explicit temperature reconstruction and need for retraining or parameter conditioning when $C_{0,\text{simpl}}$, D_s or the operating regime changes substantially.

20 Data-Driven Reconstruction of the Thermal Dynamics: SINDyC and Feedforward Neural Networks

The hybrid concentration model developed in the previous sections corrects the concentration field predicted by a simplified Single Particle Model. However, the simplified model assumes a constant diffusion coefficient evaluated at the reference concentration and ambient temperature. Its thermal dynamics are therefore omitted and the particle temperature remains equal to the ambient value.

To reconstruct the missing temperature evolution without supplying the thermal balance equation used internally by the reference solver, two data-driven models were investigated. The first is Sparse Identification of Nonlinear Dynamics with Control (SINDyC), which searches for an explicit and parsimonious differential equation in a prescribed library of candidate functions. The second is a feedforward neural network (FFN), which directly approximates the temperature derivative as a nonlinear function of the instantaneous thermal state and applied current. SINDyC extends the sparse identification of nonlinear dynamics framework by including external forcing and control variables [Brunton et al., 2016a,b].

The comparison is intended to separate two different modelling properties. SINDyC favours interpretability and structural discovery when the relevant candidate functions are present in the library. The FFN requires no explicit choice of candidate terms and is therefore more flexible, but it does not automatically recover a sparse or physically interpretable equation. Both models are trained and evaluated using the same collection of thermal trajectories and are reconstructed through the same numerical rollout procedure.

20.1 Thermal state and controlled dynamics

The temperature deviation from the ambient condition is adopted as the scalar state,

$$\theta(t) = T(t) - T_{\text{amb}}, \quad T_{\text{amb}} = 298.15 \text{ K}, \quad (49)$$

while the applied current is treated as an external control input,

$$u(t) = I(t). \quad (50)$$

Both data-driven models therefore seek an approximation of the unknown controlled dynamics

$$\dot{\theta}(t) = f(\theta(t), I(t)). \quad (51)$$

The use of θ rather than the absolute temperature improves both conditioning and physical interpretation. In particular, a negative term proportional to θ represents relaxation towards the ambient condition, for which $\theta = 0$.

20.2 Generation of multiple thermal trajectories

The original reference dataset contains one row for every time–radius pair. Because the reference model adopts a lumped thermal description, temperature and current are independent of the radial coordinate and are repeated for all radial nodes. For each simulation, redundant radial entries were removed by retaining one sample per physical time and sorting the resulting signals in ascending temporal order:

$$\mathcal{D}_T^{(j)} = \left\{ t_k^{(j)}, T_k^{(j)}, I_k^{(j)} \right\}_{k=1}^{N_t}, \quad j = 0, \dots, 7. \quad (52)$$

Eight full-reference simulations were generated by varying the peak current and the durations of the ramp-up, constant-current, ramp-down and cooling stages. Each trajectory contains $N_t = 601$ unique temporal samples. The main characteristics and assigned roles are reported in Table 21.

Table 21: Thermal trajectories used for identification, validation and out-of-sample testing.

Dataset	Role	t_f [s]	$I_{\max}$ [A]	$T_{\max}$ [K]
0	Training	7200	1.2420	299.693
1	Training	6000	1.6560	300.892
2	Training	6000	1.6560	300.891
3	Training	5200	2.0700	302.429
4	Training	4800	2.4840	304.294
5	Validation	6400	1.4904	300.371
6	Interpolation test	5000	2.2356	303.138
7	Extrapolation test	4500	2.7324	305.557

Datasets 0–4 are used jointly for model identification or FFN training. Dataset 5 is used exclusively for SINDyC threshold selection and FFN early stopping. Dataset 6 is an interpolative test because its peak current and thermal excursion lie inside the envelope covered by the training trajectories. Dataset 7 is an extrapolative test because its peak current, $I_{\max} = 2.7324$ A, exceeds the largest training value, 2.4840 A, and produces a temperature beyond the observed training range. Neither test trajectory is used for parameter estimation, model selection or normalization.

20.3 SINDyC formulation and sparse regression

For a nonlinear system influenced by an external input, SINDyC approximates the unknown dynamics as a sparse linear combination of candidate nonlinear functions:

$$\dot{\mathbf{X}} = \mathbf{\Theta}(\mathbf{X}, \mathbf{U})\mathbf{\Xi}, \quad (53)$$

where $\mathbf{\Theta}$ is the candidate-function library and $\mathbf{\Xi}$ contains the coefficients to be identified. Sparsity is imposed under the assumption that only a small subset of the candidate terms is required to describe the underlying dynamics [Brunton et al., 2016a].

For the present scalar state and control input, a second-order polynomial library was adopted:

$$\mathbf{\Theta}(\theta, I) = [1 \quad \theta \quad I \quad \theta^2 \quad \theta I \quad I^2]. \quad (54)$$

This library can represent linear heat removal, current-dependent heat generation and possible second-order coupling, while remaining sufficiently small to preserve interpretability. Temperature derivatives were estimated using a second-order finite-difference method, with endpoint samples excluded from the sparse regression.

The coefficients were identified through Sequentially Thresholded Least Squares (STLSQ), with

$$\alpha_{\text{STLSQ}} = 10^{-8} \quad (55)$$

and column normalization enabled. The ridge parameter α_{STLSQ} improves numerical conditioning, while sparsity is controlled primarily through the threshold λ .

The following threshold values were evaluated:

$$\lambda \in \{10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}\}. \quad (56)$$

For each candidate, a single SINDyC model was fitted jointly to training trajectories 0–4 and integrated over validation trajectory 5. A candidate was accepted when its validation RMSE was no greater than 1.15 times the minimum validation RMSE. Among the acceptable candidates, the model with the smallest number of active terms was retained; validation RMSE was used as the secondary criterion.

All examined thresholds produced the same two-term structure and a validation temperature RMSE of 1.4640×10^{-4} K. The first threshold, $\lambda = 10^{-3}$, was retained by the tie-breaking rule. The resulting multi-trajectory equation is

$$\boxed{\frac{d\theta}{dt} = -2.4685252 \times 10^{-3} \theta + 2.4685255 \times 10^{-3} I^2}. \quad (57)$$

Equivalently,

$$\frac{dT}{dt} = -2.4685252 \times 10^{-3} (T - T_{\text{amb}}) + 2.4685255 \times 10^{-3} I^2. \quad (58)$$

The first term represents effective heat rejection to the environment, whereas the second represents current-induced heat generation. The recovered equation is therefore both sparse and physically interpretable, even though the thermal balance employed by the reference model was not supplied to the algorithm.

20.4 Feedforward neural-network model

The FFN provides a black-box alternative to sparse equation identification. It approximates the same controlled vector field through

$$\hat{\theta} = \mathcal{N}_{\phi}(\theta, I), \quad (59)$$

where ϕ denotes the trainable parameters. The network has two inputs, two hidden layers containing 32 neurons each, hyperbolic-tangent activation functions and one linear output. Its architecture can be written as

$$2 \longrightarrow 32 \longrightarrow 32 \longrightarrow 1, \quad (60)$$

corresponding to 1185 trainable parameters.

The derivative targets are constructed independently for every trajectory:

$$\dot{\theta}_k^{(j)} \approx \text{gradient} \left(\theta^{(j)}, t^{(j)} \right)_k, \quad (61)$$

using a second-order numerical gradient. The two input variables and derivative target are standardized. To prevent information leakage, both input and target scalers are fitted only on the union of training trajectories 0–4 and are subsequently applied without refitting to validation and test trajectories.

The FFN is trained by minimizing the mean-squared error in standardized derivative space,

$$\mathcal{L}_{\text{FFN}} = \frac{1}{N_{\text{tr}}} \sum_{n=1}^{N_{\text{tr}}} \left| \tilde{\theta}_n - \mathcal{N}_{\phi} \left(\tilde{\theta}_n, \tilde{I}_n \right) \right|^2. \quad (62)$$

Optimization is performed with Adam, a learning rate of 10^{-3} , a batch size of 512 and a maximum of 2000 epochs. The parameters associated with the lowest validation loss are restored, and training is stopped if the validation loss does not improve for 200 consecutive epochs.

No thermal balance, sparse-library structure or explicit physics residual is included in Eq. (62). The quantity denoted as physical derivative MSE in the numerical implementation is the derivative error after inverse scaling, expressed in $\text{K}^2 \text{s}^{-2}$; it is a diagnostic metric rather than an additional physics-informed loss.

20.5 Common rollout and temperature reconstruction

Pointwise derivative accuracy does not by itself guarantee an accurate temperature trajectory. Consequently, both identified vector fields are evaluated through an autonomous rollout driven only by the prescribed current profile:

$$\frac{d\hat{\theta}}{dt} = \hat{f}(\hat{\theta}(t), I(t)), \quad \hat{\theta}(0) = T(0) - T_{\text{amb}}. \quad (63)$$

Here, $\hat{f}$ is either the sparse SINDyC equation or the trained FFN. The discrete current samples are linearly interpolated during integration. Both models are integrated with LSODA using relative and absolute tolerances of 10^{-8} and 10^{-10} , respectively. The FFN receives the current predicted state at every solver evaluation; the reference temperature is used only to set the initial condition and to calculate the final metrics.

Because the thermal model is lumped, the reconstructed scalar trajectory is replicated over the radial grid:

$$\hat{T}(\rho_i, t_k) = \hat{T}(t_k), \quad i = 1, \dots, N_r. \quad (64)$$

The resulting maps are visual representations of a spatially uniform thermal state and should not be interpreted as predictions of radial heat conduction.

20.6 Evaluation metrics

The trajectory error is characterized through RMSE, MAE and maximum absolute error. Because the absolute temperature is dominated by the ambient offset, the relative L^2 error is evaluated on the temperature deviation:

$$\varepsilon_{L^2, \theta} = \frac{\|\hat{\theta} - \theta_{\text{ref}}\|_2}{\|\theta_{\text{ref}}\|_2}. \quad (65)$$

Additional diagnostics include the peak-temperature error, peak-time error, RMSE during maximum-current operation, cooling-stage RMSE, final-temperature error, and RMSE and MAE of the temperature derivative. Training or identification time, rollout inference time and model complexity are also reported. The SINDyC complexity is the number of active library terms, whereas the FFN complexity is the number of trainable parameters.

20.7 Interpolative test results

Table 22 reports the results on unseen trajectory 6. Both models reconstruct the temperature with small absolute errors. The FFN reaches an RMSE of 6.91×10^{-3} K and a maximum error

of 1.16×10^{-2} K, demonstrating accurate interpolation. SINDyC is nevertheless more accurate for every reported metric, with a temperature RMSE approximately 20.3 times smaller than that of the FFN.

Table 22: Thermal-model metrics on the interpolative test trajectory.

Metric	SINDyC	FFN
Temperature RMSE [K]	3.3981×10^{-4}	6.9088×10^{-3}
Temperature MAE [K]	2.4985×10^{-4}	5.0773×10^{-3}
Maximum error [K]	8.7669×10^{-4}	1.1593×10^{-2}
Relative L^2 on θ	1.0263×10^{-4}	2.0866×10^{-3}
Peak-temperature error [K]	6.2663×10^{-6}	9.0167×10^{-3}
Peak-time error [s]	0	0
Peak-current RMSE [K]	4.1724×10^{-4}	9.7556×10^{-3}
Cooling RMSE [K]	2.2675×10^{-4}	6.5103×10^{-4}
Final-temperature error [K]	7.9096×10^{-5}	1.2358×10^{-3}
Derivative RMSE [K s^{-1}]	3.0900×10^{-6}	2.2267×10^{-5}
Derivative MAE [K s^{-1}]	1.2570×10^{-6}	1.6928×10^{-5}

The zero peak-time error indicates that the predicted and reference maxima occur at the same sampled time. It should be interpreted relative to the temporal resolution of the dataset and not as proof that the continuous peak times are exactly identical.

20.8 Extrapolative test results

Trajectory 7 evaluates the models beyond the maximum current and temperature covered by the training data. The resulting metrics are given in Table 23. SINDyC maintains approximately the same relative accuracy observed in interpolation. Its relative L^2 error changes from 1.0263×10^{-4} to 1.0844×10^{-4} , an increase of only approximately 5.7%.

The FFN remains stable and produces a physically plausible rollout, but its temperature RMSE increases from 6.91×10^{-3} to 7.95×10^{-2} K, a factor of approximately 11.5. The largest discrepancy occurs during the high-current heating stage, where the FFN slightly underestimates the temperature rise. During cooling, the prediction returns close to the reference trajectory.

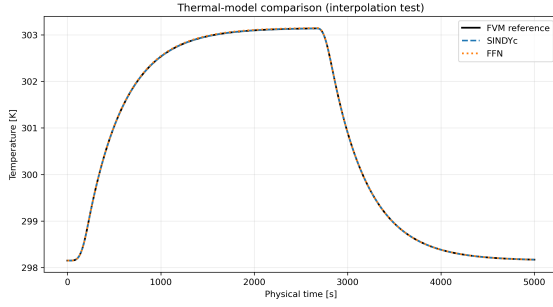
The direct trajectory comparisons are shown in Figure 31. The three curves are nearly indistinguishable in interpolation. In extrapolation, the SINDyC prediction remains superimposed on the reference, while the small FFN underprediction is visible around the maximum-temperature region.

The temperature maps reconstructed through SINDyC are shown in Figure 32. In both cases, the absence of radial variation is the expected consequence of the lumped thermal formulation. The interpolative trajectory reaches approximately 303.14 K, whereas the extrapolative trajectory reaches approximately 305.56 K because of its larger applied current. Despite this higher thermal excursion, the SINDyC rollout remains almost indistinguishable from the reference trajectory, as quantified by the sub-millikelvin errors in Table 23. The maps therefore illustrate generalization in time and operating conditions, not the emergence of a spatial temperature distribution.

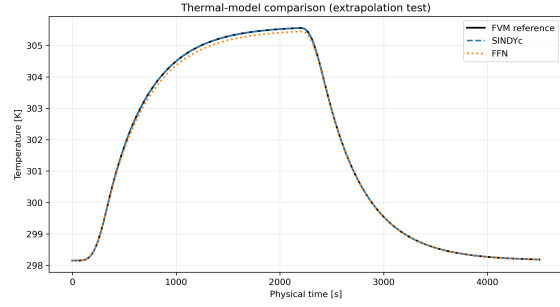
The corresponding FFN temperature maps are reported in Figure 33. Their uniformity along ρ follows from Eq. (64); only their temporal evolution is predicted by the network.

Table 23: Thermal-model metrics on the extrapolative test trajectory.

Metric	SINDyC	FFN
Temperature RMSE [K]	4.9426×10^{-4}	7.9459×10^{-2}
Temperature MAE [K]	4.0328×10^{-4}	5.4771×10^{-2}
Maximum error [K]	1.0331×10^{-3}	1.3522×10^{-1}
Relative L^2 on θ	1.0844×10^{-4}	1.7432×10^{-2}
Peak-temperature error [K]	7.0909×10^{-5}	1.1227×10^{-1}
Peak-time error [s]	0	0
Peak-current RMSE [K]	5.9842×10^{-4}	1.2084×10^{-1}
Cooling RMSE [K]	4.0332×10^{-4}	5.0187×10^{-3}
Final-temperature error [K]	8.1410×10^{-5}	8.2387×10^{-4}
Derivative RMSE [K s^{-1}]	2.0568×10^{-6}	2.3533×10^{-4}
Derivative MAE [K s^{-1}]	1.1682×10^{-6}	1.5873×10^{-4}

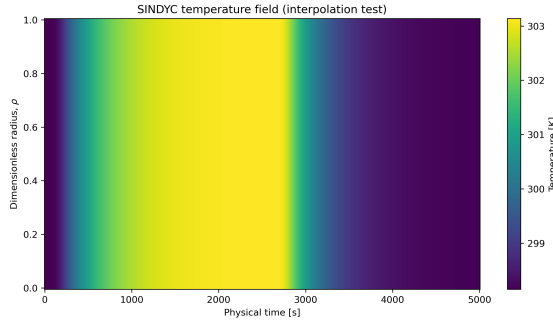


(a) Interpolative trajectory.

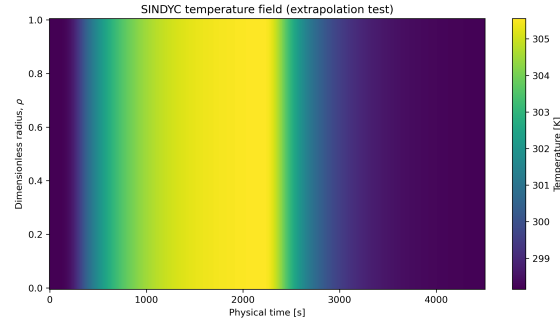


(b) Extrapolative trajectory.

Figure 31: Temperature trajectories reconstructed by SINDyC and the FFN compared with the full-reference thermal state.



(a) Interpolative test.



(b) Extrapolative test.

Figure 32: Spatially uniform temperature maps reconstructed by integrating the sparse thermal equation identified through SINDyC.

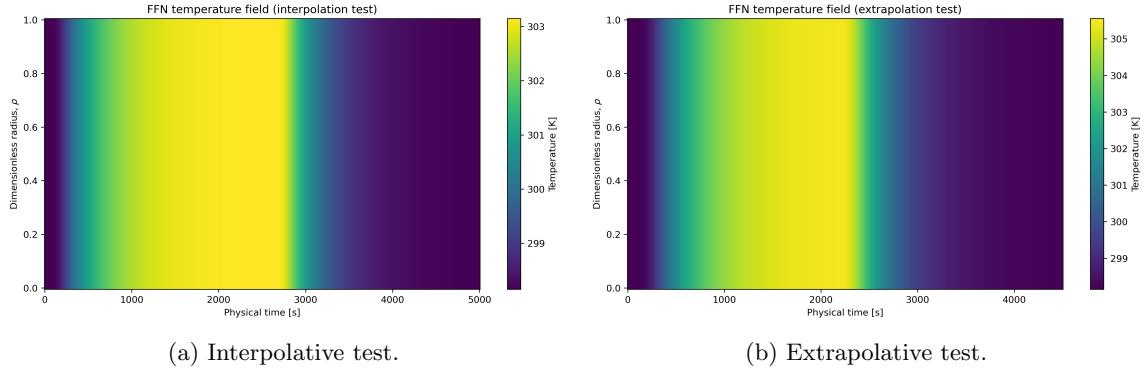


Figure 33: Spatially uniform temperature maps reconstructed through the FFN thermal rollout.

20.9 Computational cost and model complexity

Table 24 compares the measured computational costs. Identification or training was repeated when producing the two test outputs; the reported range therefore reflects the two observed runs. SINDyC requires approximately 5.2 s for sparse identification, whereas FFN training requires approximately 34 s. The sparse model also provides the faster rollout.

Table 24: Computational cost and complexity of the thermal models.

Quantity	SINDyC	FFN
Identification/training time [s]	5.21–5.23	33.87–33.98
Interpolation rollout time [s]	1.034	2.427
Extrapolation rollout time [s]	1.294	2.873
Active terms/trainable parameters	2	1185

These timings are implementation- and hardware-dependent. SINDyC evaluates a two-term algebraic expression on the CPU, whereas the FFN was executed on the Apple MPS device and repeatedly called by an adaptive scalar ODE solver. For a network of this size, device-transfer and kernel-launch overhead can represent a substantial part of the FFN rollout time. The measured values therefore describe the present implementation and should not be interpreted as a universal timing comparison between sparse models and neural networks.

20.10 Comparative assessment and limitations

The multi-trajectory experiment provides a substantially stronger assessment than a reconstruction performed on the same trajectory used for identification. Both models are now tested on completely held-out current profiles, and the extrapolative trajectory explicitly leaves the training range.

For the current toy problem, SINDyC provides the most effective thermal model. It is more accurate in both test regimes, generalizes almost unchanged beyond the training current, requires fewer computational resources and yields an explicit equation containing only heat-removal and Joule-heating terms. This outcome is consistent with the simple structure of the reference thermal

balance and with the fact that the second-order polynomial library contains the correct terms θ and I^2 .

The FFN should nevertheless not be regarded as unsuccessful. Its interpolation errors are of millikelvin order, and even in extrapolation its maximum error remains below 0.136 K. Moreover, the FFN does not require the user to enumerate candidate functions in advance. This flexibility may become advantageous when the thermal dynamics include higher-dimensional states, non-polynomial effects, hysteresis or partially unknown couplings that cannot be represented by a compact predefined library.

Conversely, the present results do not establish that SINDyC will retain its advantage for more complex systems. Sparse identification depends on the quality of derivative estimates and on the inclusion of appropriate functions in the candidate library. When the required terms are absent, the resulting equation can be sparse but structurally misspecified. A broader conclusion on the relative merits of the two approaches would therefore require noisy data, additional operating conditions and reference thermal models of increasing complexity.

Within the present scope, SINDyC is retained as the preferred thermal extension because it recovers a compact, interpretable and highly generalizable equation. The FFN is retained as a complementary black-box benchmark that quantifies the accuracy achievable without prior selection of a function library. Together, the two approaches clarify the trade-off between structural interpretability and representational flexibility in the data-driven reconstruction of physics omitted from the simplified electrochemical model.

21 Post-Training Residual Correction of the Physics-Informed Neural Network

21.1 Motivation and Position within the Modelling Framework

The standalone PINN in Sections 15 and 16 provides a continuous approximation $\widehat{C}_\theta(\rho, \tau)$ of the concentration field while retaining the exact initial condition and centre symmetry. Its remaining error is nevertheless systematic: it grows when the constant surface flux imposed in the simplified PINN differs from the time-dependent operating flux of the thermo-diffusive reference model.

This section introduces a post-training residual correction of that PINN. The selected PINN checkpoint is loaded, placed in evaluation mode and frozen. Its parameters are never updated during the corrective stage. A separate feedforward neural network is then trained to predict the concentration discrepancy. Correcting a misspecified PINN with an additional neural representation of its discrepancy has direct precedent in the model-correction literature [Zou et al., 2024]:

$$\Delta C_{\text{ref}}(\rho, \tau) = C_{\text{FVM}}(\rho, \tau) - \widehat{C}_\theta(\rho, \tau). \quad (66)$$

The resulting Hybrid-PINN is

$$C_{\text{HP}}(\rho, \tau) = \widehat{C}_\theta(\rho, \tau) + \widehat{\Delta C}_\phi(\rho, \tau, \widehat{C}_\theta(\rho, \tau)). \quad (67)$$

This separation makes the workflow modular. The expensive PINN optimization is not repeated when the residual model is trained, tested or replaced. It also preserves the standalone PINN as an unchanged baseline, enabling a direct comparison between the uncorrected and corrected fields. More generally, physics-guided hybrid models seek this division of labour: the mechanistic component supplies structure, whereas the learning component absorbs systematic inadequacy [Jia et al., 2020; Garg et al., 2022].

The present Hybrid-PINN is different from the FVM–FNN hybrid model of Section 17. The latter corrects a simplified finite-volume solver and returns an entire radial vector. The present model instead corrects a frozen, coordinate-based PINN and returns one scalar residual for each (ρ, τ) pair.

21.2 Frozen PINN Baseline and Residual Dataset

The residual-learning script restores the selected PINN checkpoint and applies

$$\text{requires_grad}(\theta) = \text{False}. \quad (68)$$

The reference concentration dataset contains the common $101 \times 201 = 20\,301$ structured space–time points. The same 70%–15%–15% pointwise split and fixed random seed used for the PINN data are retained. For each subset, the frozen PINN is evaluated without gradient tracking and the residual samples are assembled as

$$\mathbf{x}_{\text{res}} = [\rho, \tau, \widehat{C}_\theta(\rho, \tau)], \quad (69)$$

$$y_{\text{res}} = C_{\text{FVM}}(\rho, \tau) - \widehat{C}_\theta(\rho, \tau). \quad (70)$$

The PINN concentration is included as a state-aware feature. Therefore, the residual network is not asked to infer the baseline response again from the coordinates alone; it receives the value that must be corrected at the same point.

Training batches are shuffled, whereas validation and test batches retain deterministic ordering. The adopted batch size is 512. Because the split is pointwise and all subsets belong to the same operating trajectory, the held-out subset measures interpolation within the available trajectory. It does not establish generalization to unseen current profiles, initial concentrations or thermal conditions.

21.3 Residual-Network Architecture and Optimization

The residual network is a compact fully connected neural network. Its effective input is

$$\left[\rho^2, \tau, \widehat{C}_\theta\right], \quad (71)$$

followed by four 32-neuron hidden transformations with hyperbolic-tangent activation and a scalar linear output. The architecture contains 3,329 trainable parameters, compared with 30,701 parameters in the frozen PINN. Their roles are compared in Table 25.

Table 25: Structural comparison of the frozen PINN baseline and the trainable residual FFN.

Property	Frozen PINN	Residual FFN
Effective input	(ρ^2, τ)	$(\rho^2, \tau, \widehat{C}_\theta)$
Output	Concentration C_P	Signed correction $\widehat{\Delta C}_\phi$
Hidden architecture	Four layers, 100 neurons each	Four layers, 32 neurons each
Activation	tanh	tanh
Parameters	30,701, frozen	3,329, trainable
Training objective	Data, PDE and boundary terms	Residual MSE and soft box penalty

If the raw network output is denoted by $\mathcal{N}_\phi$, the applied residual is

$$\widehat{\Delta C}_\phi = \tau \mathcal{N}_\phi \left(\rho^2, \tau, \widehat{C}_\theta \right). \quad (72)$$

The multiplicative factor τ forces the correction to vanish at the initial time. The use of ρ^2 , together with the identical transformation used by the PINN, makes the radial derivative of the corrected field vanish at the particle centre.

The network is optimized with Adam using an initial learning rate of 10^{-3} , a maximum of 3,000 epochs and early-stopping patience of 300 epochs. The retained parameters are those associated with the smallest validation objective. The residual-fitting term is

$$\mathcal{L}_{\text{res}} = \frac{1}{N} \sum_{i=1}^N \left[\widehat{\Delta C}_{\phi,i} - \Delta C_{\text{ref},i} \right]^2. \quad (73)$$

To discourage normalized concentrations outside the physical interval $[0, 1]$, a soft box-constraint penalty is added to the raw corrected prediction $C_{\text{raw}} = \widehat{C}_\theta + \widehat{\Delta C}_\phi$:

$$\mathcal{L}_{\text{box}} = \frac{1}{N} \sum_{i=1}^N [\text{ReLU}(-C_{\text{raw},i})^2 + \text{ReLU}(C_{\text{raw},i} - 1)^2]. \quad (74)$$

The optimized objective is

$$\mathcal{L}_{\text{HP}} = \mathcal{L}_{\text{res}} + \mathcal{L}_{\text{box}}. \quad (75)$$

The residual output is signed: it can either increase or decrease the PINN concentration. A rectifier is therefore used only to measure violations in the box penalty; it is not applied to $\widehat{\Delta C}_\phi$ itself.

This penalty is differentiable almost everywhere and leaves admissible predictions unchanged. It is nevertheless a soft term and does not mathematically guarantee the bounds. No hard clipping is applied to the residual or to the final Hybrid-PINN concentration.

21.4 Inherited and Non-Inherited Physical Structure

The transformations in the two neural components imply

$$C_{\text{HP}}(\rho, 0) = C_0, \quad \left. \frac{\partial C_{\text{HP}}}{\partial \rho} \right|_{\rho=0} = 0. \quad (76)$$

These two properties are inherited exactly from the architecture. Other physical properties are not inherited automatically. In particular, pointwise residual fitting does not enforce:

- global lithium conservation under the reference flux history;
- the full-physics flux condition at $\rho = 1$;
- the variable-diffusivity thermo-diffusive PDE;
- strict concentration bounds for every possible input.

These quantities are therefore evaluated as diagnostic checks after training. They are not added to the residual-network loss in the present implementation. This distinction is essential: agreement with the checks is empirical evidence of physical consistency, not a hard guarantee supplied by the corrective architecture.

21.5 Direct Comparison with the Standalone PINN

The numerical values below correspond to the retained reference run. After residual training, the train, validation and test coordinates are reassembled in their original structured-grid order. The dense-grid errors therefore quantify reconstruction over the complete available trajectory; they must not be read as independent out-of-distribution test errors.

Table 26 compares the frozen PINN and Hybrid-PINN with the full-physics FVM field.

The relative full-field error decreases from 5.0674×10^{-2} to 9.5238×10^{-4} . Defining

$$\text{RER}_{\text{HP}} = \left(1 - \frac{\varepsilon_{L^2, \text{HP}}}{\varepsilon_{L^2, \text{PINN}}} \right) 100, \quad (77)$$

Table 26: Dense-grid errors of the standalone PINN and the post-training Hybrid-PINN with respect to the full-physics reference.

Model	Quantity	Relative L^2	RMSE	MAE	Maximum absolute error
PINN	Complete field	5.0674×10^{-2}	3.7757×10^{-2}	2.7721×10^{-2}	1.6061×10^{-1}
	Particle centre	4.5458×10^{-2}	3.3251×10^{-2}	2.6263×10^{-2}	7.3184×10^{-2}
	Particle surface	6.8609×10^{-2}	5.2938×10^{-2}	3.4088×10^{-2}	1.6061×10^{-1}
	Volume average	5.3791×10^{-2}	4.0645×10^{-2}	2.8642×10^{-2}	1.2448×10^{-1}
Hybrid-PINN	Complete field	9.5238×10^{-4}	7.0962×10^{-4}	3.9662×10^{-4}	1.2054×10^{-2}
	Particle centre	6.1086×10^{-4}	4.4682×10^{-4}	3.6112×10^{-4}	1.1427×10^{-3}
	Particle surface	2.9153×10^{-3}	2.2495×10^{-3}	1.2578×10^{-3}	1.2054×10^{-2}
	Volume average	7.4503×10^{-4}	5.6295×10^{-4}	3.8110×10^{-4}	2.4472×10^{-3}

the reduction is 98.12%. Equivalently, the relative error is reduced by a factor of approximately 53.2. The largest remaining discrepancy is at the particle surface, which is also the location where the omitted time-dependent boundary physics acts directly.

Table 27 reports the full-field error over the four operating phases.

Table 27: Full-field error by operating phase before and after the post-training PINN correction.

Phase	PINN		Hybrid-PINN	
	Relative L^2	RMSE	Relative L^2	RMSE
Ramp-up	9.5419×10^{-3}	4.7915×10^{-3}	4.6664×10^{-3}	2.3433×10^{-3}
Constant current	3.4402×10^{-2}	2.3050×10^{-2}	7.1251×10^{-4}	4.7739×10^{-4}
Ramp-down	5.9515×10^{-2}	5.0262×10^{-2}	9.0556×10^{-4}	7.6476×10^{-4}
Rest	6.2203×10^{-2}	5.3651×10^{-2}	4.7456×10^{-4}	4.0932×10^{-4}

The correction is particularly effective during ramp-down and rest, where the standalone PINN continues to respond to its constant-flux boundary condition while the reference current decreases and then vanishes. The centre, surface and volume-average histories, together with representative radial profiles, are shown in Figure 35.

21.6 Physical-Consistency Diagnostics

21.6.1 Concentration Bounds and Exact Constraints

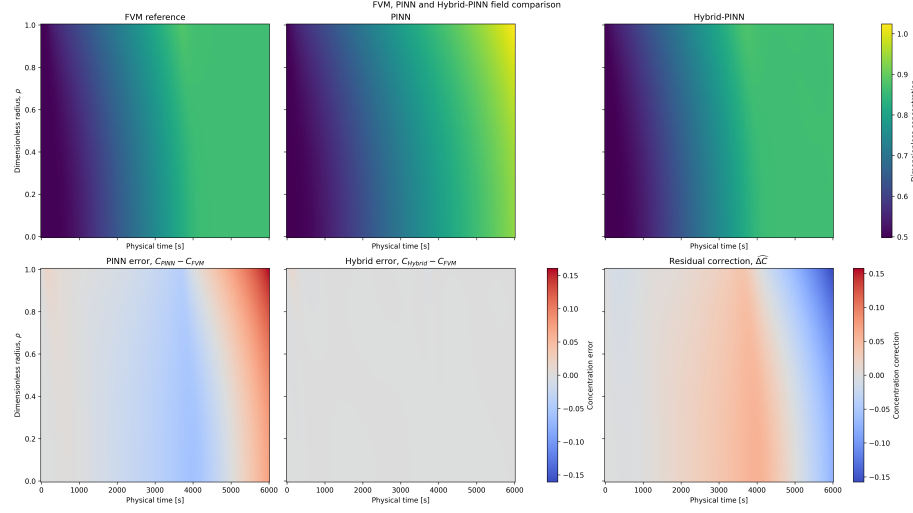
The corrected concentration range on the complete grid is

$$0.497915 \leq C_{\text{HP}} \leq 0.876149. \quad (78)$$

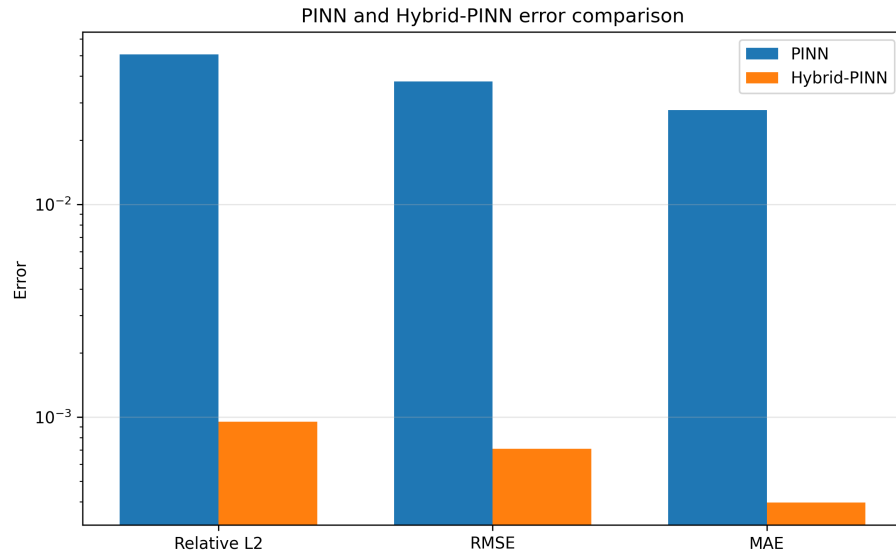
It therefore remains inside the normalized physical interval $[0, 1]$ for the retained run. The standalone PINN instead reaches 1.024085. The Hybrid-PINN result is a successful empirical outcome of the residual fit and soft box penalty, not an unconditional guarantee for unseen inputs.

The maximum initial-condition error and centre derivative are both zero to the reported numerical precision:

$$\max_{\rho} |C_{\text{HP}}(\rho, 0) - C_0| = 0, \quad \max_{\tau} \left| \frac{\partial C_{\text{HP}}}{\partial \rho} \right|_{\rho=0} = 0. \quad (79)$$

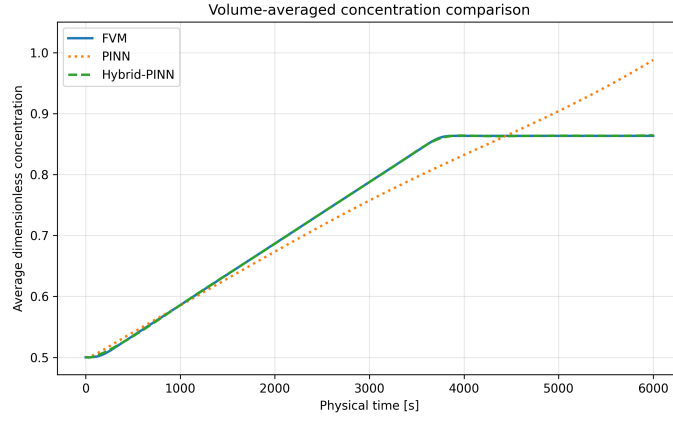


(a) Full concentration fields.

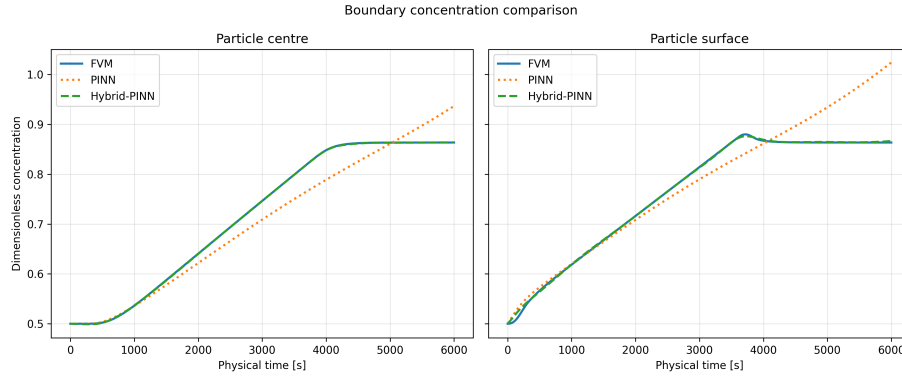


(b) Error metrics.

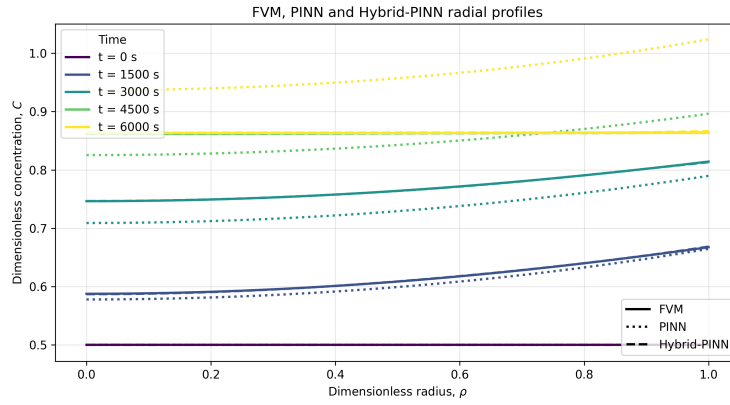
Figure 34: Direct comparison of the frozen PINN and post-training Hybrid-PINN with the full-physics reference model.



(a) Volume-averaged concentration.



(b) Centre and surface concentrations.



(c) Radial profiles at selected times.

Figure 35: Integral, boundary and radial comparison of the PINN and Hybrid-PINN predictions with the reference solution.

These two checks confirm the exact architectural properties in Eq. (76).

21.6.2 Global Mass Balance

For all three fields, mass consistency is evaluated against the time-dependent dimensionless flux stored in the full reference dataset:

$$\overline{C}(\tau) = 3 \int_0^1 C(\rho, \tau) \rho^2 d\rho, \quad (80)$$

$$\overline{C}_{\text{expected}}(\tau) = C_0 - 3 \int_0^\tau \delta(s) ds, \quad (81)$$

$$R_M(\tau) = \overline{C}(\tau) - \overline{C}_{\text{expected}}(\tau). \quad (82)$$

Table 28 reports the resulting checks.

Table 28: Mass-balance diagnostics evaluated using the full-reference time-dependent flux.

Field	Maximum $ R_M $	RMSE	Final residual	Relative mass-change error
FVM reference	4.4670×10^{-5}	1.0115×10^{-5}	1.2627×10^{-6}	3.4739×10^{-6}
PINN	1.2448×10^{-1}	4.0642×10^{-2}	1.2448×10^{-1}	3.4248×10^{-1}
Hybrid-PINN	2.4182×10^{-3}	5.5686×10^{-4}	1.0487×10^{-3}	2.8852×10^{-3}

The residual correction reduces the mass discrepancy by approximately two orders of magnitude relative to the PINN, even though conservation is not imposed in the FFN loss. The maximum residual remains larger than the 10^{-4} diagnostic tolerance and the Hybrid-PINN therefore does not pass the strict mass-balance check. This result must be reported as an empirical improvement rather than exact conservation.

21.6.3 Full-Physics Surface-Flux Condition

The full thermo-diffusive boundary condition is evaluated using

$$\widehat{D}(C, T) = \exp[\alpha(C - C_0)] \exp\left[-\frac{E_a}{R_g} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}}\right)\right], \quad (83)$$

with $\alpha = 1$, $E_a = 35\,000 \text{ J mol}^{-1}$, $R_g = 8.314462618 \text{ J mol}^{-1}\text{K}^{-1}$ and $T_{\text{ref}} = 298.15 \text{ K}$. The residual is

$$R_{\text{BC}}(\tau) = -\widehat{D}(C_{\text{HP},s}, T) \frac{\partial C_{\text{HP}}}{\partial \rho} \Big|_{\rho=1} - \delta(\tau). \quad (84)$$

Automatic differentiation is used for the surface gradient, while temperature and dimensionless flux are read from the full reference dataset. The global RMSE is 2.4816×10^{-2} , the maximum absolute residual is 1.1902×10^{-1} , and the final residual is -3.7961×10^{-2} . The phase-resolved values are reported in Table 29.

The concentration correction is therefore highly accurate without exactly recovering the full surface differential law. This is consistent with its training objective, which contains concentration targets but no surface-flux residual.

Table 29: Hybrid-PINN full-physics surface-flux residual by operating phase.

Phase	RMSE	Maximum $ R_{BC} $
Ramp-up	7.1101×10^{-2}	1.1902×10^{-1}
Constant current	1.4760×10^{-2}	3.8614×10^{-2}
Ramp-down	3.4270×10^{-2}	4.5953×10^{-2}
Rest	2.1912×10^{-2}	3.9015×10^{-2}

21.6.4 Full and Simplified PDE Consistency

The Hybrid-PINN is evaluated against the full reference equation

$$R_{\text{PDE}}^{\text{full}} = \frac{\partial C_{\text{HP}}}{\partial \tau} - \frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left[\rho^2 \hat{D}(C_{\text{HP}}, T) \frac{\partial C_{\text{HP}}}{\partial \rho} \right], \quad (85)$$

and, separately, against the constant-diffusivity equation used by the standalone PINN:

$$R_{\text{PDE}}^{\text{simpl}} = \frac{\partial C_{\text{HP}}}{\partial \tau} - \frac{\partial^2 C_{\text{HP}}}{\partial \rho^2} - \frac{2}{\rho} \frac{\partial C_{\text{HP}}}{\partial \rho}. \quad (86)$$

The 19,800 points satisfying $0 < \rho < 1$ and $\tau > 0$ are used. The centre and surface are excluded from the direct evaluation to avoid the apparent spherical-coordinate singularity and to keep the boundary checks separate. Results are summarized in Table 30.

Table 30: Hybrid-PINN differential-equation consistency diagnostics.

Residual	Phase	RMSE	Maximum absolute value
Full physics	Complete interior	5.9839×10^{-2}	5.3836×10^{-1}
	Ramp-up	1.2435×10^{-1}	5.3836×10^{-1}
	Constant current	5.0012×10^{-2}	2.6283×10^{-1}
	Ramp-down	5.5133×10^{-2}	3.1563×10^{-1}
	Rest	6.1003×10^{-2}	3.2896×10^{-1}
Simplified physics	Complete interior	1.3705×10^{-1}	4.9865×10^{-1}
	Ramp-up	1.1351×10^{-1}	4.9865×10^{-1}
	Constant current	1.6677×10^{-1}	3.8466×10^{-1}
	Ramp-down	1.9448×10^{-1}	2.8531×10^{-1}
	Rest	5.3412×10^{-2}	2.7771×10^{-1}

Neither residual is a training constraint for the FFN. The full-physics residual is smaller in global RMSE than the simplified residual, which is compatible with the correction moving the concentration field toward the reference dynamics. However, the non-zero values prevent claiming that the Hybrid-PINN is a direct solution of the full PDE. They quantify differential consistency of a data-corrected surrogate.

21.6.5 Numerical Regularity

All Hybrid-PINN concentrations, evaluated derivatives and PDE residuals are finite. The maximum derivative magnitudes over the interior evaluation set are

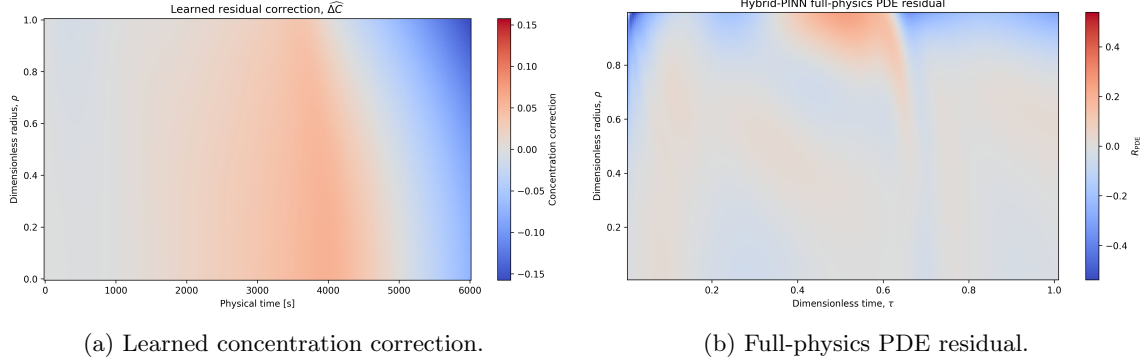


Figure 36: Space-time maps of the Hybrid-PINN residual correction and its a posteriori full-physics differential residual.

$$\max \left| \frac{\partial C_{\text{HP}}}{\partial \rho} \right| = 0.17963, \quad \max \left| \frac{\partial C_{\text{HP}}}{\partial \tau} \right| = 0.92585, \quad \max \left| \frac{\partial^2 C_{\text{HP}}}{\partial \rho^2} \right| = 1.1283. \quad (87)$$

These values are not independent physical constraints, but the absence of NaN, infinite derivatives and numerical blow-up confirms that the smooth tanh correction does not introduce an obvious instability on the evaluated grid.

21.7 Computational Cost and Corrected Dataset

Table 31 reports the synchronized timings of the retained run.

Table 31: Timing of the post-training Hybrid-PINN correction pipeline.

Operation	Time [s]
Frozen PINN inference for residual-dataset construction	0.173452
Residual-network training	147.040652
Residual inference and full-grid correction	0.033015
Total correction pipeline	147.247118

The timing excludes the original PINN training because that model is loaded from an existing checkpoint. It also excludes plotting and the a posteriori physical checks. Consequently, the reported total measures the additional cost required to construct the residual data, train the FFN and generate the corrected field.

The corrected dataset preserves the physical quantities stored by the reference FVM— ρ , τ , physical time, current, intercalation flux, dimensionless flux and temperature—and appends:

- the reference and PINN concentrations;
- the exact reference residual and learned residual correction;
- the Hybrid-PINN concentration;

- the full-physics and simplified-physics PDE residuals at the admissible interior points.

This stored dataset decouples visualization and comparison from training. Hybrid-PINN plots can therefore be regenerated without retraining either neural model.

21.8 Assessment and Limitations

The post-training correction achieves the intended numerical objective. It reduces the full-field relative error by 98.12%, removes the late-time concentration overshoot, reconstructs centre, surface and volume-average quantities, and preserves the initial and centre conditions exactly. The result demonstrates that a frozen PINN can be treated as a reusable physics-informed baseline rather than retrained as part of a monolithic hybrid script.

The physical checks also refine the interpretation of this accuracy. The corrected field remains within the normalized concentration bounds and improves mass balance substantially, but conservation is not exact. Similarly, the full surface-flux and PDE residuals remain non-zero because they are diagnostic quantities rather than terms in the FFN objective. The Hybrid-PINN should therefore be described as a high-accuracy, data-corrected surrogate with inherited hard constraints, not as a network that exactly solves the complete thermo-diffusive problem.

Finally, the residual model is trained and evaluated on samples from one operating trajectory. The random pointwise split supports interpolation inside that trajectory but does not demonstrate transfer to a new current history. A stronger generalization study would require several independent operating cycles and trajectory-wise splitting, with entire current profiles reserved for validation and testing. Within the scope of the present experiment, the Hybrid-PINN provides a clear and quantified extension of the standalone simplified-physics PINN.

22 Operational Benchmark and Model-Selection Guidance

This final part consolidates the numerical evidence produced throughout the study and translates it into operational model-selection guidance. Its purpose is not to identify a universally superior algorithm, but to determine which strategy is appropriate under different requirements concerning accuracy, physical reliability, offline cost, online latency, robustness, and engineering effort.

The benchmark provides the quantitative evidence required to answer the six research questions introduced in Section 1. In particular, it evaluates:

- whether the higher-fidelity simulator is sufficiently expensive to justify adopting a reduced representation;
- whether an explicit simplified physical model or a physics-informed learned surrogate provides the more appropriate baseline;
- how accurately each model reproduces the reference solution and which physical properties are satisfied exactly or only empirically;
- the complete offline cost, online latency, speedup, and computational break-even point of each applicable strategy;
- the accuracy gained by residual correction relative to its added training, deployment, and validation cost;
- the sensitivity of the learned strategies to stochastic initialization, parameter changes, and operating conditions outside the nominal training case.

These dimensions correspond to evidence categories rather than independent model rankings. A strategy is considered operationally preferable only when its combined accuracy, physical admissibility, latency, robustness, and lifecycle cost satisfy the requirements of the intended application.

The numerical values reported in the preceding technical chapters document representative development runs and explain the behaviour of the individual methods. Final operational comparisons must instead be based on the aggregated benchmark statistics obtained with the controlled protocol, declared hardware, and recorded software environment described below.

Robustness evidence is necessarily model-specific. The FVM-FNN branch is tested under perturbations of the simplified physical parameters and initial state; the thermal models are tested on interpolation and extrapolation current trajectories; and the standalone PINN and Hybrid-PINN are evaluated primarily within a single operating trajectory. The benchmark therefore does not claim the same type or extent of out-of-domain validation for every model.

22.1 Benchmark Protocol and Scope

Each benchmark execution records the effective configuration, random seed, repeat index, computational device, software environment, dataset provenance, and execution timestamp. Accuracy and physical-consistency metrics are computed on aligned model outputs, whereas latency is measured through dedicated warm-up and timing repetitions. Every seed-repeat combination is executed in a separate Python process so that imports, model state, device memory, and temporary checkpoints from one execution do not affect the next one. The repeat index identifies a new execution of the same configured seed; it does not replace or modify that seed.

Models addressing different outputs are not ranked as if they were direct substitutes. Concentration models are compared against the thermo-diffusive reference concentration field; thermal models are compared on temperature derivative identification and trajectory rollout. Similarly, correction-only latencies are not compared with complete-solver latencies unless the complete end-to-end pipeline is also reported.

22.2 Measurement Metrics

The benchmark results are generated using several task-specific metrics, each associated with a defined evaluation purpose. The metrics assess not only predictive accuracy, but also physical consistency, computational cost, inference speed, robustness, and sensitivity to stochastic training components.

Let y_i denote a value produced by the higher-fidelity reference simulator, $\hat{y}_i$ the corresponding model prediction, and N the total number of evaluated points. The pointwise prediction error is defined as

$$e_i = \hat{y}_i - y_i. \quad (88)$$

All accuracy metrics are evaluated on aligned spatial and temporal grids.

22.2.1 Prediction Accuracy

Four complementary error metrics are used to quantify the difference between a model prediction and the reference solution.

Relative L_2 error. The relative L_2 error is defined as

$$\varepsilon_{L_2} = \frac{\|\hat{\mathbf{y}} - \mathbf{y}\|_2}{\|\mathbf{y}\|_2}. \quad (89)$$

This dimensionless metric measures the global magnitude of the prediction error relative to the magnitude of the reference solution. It is particularly useful when comparing errors associated with quantities or datasets having different numerical scales. A value close to zero indicates that the predicted solution is globally close to the reference one.

Root Mean Squared Error. The Root Mean Squared Error (RMSE) is defined as

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2}. \quad (90)$$

The RMSE measures the characteristic magnitude of the prediction error in the same units as the predicted variable. Since errors are squared before averaging, this metric assigns greater importance to large local deviations. It is therefore adopted as the primary accuracy metric in the comparison between the different concentration models.

Mean Absolute Error. The Mean Absolute Error (MAE) is defined as

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i|. \quad (91)$$

Unlike the RMSE, the MAE weights all deviations linearly and is less sensitive to isolated large errors. It provides an interpretable estimate of the typical absolute prediction error and complements the information provided by the RMSE.

Maximum absolute error. The maximum absolute error is defined as

$$e_{\max} = \max_i |\hat{y}_i - y_i|. \quad (92)$$

This metric identifies the worst prediction error over the entire evaluation domain. It is relevant in industrial applications in which a model may be unacceptable even when its average error is small, if a large local deviation occurs under a critical operating condition.

22.2.2 Spatially Resolved Accuracy

The concentration error metrics are evaluated at different levels of spatial aggregation:

- **Full-field accuracy:** the metrics are computed over the complete space–time concentration field $c(\rho, \tau)$.
- **Centre accuracy:** the metrics are evaluated at the centre of the particle, where $\rho = 0$.
- **Surface accuracy:** the metrics are evaluated at the particle surface, where $\rho = 1$.
- **Average-concentration accuracy:** the metrics are computed using the volume-averaged concentration.

The full-field metrics provide a global evaluation of the solution, whereas the centre and surface metrics identify local inaccuracies that may be hidden by spatial averaging. Surface accuracy is particularly relevant because the surface concentration is directly connected to the applied flux and may influence other electrochemical quantities.

For a spherically symmetric particle, the volume-averaged normalized concentration is computed as

$$\bar{c}(\tau) = \frac{\int_0^1 c(\rho, \tau) \rho^2 d\rho}{\int_0^1 \rho^2 d\rho}. \quad (93)$$

The average-concentration error is used to assess whether a model correctly reproduces the global lithium inventory. However, this metric must be considered together with the local field error, since an accurate average value does not necessarily imply an accurate radial concentration distribution.

22.2.3 Training, Test, and End-to-End Accuracy

Different accuracy evaluations are stored depending on the model architecture:

- **Global accuracy** evaluates the predicted solution over the complete benchmark grid.
- **Held-out or test accuracy** is computed using samples excluded from the training process and provides an estimate of generalization performance.
- **End-to-end accuracy** evaluates the complete prediction pipeline rather than the neural component alone.

For example, the end-to-end FVM-FNN evaluation includes both the simplified physical solver and the neural correction. This distinction prevents the accuracy of an isolated neural component from being reported as the accuracy of the complete software pipeline.

22.2.4 Physical Consistency

Low prediction error does not necessarily imply that a model satisfies the governing physical constraints. Consequently, a separate set of physics-based metrics is evaluated.

Mass-balance residual. For the normalized spherical diffusion problem, the expected average concentration is computed from the imposed dimensionless surface flux $j(\tau)$ as

$$\bar{c}_{\text{expected}}(\tau) = c_0 - 3 \int_0^\tau j(s) ds, \quad (94)$$

where c_0 is the initial normalized concentration. The mass-balance residual is then defined as

$$r_m(\tau) = \bar{c}_{\text{predicted}}(\tau) - \bar{c}_{\text{expected}}(\tau). \quad (95)$$

Three quantities are derived from this residual:

- **Maximum absolute mass-balance residual:**

$$r_{m,\max} = \max_{\tau} |r_m(\tau)|. \quad (96)$$

It measures the worst conservation violation during the simulation.

- **Mass-balance RMSE:**

$$\text{RMSE}_m = \sqrt{\frac{1}{N_\tau} \sum_{k=1}^{N_\tau} r_m(\tau_k)^2}. \quad (97)$$

It measures the average conservation error over the complete trajectory.

- **Final mass-balance residual:**

$$r_{m,f} = r_m(\tau_f). \quad (98)$$

It quantifies the accumulated inventory error at the end of the simulation.

A Boolean mass-balance indicator is also recorded. The test is passed when

$$r_{m,\max} < \varepsilon_m, \quad (99)$$

where the benchmark currently adopts a tolerance of $\varepsilon_m = 10^{-4}$. This indicator provides an immediate acceptance criterion, while the continuous residual metrics indicate the magnitude of a possible violation.

Physical concentration bounds. Since the concentration is normalized, a physically admissible solution should satisfy

$$0 \leq c(\rho, \tau) \leq 1. \quad (100)$$

The benchmark therefore records:

- the minimum predicted concentration;
- the maximum predicted concentration;
- the number of points outside the physical interval;
- the fraction of the complete field violating the bounds.

The number of violations is useful for identifying isolated non-physical predictions, while the violation fraction indicates whether the problem is local or widespread.

Numerical finiteness. A Boolean finiteness check verifies that every predicted value is finite. This check detects the presence of `NaN` or infinite values, which would indicate numerical instability or failure of the prediction pipeline.

22.2.5 Training Cost

The primary training-cost metric is the wall-clock training time,

$$t_{\text{train}}, \quad (101)$$

measured in seconds. Wall-clock time is preferred to the number of epochs alone because it directly represents the computational delay experienced by an industrial user and includes the effects of the selected hardware and software implementation.

The benchmark also records:

- requested number of epochs;
- number of completed epochs;
- epoch associated with the best validation result;
- best validation loss;
- number of validation evaluations;

- activation of early stopping;
- optimizer used at the end of training.

These quantities explain why a particular training time was obtained. For example, two runs may have the same maximum number of epochs but terminate at different times because of seed-dependent early stopping.

Additional offline costs are measured when required by the model pipeline:

- **Dataset-generation time**, representing the time required to run the physical solver and generate the training data.
- **Dataset-assembly time**, representing the cost of converting the numerical solution into the required tabular structure.
- **Residual-dataset generation time**, representing the cost of evaluating the frozen PINN and constructing the dataset used by the residual model.
- **Correction-pipeline time**, defined as the sum of residual-dataset generation and residual-model training.

These measurements distinguish the cost of training a model from the complete offline cost required to make the model operational.

22.2.6 Inference and Online Computational Cost

The benchmark distinguishes several definitions of online latency:

- **Online inference time**: time required to evaluate a trained model.
- **Correction-only time**: time required by the neural correction without including the simplified physical solver.
- **End-to-end time**: time required by the complete operational pipeline.
- **Rollout time**: time required to generate a complete temporal trajectory by integrating a learned dynamical model.

For the FVM-FNN model, the end-to-end time includes both the simplified FVM simulation and the neural correction. This is the appropriate metric for comparison with the reference FVM because it represents the latency experienced by the final user.

Each timed operation is first executed for a predefined number of warm-up iterations. The measured operation is then repeated multiple times. Accelerator devices are explicitly synchronized before and after each measurement to prevent asynchronous execution from producing artificially low timings.

Given a set of N_t timing measurements t_i , the following statistics are reported:

- mean execution time;
- standard deviation;

- median;
- minimum and maximum;
- 5th and 95th percentiles;
- number of timing repetitions.

The mean provides the expected execution time, whereas the 95th percentile is useful as an operational upper-bound indicator: approximately 95% of the measured executions are completed within this time. The standard deviation quantifies timing variability due to system activity and runtime effects.

When applicable, latency is also normalized by the number of predicted profiles or coordinates. These normalized quantities make it possible to evaluate how the implementation may scale with the requested output size.

22.2.7 Speedup

The speedup of a model relative to the reference FVM is defined as

$$S = \frac{t_{\text{reference}}}{t_{\text{model}}}, \quad (102)$$

where both times must represent comparable end-to-end operations.

A value $S > 1$ indicates that the alternative model is faster than the reference solver. For example, $S = 10$ indicates that the alternative approach is ten times faster. Speedup is used to provide an intuitive measure of the computational benefit obtained by adopting a reduced-order or data-driven strategy.

22.2.8 Computational Break-Even Point

A trained surrogate requires an initial computational investment that must be recovered through faster online evaluations. The compute-only break-even number of calls is defined as

$$N_{\text{BE}} = \frac{t_{\text{train}}}{t_{\text{reference}} - t_{\text{model}}}, \quad (103)$$

provided that $t_{\text{model}} < t_{\text{reference}}$.

This metric estimates how many online simulations must be performed before the time saved during inference compensates for the initial training cost. A low break-even value is desirable in applications involving repeated simulations, optimization, control, uncertainty quantification, or sensitivity analysis.

The current definition is intentionally limited to computational time. It does not include human development effort, software maintenance, model validation, or the generation of every possible reference dataset. Moreover, for the Hybrid-PINN, the reported break-even is incremental because the cost of training the frozen PINN is not included in the residual-model training time.

22.2.9 Robustness to Trajectory Perturbations

The robustness analysis evaluates whether the FVM–FNN correction remains beneficial when the input trajectory or the simplified physical model differs from the nominal training condition.

For each variant, the benchmark records the RMSE of the uncorrected simplified solution,

$$\text{RMSE}_{\text{simplified}}, \quad (104)$$

and the RMSE obtained after applying the neural correction,

$$\text{RMSE}_{\text{hybrid}}. \quad (105)$$

The relative RMSE reduction is defined as

$$\Delta_{\text{RMSE}} = 100 \left(1 - \frac{\text{RMSE}_{\text{hybrid}}}{\text{RMSE}_{\text{simplified}}} \right). \quad (106)$$

Its interpretation is as follows:

- $\Delta_{\text{RMSE}} > 0$: the hybrid correction improves the simplified model;
- $\Delta_{\text{RMSE}} = 0$: the correction provides no measurable benefit;
- $\Delta_{\text{RMSE}} < 0$: the correction worsens the simplified prediction.

The initial simplified concentration and final dimensionless time of each variant are also recorded to describe the perturbation. Physical-bound and mass-balance checks are then repeated on the corrected trajectories.

This analysis is used to determine whether the neural correction has learned a general relationship or only a local correction valid near the nominal training trajectory.

22.2.10 Thermal-Model Metrics

The thermal models are evaluated separately under interpolation and extrapolation conditions. Interpolation tests conditions similar to the training domain, whereas extrapolation evaluates a trajectory outside the observed range.

The following temperature metrics are computed:

- **Temperature RMSE**: global quadratic error over the complete temperature trajectory.
- **Temperature MAE**: mean absolute temperature error.
- **Maximum temperature error**: worst absolute temperature deviation.
- **Relative L_2 temperature-rise error**: relative error computed on

$$\theta(t) = T(t) - T_{\text{amb}}, \quad (107)$$

rather than on the absolute Kelvin temperature. This prevents the ambient-temperature offset from artificially reducing the relative error.

- **Peak-temperature error**: difference between the predicted and reference maximum temperatures.

- **Peak-time error:** difference between the predicted and reference times at which the maximum temperature occurs.
- **Peak-current RMSE:** temperature RMSE evaluated during the maximum-current phase.
- **Cooling RMSE:** temperature RMSE evaluated after the active current phase.
- **Final-temperature error:** absolute error at the final simulation time.

The benchmark also evaluates the learned temperature derivative through normalized and physical derivative errors, including derivative MSE, RMSE, and MAE. Derivative metrics determine whether the model has correctly identified the local dynamics, while rollout metrics determine whether small local errors accumulate into a significant trajectory error.

This distinction is important because accurate derivative predictions do not automatically guarantee an accurate long-term integration.

22.2.11 Model Complexity

To complement timing and accuracy results, the benchmark records structural model information, including:

- number of trainable parameters;
- number of frozen and total parameters for the Hybrid-PINN;
- number of active terms in the sparse SINDy model;
- number of radial nodes;
- number of temporal points;
- total number of evaluated field points.

These quantities provide a hardware-independent indication of model complexity. They are used to interpret memory requirements, training cost, deployment difficulty, and model interpretability.

22.2.12 Stability Across Seeds and Repeated Runs

For every numerical metric, the benchmark collects the values produced by all successfully completed seed-repeat executions of the corresponding model. Different seeds change the stochastic initialization and data-loader order of the learned models. Repeats execute the complete worker again with the same assigned seed and therefore measure the reproducibility of that configured run, including variations in wall-clock and inference timing.

Let N_r denote the number of successful seed-repeat results in which a given metric is available, and let x_i denote the value stored for run i . The reported mean is

$$\bar{x} = \frac{1}{N_r} \sum_{i=1}^{N_r} x_i, \quad (108)$$

and the sample standard deviation is

$$s_x = \sqrt{\frac{1}{N_r - 1} \sum_{i=1}^{N_r} (x_i - \bar{x})^2}. \quad (109)$$

The mean describes the average value observed across the completed benchmark runs, whereas the standard deviation quantifies their run-to-run dispersion. For prediction metrics, this dispersion primarily reflects changes across seeds; repeated executions using the same deterministic seed may produce the same prediction. For computational times, it also captures differences in runtime conditions. Within each run, online latency is additionally measured through the dedicated timing repetitions described above.

The benchmark additionally reports:

- median;
- minimum and maximum;
- 5th and 95th percentiles;
- approximate 95% interval around the run-level mean.

The reported interval is computed using the normal approximation

$$\bar{x} \pm 1.96 \frac{s_x}{\sqrt{N_r}}. \quad (110)$$

This interval is reported as a descriptive summary of the executed benchmark runs. The number N_r stored in the output may vary between metrics when a quantity is not produced by every successful run.

A model with slightly better mean accuracy but high run-to-run variability may be less suitable for industrial deployment than a marginally less accurate but more reproducible alternative.

22.2.13 Engineering Effort

The computational benchmark is complemented by the recorded engineering effort. The primary quantities are:

- person-hours associated with each development phase;
- total person-hours;
- equivalent person-days, assuming eight working hours per day;
- documentary evidence associated with the recorded activity.

Engineering effort is kept separate from computational training time. Training time measures machine cost after the pipeline has been implemented, whereas engineering effort represents the human cost required to formulate, develop, validate, and document the solution.

This distinction is essential in industrial decision-making: a model may be inexpensive to train but expensive to develop, integrate, and maintain.

22.2.14 Reproducibility Information

Finally, each benchmark execution stores the random seed, effective configuration, software environment, selected computational device, dataset hashes, Git revision, and execution timestamp.

These quantities are not performance metrics themselves, but they establish the provenance of the reported results. They make it possible to determine whether differences between experiments arise from the model, the data, the hardware, or the software environment.

Overall, the combined metric set enables the models to be compared according to complementary technical and operational dimensions:

$$\text{Suitability} = f(\text{accuracy, physical consistency, computational cost, robustness, engineering effort}) \quad (111)$$

Consequently, model selection is not based exclusively on the minimum prediction error or maximum speedup, but on the balance that best satisfies the accuracy, reliability, latency, and development requirements of the considered industrial application.

Tables 32 and 33 summarize the key metrics used.

22.3 Benchmark Results

This subsection reports the numerical outputs of the publication benchmark. Interpretation, model-selection recommendations, and scenario-dependent considerations are deliberately deferred to the following discussion.

PINN clarification. The PINN results should be interpreted in light of its deliberately incomplete training physics: constant diffusivity and constant surface flux, whereas the reference model includes state-dependent diffusivity and a time-dependent current. This mismatch creates conflicting data and physics targets, particularly during current reduction and relaxation. The observed physical inconsistencies therefore reflect this training compromise, together with the approximate enforcement of the PDE and surface boundary condition; the initial condition and centre symmetry remain exactly satisfied. The benchmark mass residuals refer to the reference current profile. Consequently, the operational conclusions remain valid for the implemented configuration but should not be interpreted as a general limitation of PINNs but inside such context, assumptions and configuration.

22.3.1 Execution Protocol and Completeness

The publication preset evaluated seven implemented strategies: the thermo-diffusive reference FVM, the constant-diffusivity simplified FVM, the FVM-FNN model, the standalone PINN, the post-training Hybrid-PINN, SINDyC, and the thermal FFN. Five random seeds were used,

$$\{7, 19, 26, 41, 73\},$$

with two complete executions for each seed and model. The resulting benchmark contained 70 runs. All 70 runs completed successfully; no run failed and no model was skipped.

Table 32: Summary of the key benchmark metrics: accuracy, physics, and computational performance.

Metric	Definition	Evaluation purpose	Preferred
Relative L_2 error	$\varepsilon_{L_2} = \frac{\ \hat{\mathbf{y}} - \mathbf{y}\ _2}{\ \mathbf{y}\ _2}$	Measures the global prediction error relative to the magnitude of the reference solution. It enables dimensionless comparisons between models.	Lower
RMSE	$\sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2}$	Primary accuracy indicator. It penalizes large local deviations and is expressed in the same units as the predicted quantity.	Lower
MAE	$\frac{1}{N} \sum_{i=1}^N \hat{y}_i - y_i $	Measures the typical absolute prediction error and is less sensitive to isolated large deviations than RMSE.	Lower
Maximum absolute error	$e_{\max} = \max_i \hat{y}_i - y_i $	Identifies the worst local prediction error and supports safety-critical acceptance decisions.	Lower
Mass-balance residual	$r_m(\tau) = \bar{c}_{\text{predicted}}(\tau) - \bar{c}_{\text{expected}}(\tau)$	Measures whether the predicted concentration satisfies the global conservation law.	Lower
Maximum mass-balance residual	$r_{m,\max} = \max_{\tau} r_m(\tau) $	Quantifies the worst conservation violation over the complete simulation and determines whether the physical tolerance is satisfied.	Lower
Physical-bound violations	Number or fraction of points for which $c(\rho, \tau) \notin [0, 1]$	Detects non-physical normalized concentration predictions.	Zero
Training time	t_{train}	Wall-clock time required to optimize the model parameters. It represents the main model-specific offline computational cost.	Lower
Offline pipeline time	Dataset generation, dataset assembly, residual generation, and training time	Represents the complete computational effort required before the model can be deployed.	Lower
Mean online latency	$\bar{t} = \frac{1}{N_t} \sum_{i=1}^{N_t} t_i$	Measures the expected execution time of the model during operational use.	Lower
95th-percentile latency	$P_{95}(t)$	Provides an operational upper-bound indicator: approximately 95% of the measured executions are completed within this time.	Lower
End-to-end latency	Time required by the complete prediction pipeline	Enables a fair comparison between models by including every online component, such as the simplified solver and neural correction.	Lower
Speedup	$S = \frac{t_{\text{reference}}}{t_{\text{model}}}$	Quantifies the online computational advantage relative to the reference FVM solver.	Higher
Break-even calls	$N_{\text{BE}} = \frac{t_{\text{train}}}{t_{\text{reference}} - t_{\text{model}}}$	Estimates how many online simulations are required to recover the initial training cost.	Lower

Table 33: Summary of the key benchmark metrics: robustness, thermal performance, stability, complexity, and engineering effort.

Metric	Definition	Evaluation purpose	Preferred
RMSE reduction	$100 \left(1 - \frac{\text{RMSE}_{\text{hybrid}}}{\text{RMSE}_{\text{simplified}}} \right)$	Measures whether the hybrid correction improves or worsens the simplified model under nominal and perturbed conditions.	Higher
Temperature RMSE	RMSE between predicted and reference temperature trajectories	Measures the global accuracy of the thermal-model rollout under interpolation and extrapolation conditions.	Lower
Peak-temperature error	$ \hat{T}_{\max} - T_{\max} $	Evaluates the accuracy of the maximum predicted temperature, which may represent a critical operating condition.	Lower
Peak-time error	$ \hat{t}_{\max} - t_{\max} $	Measures whether the thermal model correctly predicts when the maximum temperature occurs.	Lower
Cooling RMSE	Temperature RMSE evaluated after the active current phase	Evaluates whether the learned thermal dynamics reproduce the cooling behaviour of the system.	Lower
Run-level standard deviation	$\sqrt{\frac{1}{N_r-1} \sum_{i=1}^{N_r} (x_i - \bar{x})^2}$	Quantifies dispersion across the completed seed-repeat executions.	Lower
95% confidence interval	$\bar{x} \pm 1.96 \frac{s_x}{\sqrt{N_r}}$	Provides the normal-approximation interval reported around the run-level mean.	Narrower
Trainable parameters	Total number of optimized model coefficients	Provides a hardware-independent indicator of model complexity, memory requirements, and deployment cost.	Context-dependent
Engineering effort	Person-hours and equivalent person-days	Quantifies the human effort required for problem formulation, implementation, validation, and documentation.	Lower

Each seed-repeat combination was executed in an isolated Python process. Deterministic algorithms were requested. Timing measurements used five warm-up executions, 30 repetitions for the finite-volume solvers, 100 repetitions for neural inference, and 30 repetitions for end-to-end pipelines. The numerical FVM models, the FVM-FNN model, and SINDyC were executed on CPU. The standalone PINN, Hybrid-PINN, and thermal FFN were executed on Apple MPS. The reported times therefore describe the implemented deployment configurations and include their respective hardware backends.

The run used Python 3.11.14, NumPy 2.4.2, pandas 3.0.5, and PyTorch 2.10.0 on an Apple Silicon arm64 system. The complete sequential benchmark required 4.63 h between creation and completion. The manifest stores the effective configuration, dataset hashes, environment, Git commit, and Git working-tree status associated with the execution.

The standalone PINN branch trained a new PINN for every seed-repeat run. The Hybrid-PINN branch instead used the same frozen PINN checkpoint in all ten runs and varied the seed used to train the residual network. Consequently, the standalone PINN and Hybrid-PINN rows below describe two separately executed benchmark branches rather than paired before-and-after realizations of the same seed-specific PINN.

22.3.2 Concentration-Field Accuracy

Table 34 reports the complete-grid accuracy relative to the stored thermo-diffusive reference field. Values are the mean and sample standard deviation across the ten successful seed-repeat results. The near-zero reference-FVM error is a numerical self-consistency result obtained by rerunning the data-generating model with the recorded configuration; it is not an error relative to experimental data.

Table 34: Complete-grid concentration accuracy in the publication benchmark (mean $\pm$ standard deviation across ten seed-repeat runs).

Model	Relative L_2 error	Field RMSE	Maximum absolute error
Reference FVM	$2.699 \times 10^{-14} \pm 0$	$2.011 \times 10^{-14} \pm 0$	numerical round-off
Simplified FVM	$1.102 \times 10^{-2} \pm 0$	$8.208 \times 10^{-3} \pm 0$	$2.189 \times 10^{-2} \pm 0$
FVM-FNN	$(3.978 \pm 1.219) \times 10^{-4}$	$(2.964 \pm 0.908) \times 10^{-4}$	$(1.985 \pm 0.645) \times 10^{-3}$
Standalone PINN	$(5.194 \pm 0.036) \times 10^{-2}$	$(3.870 \pm 0.027) \times 10^{-2}$	$(1.696 \pm 0.032) \times 10^{-1}$
Hybrid-PINN	$(1.666 \pm 0.305) \times 10^{-3}$	$(1.241 \pm 0.227) \times 10^{-3}$	$(1.588 \pm 0.131) \times 10^{-2}$

The deterministic simplified FVM produced a field RMSE of 8.208×10^{-3} . The FVM-FNN field RMSE was $(2.964 \pm 0.908) \times 10^{-4}$, corresponding to a 96.39% reduction relative to that common simplified baseline. On the held-out FVM-FNN test profiles, the baseline RMSE was 8.423×10^{-3} and the corrected RMSE was $(2.668 \pm 0.587) \times 10^{-4}$.

The standalone PINN field RMSE was $(3.870 \pm 0.027) \times 10^{-2}$. Across the ten Hybrid-PINN residual-training runs, the corrected field RMSE was $(1.241 \pm 0.227) \times 10^{-3}$. The observed RMSE ranges were $[3.849 \times 10^{-2}, 3.919 \times 10^{-2}]$ for the standalone PINN, $[1.689 \times 10^{-4}, 3.916 \times 10^{-4}]$ for the FVM-FNN model, and $[8.340 \times 10^{-4}, 1.451 \times 10^{-3}]$ for the Hybrid-PINN.

22.3.3 Physical-Consistency Metrics

The physical checks in Table 35 use a mass-balance acceptance tolerance of 10^{-4} . Bounds violations count field coordinates for which the normalized concentration falls outside $[0, 1]$.

Table 35: Physical-consistency results for the concentration models (mean $\pm$ standard deviation across ten seed-repeat runs).

Model	Maximum mass residual	Bounds violations	Finite runs	Mass test
Reference FVM	4.467×10^{-5}	0	10/10	10/10 passed
Simplified FVM	4.552×10^{-5}	0	10/10	10/10 passed
FVM-FNN	$(4.49957 \times 10^{-5}) \pm (7.50 \times 10^{-10})$	0	10/10	10/10 passed
Standalone PINN	$(1.326 \pm 0.031) \times 10^{-1}$	123.8 ± 21.9	10/10	0/10 passed
Hybrid-PINN	$(5.201 \pm 1.039) \times 10^{-3}$	0	10/10	0/10 passed

The mean standalone-PINN bounds-violation fraction was 6.098×10^{-3} , corresponding to 0.610% of the 20,301 evaluated coordinates. The Hybrid-PINN predictions contained no bounds violations in the ten benchmark runs. Its mean maximum mass residual was lower than that of the standalone PINN but remained above the 10^{-4} acceptance tolerance.

22.3.4 Training and Online Timing

Table 36 reports the wall-clock measurements used by the operational comparison. The FVM-FNN latency includes both the simplified finite-volume simulation and the neural correction. The correction-only FVM-FNN evaluation required $(1.307 \pm 0.100) \times 10^{-4}$ s for the complete set of profiles, but this quantity is not used as the end-to-end deployment latency.

For each model, the reported p95 value is the mean of the run-level 95th percentiles. The speedup is computed using the corresponding mean online latency and the mean reference-FVM latency.

Table 36: Training and online wall-clock measurements in the publication benchmark. Timing values are mean $\pm$ standard deviation across ten seed-repeat runs.

Model	Training time [s]	Mean online latency [ms]	Mean run-level p95 [ms]	Speedup
Reference FVM	–	108.679 ± 1.219	112.472	1.000
Simplified FVM	–	80.767 ± 0.890	83.369	1.346
FVM-FNN	11.797 ± 3.803	83.793 ± 2.746	86.016	1.297
Standalone PINN	590.203 ± 57.388	4.648 ± 0.217	4.877	23.383
Hybrid-PINN	121.328 ± 33.288	7.031 ± 0.238	7.657	15.457

The Hybrid-PINN training value in Table 36 refers to the residual network and does not include

the previous optimization of the frozen PINN. Residual-dataset generation required 0.223 ± 0.057 s, producing a complete measured correction-pipeline time of 121.551 ± 33.285 s.

Using the compute-only definition implemented in the benchmark, the FVM-FNN break-even point was 474 calls relative to the reference FVM, the standalone PINN break-even was 5,673 calls, and the incremental Hybrid-PINN break-even was 1,194 calls. Reference-data generation, human effort, and maintenance are not included in these break-even values. The Hybrid-PINN value additionally excludes the original frozen-PINN training cost.

The completed epoch counts were $6,285 \pm 2,057$ for the FVM-FNN model, $2,156 \pm 214$ for the standalone PINN, and $2,078 \pm 603$ for the Hybrid-PINN residual network. Their respective trainable-parameter counts were 5,573, 30,701, and 3,329. The Hybrid-PINN additionally contained 30,701 frozen PINN parameters, for 34,030 total parameters.

22.3.5 Frozen FVM-FNN Evaluation on Simplified-Model Variants

The nominal FVM-FNN correction was evaluated without retraining on five stored simplified-model variants. Table 37 reports the variant descriptors and complete-field errors. Positive RMSE reduction denotes an improvement over the corresponding uncorrected variant; negative reduction denotes a larger corrected error.

Table 37: Frozen FVM-FNN results on the stored simplified-model variants (mean $\pm$ standard deviation across ten seed-repeat runs).

Variant	$C_{0,\text{simpl}}$	τ_f	Baseline RMSE	Hybrid RMSE	RMSE reduction [%]
0	0.48	1.1211	2.304×10^{-2}	2.088×10^{-2}	9.40 ± 3.87
1	0.49	1.2872	1.102×10^{-2}	9.741×10^{-3}	11.60 ± 3.77
2	0.50	1.4533	2.618×10^{-3}	8.381×10^{-3}	-220.12 ± 4.44
3	0.51	1.5529	1.185×10^{-2}	1.875×10^{-2}	-58.26 ± 6.26
4	0.52	1.6692	2.250×10^{-2}	3.015×10^{-2}	-34.00 ± 2.53

No physical-bound violation was recorded for any corrected variant. The inventory diagnostic stored for these tests was evaluated against the nominal reference inventory. Consequently, for variants having a different initial concentration, that diagnostic is not a self-conservation test relative to the variant’s own initial state.

22.3.6 Thermal-Model Results

The thermal models were evaluated on one interpolation trajectory and one extrapolation trajectory. These outputs are distinct from the concentration field and are therefore reported separately. Table 38 contains the mean and standard deviation across ten seed-repeat runs.

All SINDyC runs selected a threshold of 10^{-3} and produced two active terms. Their prediction metrics were identical across seed-repeat runs. The thermal-FFN interpolation RMSE ranged from 1.808×10^{-3} to 6.909×10^{-3} K, while its extrapolation RMSE ranged from 4.126×10^{-2} to 8.527×10^{-2} K. Both models predicted the peak time without an observed error on the two evaluated trajectories.

Table 38: Thermal-model accuracy, training cost, rollout time, and complexity in the publication benchmark.

Metric	SINDyC	Thermal FFN
Interpolation temperature RMSE [K]	$3.40 \times 10^{-4} \pm 0$	$(3.808 \pm 2.012) \times 10^{-3}$
Extrapolation temperature RMSE [K]	$4.94 \times 10^{-4} \pm 0$	$(6.549 \pm 1.646) \times 10^{-2}$
Interpolation peak-temperature error [K]	$6.27 \times 10^{-6} \pm 0$	$(4.517 \pm 3.055) \times 10^{-3}$
Extrapolation peak-temperature error [K]	$7.1 \times 10^{-5} \pm 0$	$(8.465 \pm 3.707) \times 10^{-2}$
Training/identification time [s]	5.454 ± 0.188	34.437 ± 3.048
Interpolation rollout time [s]	1.017 ± 0.010	2.722 ± 0.195
Extrapolation rollout time [s]	1.333 ± 0.034	3.179 ± 0.108
Model complexity	2 active terms	1,185 trainable parameters

22.3.7 Recorded Engineering Effort

The project effort log contains 152 person-hours, equivalent to 19 person-days at eight hours per day. Table 39 reports the recorded phase breakdown. These values describe human engineering effort and are separate from the machine-training and benchmark-execution times reported above.

Table 39: Recorded engineering effort by project phase.

Phase	Person-hours	Person-days	Share [%]
Problem definition	36	4.500	23.7
Reference-model development	16	2.000	10.5
Dataset engineering	6	0.750	3.9
Surrogate development	26	3.250	17.1
Training	3	0.375	2.0
Validation	28	3.500	18.4
Documentation	37	4.625	24.3
Total	152	19.000	100.0

Problem definition, validation, and documentation together accounted for 101 person-hours. Surrogate development accounted for 26 person-hours, while the activity recorded specifically as training accounted for 3 person-hours.

23 Conclusions and Operational Guidance

This study addressed a practical model-selection problem: when should a high-fidelity solver be replaced by a reduced model, which reduced-modelling strategy should be selected, how reliable and fast is the resulting software, and when is a separate correction model worthwhile? The benchmark does not identify a universally superior architecture. Instead, it shows that the appropriate choice depends on the required combination of online latency, admissible error, physical guarantees, expected number of evaluations, and distance from the conditions represented during development.

23.1 The observed accuracy–latency trade-off

Figure 37 summarises the concentration-field accuracy and mean online wall time of the five principal alternatives. The axes are logarithmic; consequently, their separation represents changes of orders of magnitude rather than small absolute differences.

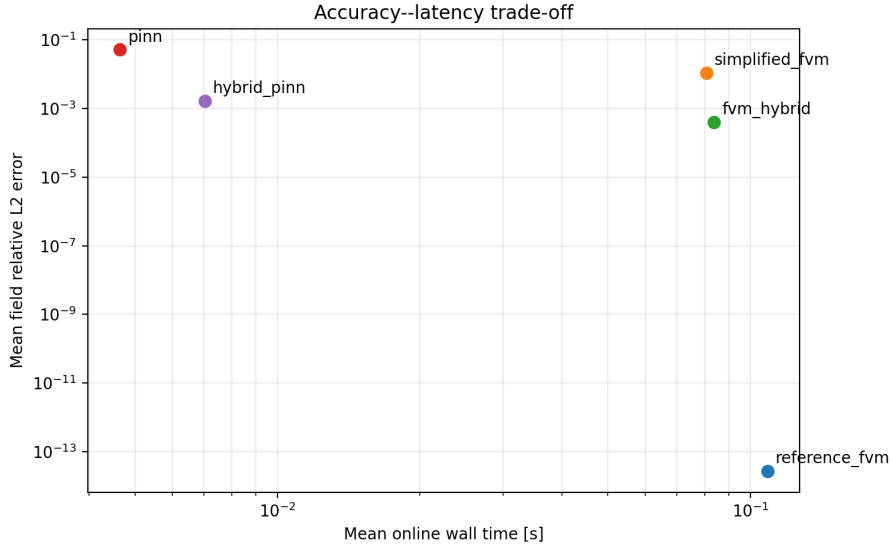


Figure 37: Measured concentration-field accuracy–latency trade-off for the publication benchmark. The error is computed with respect to the reference FVM solution. The near-zero error of `reference_fvm` is therefore a numerical self-consistency result and must not be interpreted as experimental validation. Offline cost, physical consistency and out-of-domain robustness are not represented by the two axes.

The reference FVM provides the numerical target and the strongest physical assurance, but requires approximately 108.7 ms per evaluation. The simplified FVM reduces this time only moderately, to 80.8 ms, while introducing a mean relative L^2 error of approximately 1.10×10^{-2} . Its value is therefore not extreme acceleration, but an interpretable, training-free approximation with the same conservation structure as the reference implementation.

Correcting the simplified solution with the FVM–FNN hybrid reduces the mean relative L^2 error to approximately 3.98×10^{-4} , with an end-to-end latency of 83.8 ms. On the nominal trajectory, this

corresponds to a paired RMSE reduction of 96.39% relative to the simplified FVM, while retaining the mass-balance and bound checks satisfied by the physical baseline. The correction is therefore highly effective when the operating condition is represented by its training domain, although it is not intended to replace the physical solver itself.

The neural branch occupies a different region of the trade-off. The standalone PINN is the fastest evaluated model, at approximately 4.65 ms, but it also has the largest mean relative L^2 error, approximately 5.19×10^{-2} , a maximum mass residual of approximately 1.33×10^{-1} , and bound violations in about 0.61% of the evaluated points. The Hybrid-PINN reaches approximately 7.03 ms and a substantially smaller mean relative L^2 error of 1.67×10^{-3} , with no detected bound violations. Nevertheless, its maximum mass residual, approximately 5.20×10^{-3} , remains above the acceptance threshold. Moreover, the standalone PINN runs were retrained for each seed, whereas the Hybrid-PINN residual corrector was built on a fixed PINN checkpoint. Their error difference is thus evidence about the performance of two implemented pipelines, not a paired causal estimate of the improvement produced by correcting each independently trained PINN.

If only the two axes in Figure 37 are considered, the Hybrid-PINN is both faster and more accurate than the simplified FVM. This does not make the latter operationally obsolete: the simplified FVM requires no training, preserves the imposed physical structure, and does not inherit checkpoint or training-distribution dependence. The figure must therefore be read as one layer of the decision, rather than as a complete ranking. The error reported for the reference FVM has no independent accuracy meaning, since the benchmark dataset is generated by the same solver. Its value, of the order of 10^{-14} , only reflects floating-point round-off and data-storage precision. Therefore, it must not be interpreted as evidence of physical or experimental accuracy.

23.2 Answers to the guiding questions

When should a reduced model be selected? For the present problem, the reference solver already completes one evaluation in approximately 0.11 s. Consequently, model reduction is not automatically justified for isolated simulations or small workloads. It becomes relevant when millisecond-scale response is a hard requirement, when the solver is embedded in optimisation, control, uncertainty quantification or repeated parameter studies, or when future extensions make the reference model substantially more expensive. The number of expected online calls must be considered together with the cost of producing, training and validating the reduced model.

Which reduced-modelling strategy should be selected? The simplified FVM is the most defensible baseline when interpretability, zero training cost and physical consistency are more important than maximum acceleration. The standalone PINN provides the smallest measured latency, but its present accuracy and conservation results make it unsuitable as the default reliable solver. The Hybrid-PINN is the strongest option when sub-10 ms inference is required and a monitored approximation is acceptable. The FVM-FNN hybrid is the most accurate non-reference pipeline on the nominal concentration trajectory and is preferable when an approximately 84 ms response is sufficiently fast.

How reliable are the alternatives? Reliability cannot be represented by field error alone. The reference FVM, simplified FVM and FVM-FNN hybrid satisfy the implemented mass-balance and physical-bound diagnostics. The standalone PINN violates both the selected conservation tolerance and, in a small fraction of predictions, the admissible bounds. The Hybrid-PINN removes

the observed bound violations but does not recover exact conservation. These findings support the use of explicit physics diagnostics and fallback rules in deployment; a low prediction error is not, by itself, a physical guarantee. The reference solution is also a computational benchmark rather than experimental ground truth, so the conclusions establish relative numerical fidelity within the study and not absolute validation of the underlying electrochemical model.

How fast are training and inference? The measured online times separate the alternatives into two groups: approximately 81–109 ms for the FVM-based pipelines and approximately 5–7 ms for the PINN-based pipelines. The corresponding offline costs differ substantially. FVM–FNN training required about 11.8 s, whereas standalone PINN training required about 590 s and the Hybrid-PINN correction stage about 122 s. Under the measured configuration, FVM–FNN training is amortised after approximately 474 evaluations relative to repeated use of the reference solver. The Hybrid-PINN correction stage requires approximately 1.2×10^3 evaluations to recover its incremental offline cost through online savings; this estimate excludes the original cost of producing the fixed PINN checkpoint. These break-even values are computational, not economic: data generation, validation, maintenance and engineering labour must be assessed separately.

When is a separate correction model worthwhile? Residual correction is justified when the baseline has a systematic and learnable error, remains available at inference time, and the deployment domain is sufficiently close to the data used to learn that error. This condition is clearly met by the nominal FVM–FNN experiment, in which the correction removes most of the simplified-model error without breaking the monitored constraints. It is only partially met by the PINN branch: the corrected pipeline reaches low field error, but exact conservation is not restored and the comparison with the independently retrained PINNs is not paired. A correction layer should therefore be treated as a local model of discrepancy, not as a universal repair mechanism.

How robust are the conclusions to new trajectories and stochastic training? The repeated-seed benchmark quantifies training variability, but it does not establish unrestricted generalisation. The deterministic physical baselines are insensitive to the nominal seed apart from timing noise, whereas the learned models exhibit seed-dependent training cost and predictive performance; consequently, a single favourable checkpoint is not sufficient evidence of stability. The frozen FVM–FNN corrector improves two of the five tested perturbed trajectories and worsens three, despite maintaining admissible bounds. This is the most important deployment warning produced by the study: nominal accuracy does not imply robustness to changes in initial concentration or final time. The PINN and Hybrid-PINN results are based on pointwise splits of the investigated trajectory and likewise do not demonstrate transfer to independent operating trajectories. A production implementation therefore requires an applicability-domain test, representative trajectory-level validation, and either a fallback to a physical solver or a retraining policy when the operating regime changes.

23.3 Operational model-selection guide

The results can be translated into the decision rules reported in Table 40. These rules are conditional on the present implementation, hardware and test domain and should be recalibrated whenever the governing inputs, discretisation or deployment platform changes.

For the thermal subsystem, SINDyC provides a particularly strong result under the tested conditions. It identifies a two-term model, achieves interpolation and extrapolation RMSE values

Table 40: Operational interpretation of the benchmark results.

Pipeline	Use when	Required safeguard or limitation
Reference FVM	Maximum numerical fidelity, physical assurance, low call volume, or out-of-domain operation is required.	It is the validation and fallback baseline; its near-zero benchmark error is self-consistency, not experimental truth.
Simplified FVM	A training-free, interpretable and physically consistent approximation is sufficient and an error of order 10^{-2} is acceptable.	The speed gain is modest; verify that the modelling assumptions remain valid.
FVM-FNN hybrid	High nominal accuracy and conservation are required, about 84 ms is acceptable, and repeated use can amortise training.	Check the applicability domain; fall back to or retrain against the physical solver under trajectory changes.
Hybrid-PINN	Sub-10 ms response is required and an error of order 10^{-3} is acceptable.	Monitor conservation and distribution shift; account separately for the cost and validity of the frozen PINN baseline.
Standalone PINN	Minimum latency is prioritised and the observed error and physics violations are tolerable, or the model is used as a research baseline.	It is not the recommended default for reliability-critical use in its present form.
Thermal SINDyC	The thermal dynamics remain sparse in the selected candidate library and compact, interpretable equations are desired.	Re-identify and validate the library if the thermal physics or excitation regime changes.

of approximately 3.40×10^{-4} K and 4.94×10^{-4} K, and outperforms the thermal feed-forward network in accuracy, rollout time and offline cost. This result is explained by the alignment between the underlying low-dimensional dynamics and the candidate function library; it should not be generalised into a universal superiority of symbolic regression over neural models.

23.4 Engineering implications and limitations

The effort analysis changes the interpretation of computational speed. Of the recorded 152 person-hours, 101 hours (66.4%) were devoted to problem definition, validation and documentation, whereas model training represented only 3 hours (2.0%). For an industrial project, the dominant cost is therefore not the optimiser run alone. It is the work required to define trustworthy quantities of interest, produce reference data, formulate diagnostics, establish a validity domain and document the decision process. A model with fast training can still be expensive to industrialise, while a model with a moderate offline cost may be worthwhile if its validation and maintenance requirements are controlled.

The quantitative rankings must also be interpreted within the limits of the experiment. Timing was measured for the implemented pipelines on one Apple Silicon system, with FVM-based models executed on CPU and neural models on MPS. The values describe deployable end-to-end implementations on that platform, but they are not hardware-neutral measures of algorithmic complexity. In addition, the benchmark uses the numerical reference model as ground truth, covers a limited set of trajectories and parameters, and does not quantify uncertainty due to model-form error or experimental noise.

Before industrial deployment, the next validation stage should therefore: (i) use trajectory-wise and parameter-wise test sets that are excluded from training; (ii) define quantitative applicability-domain and fallback criteria; (iii) repeat latency measurements on the target production hardware and at the intended batch size; (iv) include experimental or independently validated data where available; and (v) report lifecycle cost under explicit retraining frequencies and workload scenarios. These additions would convert the present comparative study into a qualification procedure for a production solver.

23.5 Final conclusion

The principal outcome is not that learning should replace physics, nor that the most accurate model should always be selected. The study shows that physical reduction and learned correction serve different operational needs. A simplified physical model provides a transparent and reliable baseline; a residual neural model can recover nominal accuracy when its domain is controlled; a PINN-based pipeline can move inference into the millisecond regime, but requires stronger conservation monitoring and validation; and sparse discovery is especially effective when the governing structure is genuinely contained in the chosen library.

Accordingly, the recommended software architecture is hierarchical: retain the reference solver as the qualification and fallback layer, expose the simplified physical solver as the robust low-cost baseline, and activate learned correction or PINN-based acceleration only inside a validated applicability domain. This architecture makes the accuracy–speed compromise explicit, preserves a route back to trusted physics, and turns the benchmark from a retrospective comparison into an operational guide for model selection, correction and maintenance.

References

- Steven L. Brunton, Joshua L. Proctor, and J. Nathan Kutz. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *Proceedings of the National Academy of Sciences*, 113(15):3932–3937, 2016a. doi: 10.1073/pnas.1517384113.
- Steven L. Brunton, Joshua L. Proctor, and J. Nathan Kutz. Sparse identification of nonlinear dynamics with control (SINDYc). *IFAC-PapersOnLine*, 49(18):710–715, 2016b. doi: 10.1016/j.ifacol.2016.10.249.
- Salvatore Cuomo, Vincenzo Schiano Di Cola, Fabio Giampaolo, Gianluigi Rozza, Maziar Raissi, and Francesco Piccialli. Scientific machine learning through physics-informed neural networks: Where we are and what’s next. *Journal of Scientific Computing*, 92:88, 2022. doi: 10.1007/s10915-022-01939-z.
- Shailesh Garg, Souvik Chakraborty, and Budhaditya Hazra. Physics-integrated hybrid framework for model form error identification in nonlinear dynamical systems. *Mechanical Systems and Signal Processing*, 173:109039, 2022. doi: 10.1016/j.ymssp.2022.109039.
- Chady Ghnatios and Francisco Chinesta. Discovering PDEs corrections from data within a hybrid modeling framework. *Mathematics*, 13(1):5, 2025. doi: 10.3390/math13010005.
- A. Emir Gümrükçüoğlu, James Burridge, Kieran O’Regan, and Jamie M. Foster. Inferring solid-state diffusivity in lithium-ion battery active materials: Improving upon the classical GITT method, 2024. Preprint dated 26 April 2024.
- Meng Guo and Ralph E. White. An approximate solution for solid-phase diffusion in a spherical particle in physics-based Li-ion cell models. *Journal of Power Sources*, 198:322–328, 2012. doi: 10.1016/j.jpowsour.2011.08.096.
- Xiaowei Jia, Jared Willard, Anuj Karpatne, Jordan S. Read, Jacob A. Zwart, Michael Steinbach, and Vipin Kumar. Physics-guided machine learning for scientific discovery: An application in simulating lake temperature profiles. arXiv preprint arXiv:2001.11086, 2020.
- Subarna Khanra, Vijay Kumar Kukreja, and Indu Bala. Physics-informed neural networks for differential equation solutions: A comprehensive review. *Neurocomputing*, 680:133317, 2026. doi: 10.1016/j.neucom.2026.133317.
- K. R. Niraimathi, S. Amutha, R. Dhanush, and B. Surendiran. A systematic review and numerical evaluation on physics-informed neural network for blood flow dynamics. *WIREs Data Mining and Knowledge Discovery*, 16:e70108, 2026. doi: 10.1002/widm.70108.
- Hao Tu, Scott Moura, Yebin Wang, and Huazhen Fang. Integrating physics-based modeling with machine learning for lithium-ion batteries. arXiv preprint arXiv:2112.12979, 2024.
- Zongren Zou, Xuhui Meng, and George Em Karniadakis. Correcting model misspecification in physics-informed neural networks (PINNs). *Journal of Computational Physics*, 505:112918, 2024. doi: 10.1016/j.jcp.2024.112918.

Appendices

A Algorithm[1] – Constant diffusion coefficient

Algorithm 1: Constant-diffusivity baseline finite-volume solver

Input: Physical parameters, operating profile, mesh size N , stored time instants N_τ , solver tolerances.

Output: Baseline concentration field and numerical-validation quantities.

1. Read the physical and numerical parameters.
2. Compute $t_D = R_p^2/D_s$ and generate the dimensionless grids ρ and τ .
3. Build the finite-volume mesh and initialize $C(\rho, 0) = C_0$.
4. Define the applied current profile $I(t)$ and compute

$$j(t) = -\frac{I(t)}{FS_p}, \quad \delta(\tau) = \frac{R_p j}{D_s c_{s,\max}}.$$

5. Assemble the semi-discrete finite-volume system

$$\frac{d\mathbf{C}}{d\tau} = \mathbf{f}(\tau, \mathbf{C}).$$

6. Integrate the ODE system using the BDF solver.
7. Extract $C_s(\tau)$ and $\overline{C}(\tau)$.
8. Verify the mass balance and physical bounds.
9. Construct the baseline simulation dataset

$$(\rho, \tau, t, I, j, \delta, C).$$

10. Generate the analytical-comparison, mesh-convergence, conservation, and validation figures.
-

B Algorithm[2] – Thermo-diffusive model

Algorithm 2: Thermo-diffusive finite-volume solver

Input: Physical, thermal and numerical parameters, operating profile, mesh size N , stored time instants N_τ , solver tolerances.

Output: Reference concentration field, temperature evolution, validation quantities and PINN dataset.

1. Read the physical, thermal and numerical parameters.
2. Compute the diffusion time

$$t_D = \frac{R_p^2}{D_{s,\text{ref}}},$$

generate the dimensionless grids (ρ, τ) , build the finite-volume mesh and initialize

$$\mathbf{y}(0) = [C_0, \dots, C_0, T_{\text{ref}}]^T.$$

3. Define the applied current profile $I(t)$ and compute the intercalation and dimensionless fluxes.
4. At each ODE evaluation:

- (a) recover $t = \tau t_D$;
- (b) compute $D_s(C, T)$ and normalize it,

$$\hat{D}_i = \frac{D_{s,i}}{D_{s,\text{ref}}};$$

- (c) interpolate the interface diffusivities,

$$\hat{D}_{i+\frac{1}{2}} = \frac{\hat{D}_i + \hat{D}_{i+1}}{2};$$

- (d) assemble the concentration and thermal equations.

5. Integrate the coupled thermo-diffusive system using the BDF solver.
6. Extract

$$C(\rho, \tau), C_s(\tau), \overline{C}(\tau), T(\tau).$$

7. Verify mass conservation and physical bounds.
8. Construct the reference dataset

$$(\rho, \tau, t, I, j, \delta, C, T),$$

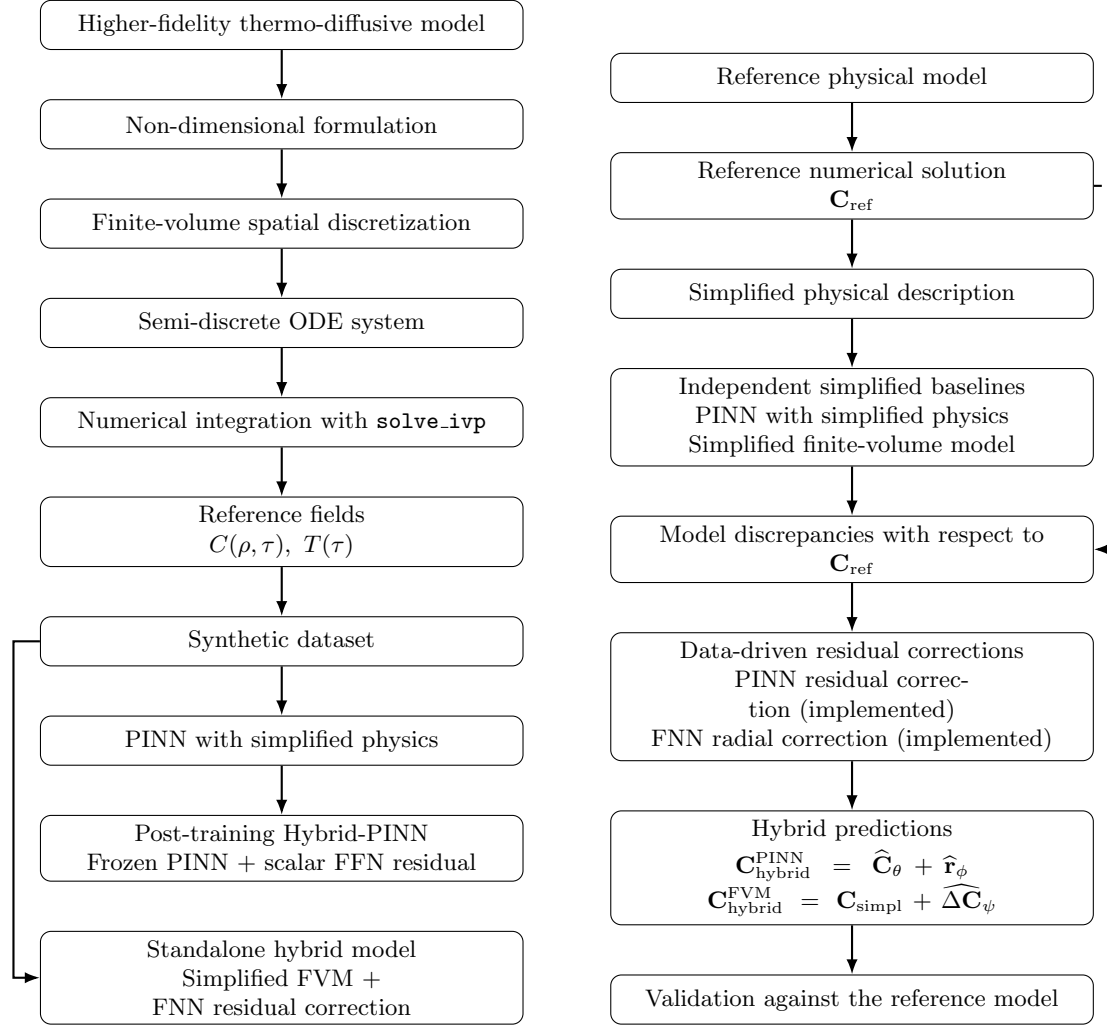
then generate the PINN dataset using only

$$(\rho, \tau, C),$$

discarding the temperature.

9. Generate the validation figures.
-

C Flowchart



(a) Reference-data generation and its use in the simplified-physics PINN and in the two residual-learning extensions.

(b) Common residual-learning methodology for the implemented Hybrid-PINN and the independently implemented standalone FVM-FNN hybrid model.

Figure 38: Overview of the numerical and methodological workflows adopted in the project. The left diagram describes the generation of the reference dataset and the resulting learning branches. The right diagram shows the common additive residual-learning structure of the implemented Hybrid-PINN and the independently implemented standalone FVM-FNN hybrid model.